

Project Final Report



Project Name: Server Enhancement

Team 7B

Supervised by:

Scott Mann

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ABSTRACT

The project is, undertaken for Captivate Connect®, a company based in Perth, Western Australia, addresses the critical need for upgrading and migrating from an outdated 2008 Windows Small Business Server (SBS) to a modern, efficient, and business-friendly server system. The project further involves the integration of Zentyal for Active Directory and user management, Google Workspace as the cloud-provider, as well as the implementation of Linode as a robust data backup solution. The overarching goal of this endeavour is to enhance system performance, reliability, security, and scalability while optimizing business operations for Captivate Connect®.

The main objectives of this project encompass thorough research into potential replacement server options, finalizing a new server solution, setting up the server for testing, integrating Zentyal as an Active Directory, implementing Linode for data backup, connecting all components, conducting comprehensive testing, and monitoring, and ensuring seamless communication and collaboration between components and the organization.

Through a well-defined project framework, the team successfully determined the major components of the solution, including Zentyal as the core server, Google Workspace for cloud-based productivity, and Linode as the backup solution. These components collectively address the project's key goals.

The project offers substantial business value by addressing the challenges of an outdated infrastructure, such as vulnerability to security threats and limited integration capabilities. The proposed solution, including Zentyal and Google Workspace, brings modern technology, regular updates, and cost-efficient solutions, resulting in enhanced productivity and security.

The implementation process involved multiple phases, including research, solution discovery, setup and configuration, extensive testing, and documentation. The team divided responsibilities among sprints, ensuring thorough coverage and analysis of each project component.

Zentyal was chosen as the core infrastructure, offering services ranging from LDAP to network management and security features. Google Workspace provides cloud-based productivity tools, including email, document storage, user management, and collaboration, while Linode acts as a secure backup solution with data redundancy and disaster recovery capabilities.

This project report details the Current system architecture, the challenges, the project management plans and strategies and finally the proposed solution and its testing.

INTRODUCTION

1.1 Purpose Statement

1.2 Audience Description

1.3 System Overview

1.4 Applicability Statement

1.5 Document Usage Description

PURPOSE STATEMENT

The purpose of the 'Project Upgrade and Migration' is to enhance the efficiency, security, and scalability of our organization's IT infrastructure by migrating from an outdated 2008 Windows Small Business Server to a modern and comprehensive solution. This project aims to seamlessly integrate Zentyal as our core server system, Google Workspace for efficient collaboration and user management, and Linode as a reliable data backup solution. By achieving these objectives, we endeavor to improve our operational performance, reduce security risks, and foster a more productive and adaptable work environment for our team and stakeholders. Ultimately, this project serves to align our technology with our business goals, ensuring our organization's continued growth and success.

AUDIENCE DESCRIPTION

The target audience for the Project is a diverse group of stakeholders involved in or impacted by the project. This audience consists of individuals and teams with varying roles and responsibilities within the organization. Here's a detailed breakdown of the audience:

Company Leadership and Decision-Makers:

C-Suite Executives: CEOs, CTOs, and other top-level executives interested in the project's impact on the organization's operations and ROI.

Department Heads: Heads of various departments concerned about how the project affects their team's daily activities.

IT Directors: Those overseeing the IT department and responsible for aligning technology with business goals.

End Users:

Employees: All employees in the organization who will be using the new systems and tools, and who need to adapt to the changes.

Clients and Stakeholders: External parties that rely on the organization's services and will be impacted by the project's success.

The project's success relies on effective communication, collaboration, and understanding among all these stakeholders. Each group has distinct needs, interests, and concerns, and it's crucial to tailor project updates and information to address these specific requirements. Clear and concise communication is vital to ensure that everyone remains informed and engaged throughout the project's lifecycle.

SYSTEM OVERVIEW

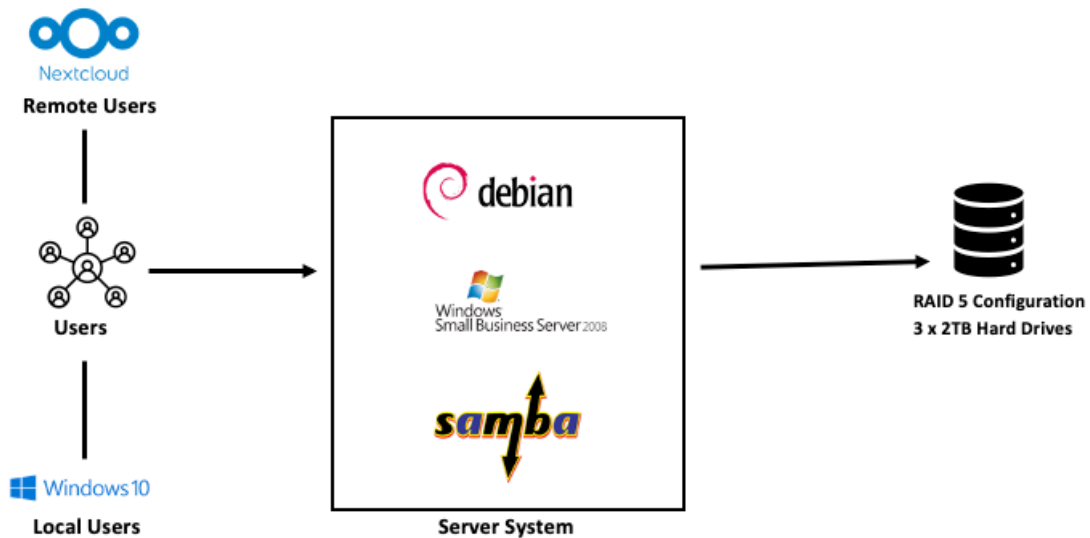


FIGURE 1

The Captivate Connect system overview offers a comprehensive picture of the organization's present technological infrastructure and operational configuration. This summary is essential for pinpointing the areas in need of modernisation, efficiency gains, and development. This is a thorough overview of the current system used by the company:

Captivate Connects Current System Overview:

Captivate Connect is a company in the telecommunications and audio services sector located in Burswood, Western Australia. The primary aim of the organisation is to offer customers a highly satisfying listening experience, especially during wait times for phone calls where they can avail themselves of content that appeals to them. Captivate Connect offers advanced telephone on hold recordings and telephony systems as part of its services.

Technical Infrastructure:

The technological infrastructure of Captivate Connect, which consists of the following components, is the backbone of its operations:

Debian 12 Server: A Debian 12 Server serves as the cornerstone of the company's internal infrastructure. This server is essential to the processing and storing of data.

Samba Software: The Debian server uses an outdated version of Samba to provide file and printer sharing. This decision was made due to operating system compatibility difficulties with the previous SBS 2008 server.

Storage Infrastructure: In RAID 5 configuration, three 2TB drives are used to store data. These engines are operated for various periods of time, ranging from 40 to 70 000 hours at power on. Data integrity and reliability are being called into question by the ageing storage infrastructure.

Backup Solution: A comprehensive data backup system has not been introduced, despite the organization's recognition of the need to provide a robust backup solution and considering options such as Black blaze.

Next cloud Instance: To improve remote access and teamwork, Captivate Connect has set up an instance of Next cloud on a different server. This instance gives workers who are located remotely access to data that is kept on the Debian server.

Demo server: Captivate Connect uses a demo server that is responsible for the production and creation of Audio files. However, it has become clear that the current way in which these audio files are shared is less optimal and may not fully meet the desired standards for accessibility and customer engagement.

Challenges and Objectives

The company currently faces several operational challenges:

Aging Infrastructure: Existing technical infrastructures need upgrading and becoming increasingly inefficient. This will pose a major risk to the security of data, availability, and overall system performance.

Lack of Backup: In case of hardware failure or other unexpected events, an organization is exposed to data loss because it does not have a reliable backup system.

Compatibility Issues: Compatibility issues with the old server from SBS 2008 are forcing you to use an older version of Samba. To ensure that operations are carried out smoothly, these compatibility concerns need to be addressed.

To address these problems, Captivate Connect has launched a project focusing on the modernization of its internal infrastructure, improvement in data management practices and increased availability of audio content. With a view to addressing data security, accessibility and performance concerns, the project intends to migrate data to more recent server infrastructures or cloud services.

This detailed introduction to the current system shall provide sufficient context in which we are able to understand the technological environment of an organisation, and why it is proposing a project. It emphasises the need for infrastructure to be upgraded in line with modern standards and industry best practices.

APPLICABILITY STATEMENT

Stimulating Client's Environment:

Server 2008: The presence of Server 2008 in the client's environment implies a legacy Windows Server operating system. Its applicability is limited due to its age and the lack of ongoing support and security updates from Microsoft. Therefore, an immediate upgrade or migration plan is highly recommended to ensure the security and functionality of the server.

Windows 10: Windows 10 is a modern, widely used client operating system with ongoing support. It is applicable for end-user workstations, offering a familiar and functional environment for daily tasks.

New Primary Domain Controller –

Zentyal: The introduction of Zentyal as a new primary domain controller indicates a transition to a more modern, Linux-based infrastructure for directory services. Zentyal can effectively manage user accounts, permissions, and network resources, making it suitable for a contemporary and cost-effective approach to directory services.

Cloud-based Solution: Google Workspace, a cloud-based suite of productivity and collaboration tools, is well-suited for businesses aiming to streamline communication, document management, and collaboration. Its applicability lies in facilitating remote work, email management, and real-time collaboration among employees.

Backup Solution: Linode, a cloud hosting provider, can be a reliable choice for backup solutions, especially for cloud-based and web applications. Its applicability is in providing secure data backup and recovery options, ensuring data integrity and availability.

Other Tools Used:

Google Credential: such as OAuth2, are widely applicable for secure authentication and authorization when integrating with various applications and services.

ForensIT: tools, known for their user profile management and migration capabilities, are applicable in scenarios where user profile migration and management are necessary, particularly during system upgrades or user transitions.

Restic: An open-source backup solution, is applicable for efficient, secure, and cross-platform backup and recovery operations, ensuring data protection and restoration.

DOCUMENT USAGE DESCRIPTION

This document has been divided into 6 sections, defined as follows:

1. Introduction:

This section provides an overall description of the Project scope, Business case and Project Management plan. Also gives details about the audience, purpose, and application of the project. Illustrates an overall description of this document and the conventions used.

2. System Architecture Design:

This section provides a detailed description about the clients' current infrastructure, the system architecture, and the challenges that the current system imposes.

3. User Stories:

This section provides detailed descriptions of all user stories for the project. Also includes unit testing.

4. Proposed Solution Architecture & Design:

This section contains diagrams and descriptions to describe the structure of the proposed solution. This section also talks about the different components in detail with steps to install and get the components running.

5. Usability Testing Report:

This section contains reports on the usability and system testing performed.

6. Special Notes:

This section provides additional material such as Glossary, that provide extra information for the use of this document.

USER STORIES

User Story Dictionary

User Story Definitions

USER STORIES DICTIONARY

As a user, I want to be able to back up my files to avoid data loss.

This user story refers to the backup strategy in the new solution that facilitates backup system snapshots or images.

As a server administrator, I want to be able to control server functions without complications.

This user story is related to the unified dashboard provided by Zentyal that displays all running activities and inactive features.

As a server administrator, I want to be able to control the server security.

This user story allows the server administrator to control the server and its features.

As a user, I want to be able to access my work system remotely.

This user story refers to the ability to access the workspace as a remote employee and still perform all activities as an onsite employee.

As a user or stakeholder, I want access to my work emails to keep updated.

This user story refers to having an email connected to the work domain that can be used outside the workspace.

As a data administrator, I want to schedule automatic backups.

This user story is to do with the backup solution outlined in the project called “Linode”.

As a user, I want to be able to access, edit and amend files from a remote location.

This user story allows users to remotely work while having the same accessibility as onsite.

As a data administrator and stakeholder, I want security and authentication measures within the system.

This user story is to provide server security and account authentication within the workspace.

As a user, I want privacy and secured access to my work accounts.

This user story refers to the “Google Credential Provider” tool incorporated in the solution.

USER STORIES DEFINITIONS

TABLE 1

User Story ID	User Story	Acceptance Criteria	Story Points
1	As a user, I want to be able to back up my files to avoid data loss.	☐ The files should be backed up into a backup system regularly to avoid data loss and to facilitate data recovery	11
2	As a server administrator, I want to be able to control server functions without complications.	☐ The Zentyal Dashboard must show all activities running and all features active and inactive.	25
3	As a server administrator, I want to be able to control the server security.	☐ The Zentyal control panel or dashboard should allow custom firewall configuration.	10
4	As a user, I want to be able to access my work system remotely.	☐ Users should be able to log in remotely using secure authentication methods. ☐ Remote access should provide the same functionality and accessibility as on-site access. ☐ Remote access should be stable.	20
5	As a user or stakeholder, I want access to my work emails to keep updated.	☐ Users should be able to log in to their work email and sync with their personal devices if required from any remote location with an internet connection. ☐ E-mail access should be encrypted and secure to protect and maintain sensitivity. ☐ Users should be able to receive and send emails in real time without any delays.	20
6	As a data administrator, I want to schedule automatic backups.	☐ The Linode backup solution should allow scheduling automatic data backups during any specified time. ☐ Backup scheduling should be configurable according to business needs.	22

		□ Automatic backups should be reliable and should run without failure.	
7	As a user, I want to be able to access, edit and amend files from a remote location.	□ Users should be able to access files and collaborate with teams effectively. □ Changes made should be updated and displayed without latency or delays	15
8	As a data administrator and stakeholder, I want security and authentication measures within the system.	□ System should implement security measures such as 2FA (Two Factor Authentication), SSO (Single Sign On), Roles and permissions. □ Accessibility should be user restricted based on roles and permissions. □ Security changes and measures should be recorded for reference and audit purposes.	10
10	As a user, I want privacy and secured access to my work accounts.	□ User accounts should be protected by strong security measures. □ Google Credential provider should work flawlessly to allow secure and smooth workflow. □ User data and communications should be encrypted to maintain privacy and security.	12

PROJECT DOCUMENTS

Project Charter

Business Case

Project Management Plan

Work Breakdown Structure (WBS)

Activity List

Gantt Chart

Change Log

Risk Management Plan

PROJECT CHARTER

PROJECT NAME: Server Enhancement

PROJECT SPONSOR: Captivate Connect Organization

PROJECT MANAGER: Madhura Dhumak

DUE DATE: 29th October 2023

PROJECT TEAM:

TABLE 2

NAME	ROLE
Bhawna Soni	Team Leader
Madhura Dhumak	Project Manager
Adib Hasheminejadfarhani	Risk Manager
Viraj Punjani	Quality Control Officer
Nagalojitthan Nagarajah	Chief of Research
Mano Beraka	Creative Director

KEY STAKEHOLDERS:

Internal Parties:

1. IT Team Members of Captivate Connect: These include networking and administration specialists, developers, and other IT professionals who will be directly involved in the project. This team may also be responsible for handling and resolving IT tickets and can play a role in aspects like troubleshooting and support.
2. Team 7B: This team will be responsible for doing the research necessary before the project is executed by the IT team. Team 7B can consist of a variety of different parts depending which can depend on the level of responsibility and technical acumen like the following:
 - SBS Researcher: This team will likely be involved in the research aspects of the project, potentially focusing on the SBS 2008 upgrade as well as another team member responsible for providing assistance as well as documentation purposes.
 - Linode and Google Workspaces Researcher: This team will focus on delivering the other aspects of the project dictated by the client. The tasks may include performing relevant research on topics such as various tools available for data migration or backup of user data.
 - Quality Assurance: This individual or team may be responsible for ensuring that the proposed solutions meet certain quality standards. They will be responsible for providing project oversight and conflict resolution. They can also be classified as the team leader.

External Parties:

1. Cloud Service Providers: These could be companies providing cloud services, such as Google Workspace or other potential alternatives.
2. Vendor/Service Provider for Server Infrastructure: If new server hardware is being considered, there may be external vendors or service providers involved in providing and setting up the necessary equipment.
3. Security Consultants (Potentially): If security is a significant concern, external security consultants or firms may be consulted to assess and advise on the best practices for securing the infrastructure.
4. Backup Service Provider (Potentially): If a new backup solution is considered, an external backup service provider may be involved in implementing and managing the backup system.
5. Potential Software Providers (e.g., Zentyal): If new software solutions are being considered, there may be external vendors involved in providing and implementing the software.
6. Consulting Team (Student Consulting Team): If an external consulting team is engaged to conduct project research, they would be considered external stakeholders.

BUSINESS CASE:

Modernising data management and addressing IT infrastructure concerns are the main goals of the Captivate Connect Project. The main goal is to ensure cost-effectiveness when moving data from outdated internal servers to more modern server architecture or cloud-based solutions.

SCOPE DESCRIPTION:

Captivate Connect is a service provider for businesses and government offices that provides advanced telephone On-Hold recordings and telephony systems. They are seeking a student consultants' team to do project research for the Captivate Connect organization on the following:

- The company's Internal infrastructure is getting old because of IT resources being focused on maintaining the availability and security of the cloud infrastructure. The team should ideally have a plan in place to move this data to a cloud-based storage system or to modern server hardware, whichever is more affordable.
- The SBS 2008 server does not function with the most recent version of Samba owing to some type of encryption incompatibility, therefore data is now saved on three 2TB drives that are configured in software RAID 5. The drives have between 40000 and 70000 hours of power on hours. The server is running Debian 12 and has an earlier version of Samba installed. It now lacks backup as well; they investigated Black Blaze as a potential solution but never used it.
- Perhaps it would be wiser to move most of the data to the cloud given that a distant worker has remote access to a next-cloud instance running on another server? Do we

think Google Workspace will be the greatest option or would you suggest anything else? What would happen if the client made an alteration while offline and another person made a modification at the same time? And how would it sync to the client machines?

- The company demo server is also located on the Cloud side. Demos are currently created and generated by a server and placed in a single location on our webserver. It has been brought to their attention that this is not the best way to share audio files, so it is up to the team how we handle this project. we can either create a portal for users to access their demos and have a way for clients to comment on their demos and suggest improvements, or we can just have it track when a demo has been listened to so the sales team knows when to move on to the next.

IN SCOPE:

They would like the team to focus more on:

- The 2008 MS SBS is aging and perhaps needs to be prioritized.
- A demo server project might be a more appropriate cup of tea when someone feels that they're developing something.
- In the company ideal, they want everyone involved to focus on their strengths and be able to look beyond the box for solutions.

OUT OF SCOPE:

- Simply suggesting Microsoft products without any other justification, as well as pricey corporate alternatives or ones that demand a lot of upkeep.
- Also, without a doubt, no Apple items. The workplace does not utilize Apple gadgets at any time since we don't want to be tied to their ecosystem.
- The objective is to teach us new skills and to find answers. The process and what we learn along the way are more important than the outcome.

PROJECT DELIVERABLE:

- Upgrading the current MS SBS 2008 server.
- Options to upgrade on the cloud.
- Suggesting a backup service.
- Providing good security measures.

MILESTONE SCHEDULE:

TABLE 3

Milestone	Cumulative Duration
Project Initiation	1 week
Infrastructure Assessment	Sprint 1
Data Migration Planning	Sprint 2

Infrastructure Upgrade	Sprint 3
Cloud Solution Evaluation	Sprint 4
Data Migration & Testing	Sprint 5
Documentation	From the Start

ACCEPTANCE CRITERIA:

- ☐ Researching on cloud services.
- ☐ Exploring options to migrate most of their data into a cloud service.
- ☐ Implementing a demo server.
- ☐ Analysing the data storage, backup, and security measures.

CONSTRAINTS:

Constraints Involved in the project are: -

1. Time
2. Resources Allocation
3. Budget Constraints

RISK ASSOCIATED:

- ☐ Server Hardware Failure
- ☐ Data Security Breach
- ☐ Data loss during migration
- ☐ Compatibility issues with SBS 2008 data

More of the risk will be further discussed in detail in the risk management plan.

PROJECT MANAGEMENT PLAN

1. Purpose

The purpose of the Project Management Plan (PMP) for the Captivate Connect project is to provide a structured framework that guides and governs all aspects of project management. This includes project initiation, planning, execution, monitoring, controlling, and closure. The specific purposes of the Project Management Plan are as follows:

1. **Project Direction:** The Project Management Plan (PMP) acts as a road map, describing the goals, parameters, and methodology of the project. It guarantees that everyone involved in the project has a common knowledge of its objectives by giving them everyone clear guidance.
2. **Resource Allocation:** It lays out the allocation and management of project resources, such as money, materials, and human resources, over the duration of the project. This guarantees the effective use of resources.
3. **Risk Management:** The Project Management Professional (PMP) recognises the hazards that may affect the project and provides methods to evaluate, reduce, and handle these risks. By being proactive, delays and disturbances are reduced.
4. **Quality Assurance:** It creates quality guidelines and protocols to guarantee that project deliverables fulfil or beyond stakeholder expectations. It has procedures for both quality assurance and verification.
5. **Documentation Repository:** Project documentation, such as deadlines, deliverables, milestones, timetables, and other vital information, is centrally stored by the PMP. It is necessary to have this document for compliance and references.
6. **Project Monitoring and Control:** Key performance indicators (KPIs) and procedures for comparing actual progress to the project schedule are defined in the plan, which offers a framework for tracking and managing project progress.
7. **Project Closure:** The project management plan (PMP) describes how to close off a project effectively and make sure that all deliverables are properly acknowledged. One way to document the lessons learnt is through a project review.
8. **Stakeholder Alignment:** The plan assists in providing all project stakeholders with a shared vision and ensures that everyone is working towards the same goals by outlining the project's strategy and objectives.

In conclusion, the Captivate Connect project's Project Management Plan (PMP) is an extensive document that offers the direction, framework, and techniques required to successfully manage and complete the project while considering any risks and modifications.

2. Definitions

TABLE 4

Term	Definition
Application	The software component of the system
SOP	Standard Operating Procedure
SEMs	SMEs stands for "Small and Medium-sized Enterprises." SMEs are businesses that are characterized by their relatively small size in terms of employees, revenue, and assets compared to large corporations.
PMP	Project Management Plan
KPIs	Key Performance Indicators

3. References

- ☐ Project Scope
- ☐ Business Case
- ☐ Client Requirements
- ☐ Activity List
- ☐ Gantt Chart
- ☐ Work Breakdown Structure
- ☐ Risk Management Plan

4. Project Organization

The project will follow a hierarchical management structure with clear roles and responsibilities assigned to each team member. The project manager will oversee the entire project and coordinate the efforts of the team members.

TABLE 5

Name	Role	Responsibilities
Bhawna Soni	Team Member	☐ Establish definite team goals and targets. ☐ Assign members of the team duties and responsibilities. ☐ Assist and mentor fellow team members. ☐ Keep an eye on and record the tasks' and projects' progress. ☐ Encourage the staff to collaborate and communicate effectively. ☐ Deal with disagreements and problems among team members. ☐ Update upper management on team progress and submit a report. ☐ Ensure that the group accomplishes its objectives and timelines.
Madhura Dhumak	Project Manager	☐ Specify the goals, deliverables, and scope of the project. ☐ Make a project plan that outlines the resources, budget, and timetables. ☐ Assign members of the team duties and responsibilities. ☐ Track project development and make required plan adjustments. ☐ Control possible risks and problems that may come up during the process. ☐ Engage in dialogue with all relevant parties, such as clients and higher management. ☐ Make that the project is completed on schedule, within budget, and to the required degree of quality. ☐ Take charge of the project team and encourage cooperation.

Adib Hasheminejadfarahani	Risk Manager	☐ Determine any dangers or hazards that might affect the organisation or project. ☐ Evaluate the possibility and effect of every risk. ☐ Create backup plans and techniques for mitigating risks. ☐ Risks should be tracked and monitored throughout the project or organisation. ☐ Share risks with the appropriate parties. ☐ Put risk management strategies into action to reduce adverse effects. ☐ Analyse how well risk management techniques are working.
Mano Pragyna Beraka	Creative Director	☐ Create a unique project vision and convey it. ☐ In charge of the creative staff, which could include of authors, illustrators, and designers. ☐ Give team members comments and creative direction. ☐ Make sure the goals of the project or brand are reflected in the creative output. ☐ Work along with other divisions, such product development and marketing. ☐ Give clients or stakeholders innovative notions and ideas. ☐ Preserve the level of creativity and consistency in your production.
Viraj Punjani	Quality Control	☐ Create and implement quality policies and guidelines. ☐ Verify that processes, goods, or services satisfy quality standards by testing and inspecting them. ☐ Determine any flaws or problems, then collaborate with the appropriate teams to fix them. ☐ Put quality improvement strategies into action. ☐ Team members should have quality control method training and education. ☐ Record and disseminate high-quality data and conclusions. ☐ Verify adherence to industry standards and quality rules.

Nagalojitthan Nagarajah	Chief Researcher	□ Initiate and manage research projects inside the company. □ Specify the aims and goals of the research. □ Create research methods and tactics. □ Control the resources and research teams. □ Analyse study data to produce new insights. □ Share study results to help in decision-making. □ Keep up on best practises, new developments in technology, and industry trends. □ Participate in the creation of tactics and ideas that are grounded in research.
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5. Product or Service Description

Captivate Connect is a service provider for businesses and government offices that provides advanced telephone On-Hold recordings and telephony systems. They are seeking a student consultants' team to do project research for the Captivate Connect organization on the following:

- The company's Internal infrastructure is getting old because of IT resources being focused on maintaining the availability and security of the cloud infrastructure. The team should ideally have a plan in place to move this data to a cloud-based storage system or to modern server hardware, whichever is more affordable.
- The SBS 2008 server does not function with the most recent version of Samba owing to some type of encryption incompatibility, therefore data is now saved on three 2TB drives that are configured in software RAID 5. The drives have between 40000 and 70000 hours of power on hours. The server is running Debian 12 and has an earlier version of Samba installed. It now lacks backup as well; they investigated Black Blaze as a potential solution but never used it.
- Perhaps it would be wiser to move most of the data to the cloud given that a distant worker has remote access to a next-cloud instance running on another server? Do we think Google Workspace will be the greatest option or would you suggest anything else? What would happen if the client made an alteration while offline and another person made a modification at the same time? And how would it sync to the client machines?
- The company demo server is also located on the Cloud side. Demos are currently created and generated by a server and placed in a single location on our webserver. It has been brought to their attention that this is not the best way to share audio files, so it is up to the team how we handle this project. we can either create a portal for users to access their demos and have a way for clients to comment on their demos and suggest improvements, or we can just have it track when a demo has been listened to so the sales team knows when to move on to the next.

6. Management processes

Project management procedures for the Captivate Connect project are essential to ensure that the project is planned, executed, and controlled effectively. The main project management practices of Captivate Connect are given in this overview:

1. Project Initiation:

Purpose: Determine the purpose, objective and original scope of the project.

Procedure:

- ☐ Determine the critical stakeholders and their respective roles.
- ☐ Develop a Project Initiation Document (PID) outlining project objectives and scope.
- ☐ Appoint a project manager and set up a project team.

2. Project Planning:

Purpose: Develop a comprehensive project plan that lays down the scope, time horizon, budget and resource allocation.

Procedure:

- ☐ Create a Project Management Plan (PMP) that outlines the resource allocation, quality standards, scope, timeline, and budget.
- ☐ To specify the tasks involved in the project, create a Work Breakdown Structure (WBS).
- ☐ Create a Gantt chart to see the timings of your projects.

3. Resource Allocation:

Purpose: Distribute project resources in an efficient manner.

Procedure:

- ☐ Determine the resources that are needed, such as personnel, tools, and supplies.
- ☐ Assign members of the team to project tasks.
- ☐ Ascertain the capability and availability of resources to fulfil project demands.

4. Risk Management:

Purpose: Identify, assess, and mitigate project risks.

Procedure:

- ☐ Create a Risk Register to keep track of possible hazards.
- ☐ Evaluate the effect and likelihood of each risk.
- ☐ Create risk response plans to control or lessen hazards that have been identified.

5. Quality Assurance and Control:

Purpose: Make that project deliverables adhere to quality requirements.

Procedure:

- ☐ Conduct quality control inspections at every stage of the project.
- ☐ Check that deliverables adhere to specified quality standards.
- ☐ It is important to quickly resolve quality concerns to uphold high standards.

6. Project Execution:

Purpose: Implement project activities according to the PMP.

Procedure:

- ☐ Assign responsibilities to group members and supply the required materials.
- ☐ Track the advancement and completion of tasks.
- ☐ To monitor the status of the project, hold daily stand-up meetings.

7. Project Monitoring and Control:

Purpose: The performance of the project shall be continuously monitored and necessary adjustments made.

Procedure:

- ☐ Check progress against the PMP on a regular basis.
- ☐ Make note of the key performance indicators (KPIs).
- ☐ To keep the project aligned, carry out modification orders as needed.

To ensure that the Captivate Connect project will be effectively planned, executed and monitored while maintaining high quality and efficient communication, these project management procedures are tailored according to the Project's specific needs. The procedures shall also provide for the mitigation of potential issues and to ensure project alignment with defined targets, risk management, change control as well as compliance with legislation.

7. Technical Processes

Below is a breakdown of the technical procedures used in the Captivate Connect project:

1. Project Design:

Purpose: To ensure that the project is planned and designed in accordance with the specified goals and specifications.

Process:

- ☐ Define the scope, objectives and specific requirements of the project.
- ☐ Strengthen the whole project plan, including allocation of resources, schedules and milestones.
- ☐ Design technical specifications for each component of a software or infrastructure.
- ☐ If necessary, you can design a user experience and interface.

2. Software and Infrastructure Implementation:

Purpose: To make a product or service come to life through implementation of the project plan.

Process:

- ☐ Set up a project team, select roles and allocate resources.
- ☐ For the required software and infrastructure components, be developed or configured as specified by the design specifications.
- ☐ The required applications or platforms can be customized or created.
- ☐ To ensure the safety of this product, security measures shall be implemented.

3. Testing and Quality Assurance:

Purpose: To verify the functionality, quality, and security of the product.

Procedure:

- ☐ To ensure that the product meets specified requirements, complete functional tests should be carried out.
- ☐ To assure that the performance and reliability are met, perform quality assurance tests.
- ☐ To identify and address vulnerabilities, in order to guarantee data protection and user privacy, perform security testing.
- ☐ Ensure that user experience and design of interface are validated for a smooth interaction between users.

4. Deployment and Rollout:

Purpose:

- To deploy the product to end-users or customers while ensuring a smooth transition.

Process:

- Identify the strategy for deployment, including user training, communications and support.
- Ensure that the deployment is carried out with minimal disruption to the operation.
- Make sure that the deployment is checked for any problems or feedback from users and provide immediate assistance.
- Help the product to perform better and meet new requirements by providing ongoing updates and maintenance.

These technical procedures are crucial to ensure that Captivate Connect is well planned, effectively implemented, thoroughly tested and successfully carried out. The project will be able to fulfil its objectives, maintaining high standards and dealing with possible technical challenges and security concerns based on these processes.

8. Schedule information

TABLE 6

Milestone	Cumulative Duration
Project Initiation	1 week
Infrastructure Assessment	Sprint 1
Data Migration Planning	Sprint 2
Infrastructure Upgrade	Sprint 3
Cloud Solution Evaluation	Sprint 4
Data Migration & Testing	Sprint 5
Documentation	From the Start

9. Revision Procedures

To ensure that the documentation of the Captivate Connect project remains accurate and up to date, the revision process of the Captivate Connect project is essential. The steps to be followed for the control and management of revisions in Project Information documents, plans or deliverables are set out in this procedure. This is an overall overview of the situation:

1. **Document Identification:** Each project document is uniquely identified, and a master document list is maintained. The document names, versions, authors and dates of their creation are included in this list.
2. **Revision request Submission:** If changes are necessary, a revision application may be submitted by any member of the Project Team, stakeholder or relevant party. These requests should be submitted to the Project Manager or the designated authority.
3. **Initial Review:** The request for a revision shall be reviewed by the project manager or designated authority to determine its validity and assess potential impact on the project, after it has been received.
4. **Impact Assessment:** An impact assessment shall be carried out when the revision request is valid. This would include looking at how the project objectives, scope, timetable, and budget are affected by the proposed changes.
5. **Revision Drafting:** The draft shall be initiated by the project manager or designated personnel when an Impact Assessment has been found to be favourable. Specific documents, plans or elements of the project that need to be amended are identified.
6. **Technical Review:** The amendments in question are being reviewed, from the point of view of technology, by subject matter expert and relevant team members. The revision is assessed to be feasible and compatible.
7. **Implementation Plan:** To determine how the approved revisions will be implemented, an implementation plan shall be developed. The assignment of tasks, setting a timeline and allocation of resources are part of this plan.
8. **Document Updates:** The maintenance of relevant project documents to take account of the agreed revisions shall be delegated to the Project Manager or assigned staff. Revision of project plans, schedules, risk assessments and other relevant documents shall also be included.

This procedure ensures that project documentation remains accurate and aligned with the project's goals while allowing for necessary changes to be implemented efficiently. Specific details and steps may vary depending on the project's nature and organization's requirements.

BUSINESS CASE

1.Introduction/ Background

The goal of Burswood, WA-based Captivate Connect, a prestigious business organisation, is to improve the on-hold experience by offering callers engaging audio. Captivate Connect is a leading provider of cutting-edge phone systems and On Hold recordings, with a solid clientele that includes government agencies and large corporations. Captivate Connect has a business strategy that involves a monthly recurring price for subscription services in a competitive market with competitors including Messages On-Hold, Ads On-Hold, and Gogo on hold. For the organisation, generating leads effectively is the biggest difficulty.

2.Business Objective-

Modernising data management and addressing IT infrastructure concerns are the main goals of the Captivate Connect Project. The main goal is to ensure cost-effectiveness when moving data from outdated internal servers to more modern server architecture or cloud-based solutions.

3.Current Situation and Problem/Opportunity Statement

The present IT infrastructure of Captivate Connect is burdened by obsolete parts and storage problems. Three 2TB drives with software RAID 5 are used to store data; these drives have 40,000–70,000 hours of power on time. Because the Debian 12 Server cannot work with the SBS 2008 server, it uses an earlier version of Samba. Moreover, no backup system is in place, and Black blaze and other earlier alternatives were never used. Remote access to this data is made possible via a next cloud instance running on a different server, further complicating issues. The project's goal is to investigate methods for moving data to a safer and more effective location.

4.Critical Assumption and Constraints

Assumption:

- ☐ Enough funding is available for the project.
- ☐ Effective collaboration between stakeholders and members of the project team.
- ☐ Technical requirements and business needs have been met by the selected solution.

Constraints:

- ☐ The scope of the project may be affected by budget constraints.
- ☐ The time frame for the projects should be consistent with broader business objectives.
- ☐ Migration of data must be managed in a way that does not compromise its integrity.

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5. Analysis of Options and Recommendation

A hybrid approach has been suggested by Captivate Connect in view of an extensive analysis of possible solutions. In combination with the integration of cloud computing solutions to store data and access, this strategy is aimed at upgrading existing infrastructure for essential services. This approach enables flexibility to be considered in the evolution of business requirements while remaining cost effective.

6. Preliminary Project Requirements

To ensure the seamless synchronization of data across devices and establish reliable backup systems, initial project requirements have included upgrade of server hardware, introduction of a Cloud Storage solution for example Google Workspace.

7. Schedule Estimate

SPRINT 1: BASIC RESEARCH ABOUT CURRENT INFRASTRUCTURE, MIGRATION OPTIONS, BACKUP AND SECURITY MEASURES.

SPRINT 2: IN DETAIL RESEARCH ABOUT THE CLOUD SERVICES AND ALTERNATIVE SERVERS

SPRINT 3: IMPLIMENTATION OF THE CLOUD SERVICES

SPRINT 4: CONFIGURING VIRTUAL ENVIROMENT TO REPLICATE EXISTING SYSTEM AND PROPOSED SYSTEM.

SPRINT 5: FINAL TESTING AND DOCUMENTATION

WORK BREAKDOWN STRUCTURE

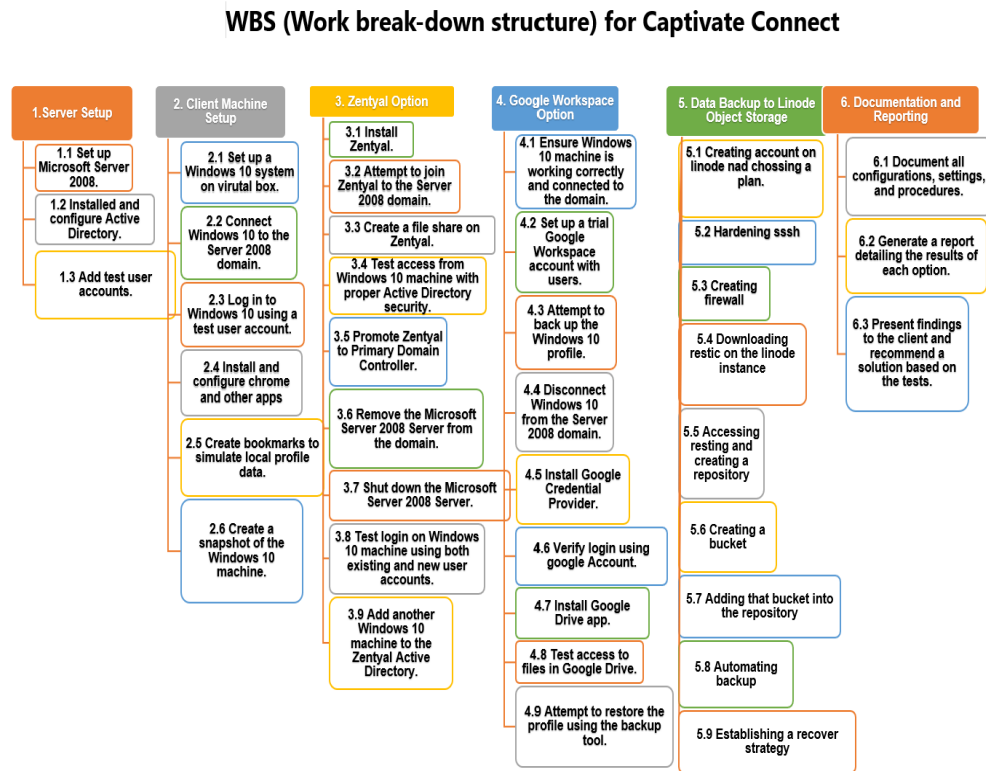


FIGURE 2

ACTIVITY LIST

PROJECT No:	7B	PROJECT NAME:	Server Enhancement
DUE DATE:	29/10/23	PROJECT MANAGER:	Madhura Chandrakant Dhumak

TABLE 7

	WBS ITEM NUMBER :	WBS ITEM NAME:	PREDECESSOR	SUCCESSOR	RESOURCES REQUIREMENT	CONSTRAINTS
1	WBS 1	Server Setup	Virtual machine	Server Configuration	Windows Server 2008 ISO	Hardware Constraints
2	WBS 1.1	Set up Microsoft Server 2008.	Server 2008 Configuration	Active Directory	Virtual Machine	Software Constraint
3	WBS 1.2	Installed and configure Active Directory	Installation of Windows server 2008	File-share system	Windows Server 2008 installation	Technical Constraints
4	WBS 1.3	Add test user accounts.	Installation of Windows server 2008	File-share system	Windows Server 2008 installation	Technical Constraints
5	WBS 2	Client Machine Setup	Windows 10 ISO	Set up Windows 10	Virtual machine software (VirtualBox), Windows 10 installation	Adequate system resources for virtualization
6	WBS 2.1	Set up a Windows 10 system on virtual box.	Basic system Configuration	Adding to a domain	Virtual machine software	Resources for virtualization
7	WBS 2.2	Connect Windows 10 to the Server 2008 domain.	Server & Active Directory Setup	Logging in through the primary domain	Windows 10 installation, Windows Server 2008	Adequate system, Technical Constraints
8	WBS 2.3	Log in to Windows 10 using a test user account.	Windows 10 Setup	Creating a profile	Windows 10 installation	System resources

9	WBS 2.4	Install and configure chrome and other apps	Windows 10	Creating a profile	Windows 10 installation	System resources
10	WBS 2.5	Create bookmarks to simulate local profile data.	Chrome account set up	Google Workspace	Windows 10 installation	Adequate system resources for virtualization
11	WBS 2.6	Create a snapshot of the Windows 10 machine	Windows 10 setup	Google Workspace	Windows 10 installation	Adequate system resources for virtualization
12	WBS 3	Zentyal Option	Server 2008 Configuration	Promoting as the primary domain controller	Zentyal installation	Resources for Configuring Zentyal
13	WBS 3.1	Install Zentyal.	Zentyal Installation	Promoting as the Main domain	Zentyal installation	Compatibility between Zentyal and Server 2008
14	WBS 3.2	Attempt to join Zentyal to the Server 2008 domain.	Client Machine Setup (Windows 10 should be set up and connected to the Server 2008 domain)	Promoting as the main domain controller	Zentyal installation	Technical Constraints
15	WBS 3.3	Create a file share on Zentyal.	Zentyal Installation	Promoting as the primary domain	Zentyal installation	Technical resources
16	WBS 3.4	Test access from Windows 10 machine with proper Active Directory security.	Windows 10	Google workspace	Windows 10	Time
17	WBS 3.5	Promote Zentyal to Primary Domain Controller.	Zentyal Setup & Server 2008	Removing Server 2008 from the network	Zentyal	Technical and Hardware resources

18	WBS 3.6	Remove the Microsoft Server 2008 Server from the domain.	Zentyal and server 2008	Removing Server 2008	Server 2008	Technical resources
19	WBS 3.7	Test login on Windows 10 machine using both existing and new user accounts.	Windows 10	Google workspace	Windows 10 & Google	Technical Constraints
20	WBS 3.8	Add another Windows 10 machine to the Zentyal Active Directory.	Zentyal & User setup	Promoting as the primary domain	Zentyal	Compatibility between Zentyal and Windows 10
21	WBS 4	Google Workspace Option	Windows 10 user setup	Google workspace	Google Workspace trial account, Google Credential Provider, Google Drive app	Internet connectivity, compatibility with Windows 10
22	WBS 4.1	Ensure Windows 10 machine is working correctly and connected to the domain.	Client Machine Setup	Google Workspace	Google Workspace trial account, Google Credential Provider, Google Drive app	Internet connectivity, compatibility with Windows 10
23	WBS 4.2	Set up a trial Google Workspace account with users.	Client Machine	Google workspace	Google Workspace trial account,	Internet connectivity
24	WBS 4.3	Attempt to back up the Windows 10 profile.	Windows 10	Google Workspace	Google Credential Provider	Technical Constraints
25	WBS 4.4	Disconnect Windows 10 from the Server 2008 domain.	Client Machine Setup & Server 2008	Google Workspace	Google Workspace trial account, Google Credential Provider, Google Drive app	Technical and time Constraints

26	WBS 4.5	Install Google Credential Provider.	Google Workspace Admin Console	Data Migration to Google workspace	Google Workspace trial account, Google Credential Provider, Google Drive app	Internet connectivity, Mon
27	WBS 4.6	Verify login using google Account.	Admin Console	Migration to google workspace	Google Workspace trial account,	Time & Budget
28	WBS 4.7	Install Google Drive app.	User Setup	Profile migration	Google Workspace	Time & Budget
29	WBS 4.8	Test access to files in Google Drive.	Google Drive	Data Migration	Google Workspace	Technical & Hardware constraints
30	WBS 4.9	Attempt to restore the profile using the backup tool.	Windows Profiles	ForsenIt tool	Google Workspace	Technical Constraints
31	WBS 5	Creating account on linode had choosing a plan.	Linode Setup	Linode Object Storage	Linode account	Linode service availability
32	WBS 5.1	Hardening ssh	Linode Account Creation	SSH Access	Linode Configuration	Internet Connectivity
33	WBS 5.2	Creating firewall	Firewall Setup	SSH Access	Linode configuration	Data centre Connectivity
34	WBS 5.3	Downloading restic on the linode instance	Configuration of LISH Console	Creating a repository	Linode account, configuration	Technical Constraints
36	WBS 5.4	Accessing resting and creating a repository	Installation of restic	Creating S3 Buckets	Restic Configuration	Technical Constraints
37	WBS 5.5	Creating a bucket	Establishing the Object Storage on Linode	Accessing the buckets	Buckets Configuration	Data Centre Availability

38	WBS 5.7	Adding that bucket into the repository	Restic & Buckets Setup	Backup Allowance	Linode account, Restic, firewall configuration	Technical Constraints
39	WBS 5.8	Automating backup	Restic Setup	Recovery Strategy	Linode account, Restic, firewall configuration	Technical Constraints
40	WBS 5.9	Establishing a recover strategy	Linode Setup	Data backup access	Linode account, Restic, firewall configuration	Technical Constraints
41	WBS 6.0	Document all configurations, settings, and procedures.	Zentyal, Google workspace & Linode setup	Project conclusion and implementation of recommendations	Documentation tools and reporting tools	Time
42	WBS 6.1	Generate a report detailing the results of each option.	Testing of all the Components	Project Implementation	Documentation tools	Time
43	WBS 6.2	Present findings to the client and recommend a solution based on the tests.	Final Documentation	Project conclusion	Presentation and final report	Time

GANTT CHART

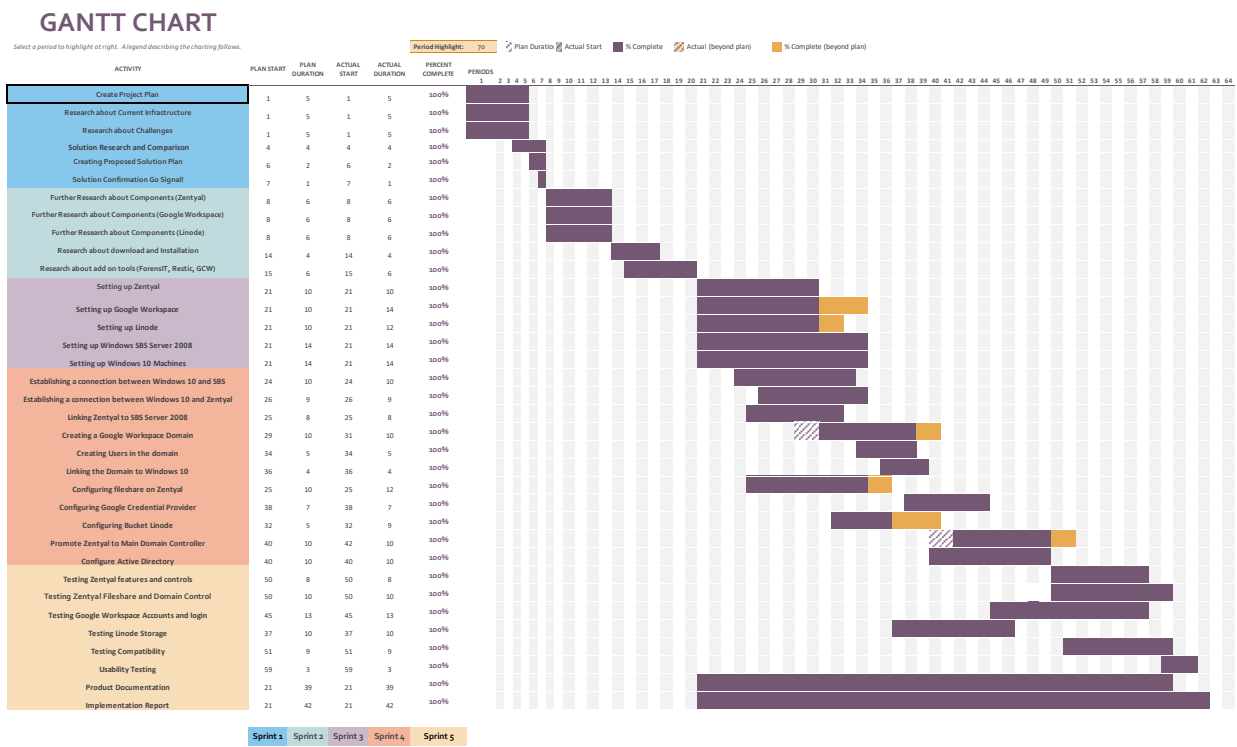


FIGURE 3

CHANGE LOG

TABLE 8

DATE	CHANGES	CHANGED BY
11 th September 2023	First release: Document Layout	Bhawna Soni
15 th September 2023	Added Project Charter Added Business Case	Mano Bereka
19 th September 2023	Added Audience Description Added Cover Page Contents	Loji
25 th September 2023	Added Contents to: System Overview Audience Description Applicability Statement Purpose Statement Document Usage	Bhawna Soni
30 st September 2023	Added Contents to: Existing System Report	Loji
4 th October 2023	Added Contents to: User story Dictionary User story Definitions	Madhura
8 th October 2023	Added Contents to: New System Report Zentyal Google Workspace Linode Implementation Reports	Adib
12 th October 2023	Added Diagrams to: New System Report Current System	Viraj
17 th October 2023	Added Content to: Usability Testing References Vendor Documents User Definitions	Bhawna
19 th October 2023	Made Updates to : Existing System Report	Adib
20 th October 2023	Added Contents to : Risk Management Plan Stakeholders	Adib
21 st October 2023	Made updates to: Implementation Report	Loji
23 rd October 2023	Added Contents to: WBS Activity List	Viraj
25 th October 2023	Added Contents to: Gantt Chart And Quality Management Plan	Madhura

RISK MANAGEMENT PLAN

Purpose

The purpose of a risk management plan is to systematically identify, assess, and address risks that could potentially impact a project, process, or organization. It is a proactive approach to minimize or mitigate the negative effects of risks and maximize opportunities for success.

The main objectives of a risk management plan include:

1. **Risk Identification:** This plan helps identify potential risks that may arise during the course of this project. By systematically identifying risks, organizations can have a comprehensive understanding of potential threats and opportunities.
2. **Risk Assessment:** This plan facilitates the assessment of identified risks in terms of their likelihood, impact, and priority. This assessment helps prioritize risks based on their significance and the potential consequences they may have on the project or organization.
3. **Risk Mitigation and Control:** A risk management plan outlines strategies and actions to mitigate or control identified risks. It includes preventive measures to reduce the likelihood of risks occurring and contingent measures to minimize the impact if they do occur. The plan may involve risk avoidance, risk transfer, risk reduction, risk acceptance, or a combination of these strategies.
4. **Risk Monitoring:** The plan establishes a framework for ongoing risk monitoring and evaluation. Regular monitoring allows organizations to track the status of identified risks, assess the effectiveness of mitigation measures, and identify new risks that may emerge during the project or organizational life cycle.
5. **Communication and Reporting:** A risk management plan provides a structure for effective communication and reporting of risks to stakeholders. It ensures that relevant information regarding risks, their potential impact, and the progress of risk mitigation efforts are shared with the appropriate individuals or groups.
6. **Decision Making:** The plan helps inform decision-making processes by providing an objective analysis of risks. It enables stakeholders to make informed decisions about resource allocation, risk tolerance, contingency planning, and overall project or organizational strategy.
7. **Continuous Improvement:** A risk management plan encourages a culture of continuous improvement. Lessons learned from past projects or risk events can be incorporated into the plan to enhance risk management practices in future endeavours.

By having a risk management plan in place, our team and Captivate Connect can be better prepared to anticipate, assess, and respond to potential risks. It helps in reducing uncertainty, protecting assets, and improving the overall resilience and success of projects or organizations.

Definitions, Abbreviations, Acronyms

TABLE 9

Term	Definition
Application	The software component of the system
AD	Active Directory - Active Directory is a directory service developed by Microsoft for Windows domain networks. It authenticates and authorizes all users and computers in a Windows domain type network—assigning and enforcing security policies for all computers and installing or updating software.
FSMO	Flexible Single Master Operation - FSMO roles are specialized tasks that are used to manage the replication of data within an Active Directory environment. These roles are assigned to specific domain controllers to ensure that data is updated correctly across all domain controllers.
FQDN	Fully Qualified Domain Name - An FQDN is the complete domain name for a specific computer, or host, on the internet. It consists of two parts: the hostname and the domain name. For example, in the FQDN "www.example.com," "www" is the hostname and "example.com" is the domain name.
RID	Relative Identifier - In the context of Active Directory, a RID is a unique number generated by the domain controller that is used to identify objects (such as users, groups, and computers) within a domain.
LDAP	Lightweight Directory Access Protocol - LDAP is an open, vendor-neutral protocol used to access and maintain distributed directory information services over an Internet Protocol (IP) network. It runs directly over TCP/IP and allows access to centrally stored information about users, machines, and other resources.
Software	An electronic program consisting of algorithms, control logic, data structures, data sets, operating system, user interface. Software may consist of critical and non-critical components and may be used as a part or as an accessory of a medical device to perform pre-defined and approved operations on medical devices. Software may itself be a medical device. Software may be used in the production of a device (e.g. programmable logic controllers used in the manufacturing of equipment.) Software may be used in implementation of the device manufacturer's quality system (e.g. Software that records and maintains the device history record.
SOP	Standard Operating Procedure
NAS	Network Attached Storage - NAS is a type of dedicated file storage device that provides local-area network (LAN) users with centralized, consolidated disk storage through a standard Ethernet connection. It allows multiple users and devices to access, retrieve, and store data.

RPIC	Risk Probability, Impact, Criteria - RPIC is a matrix used in risk assessment that helps to evaluate and prioritize risks based on their probability of occurrence and the potential impact they could have on a project or organization.
SOP	Standard Operating Procedure
Risk	For the purposes of this software Risk Management Procedure, Risk is defined as the possibility that harm (loss of data, crashing of software) might occur when exposed to a hazard.
Risk Assessment	A systematic process of evaluating the potential risks that may be involved in a projected activity or undertaking.
R&O	Risk and Opportunities
Risk Breakdown Structure (RBS)	It is a hierarchical chart that precisely breaks down the risks of a project, starting with high-level categories and moving down to sub-levels of risk.
FMECA	Failure Modes, Effects and Criticality Analysis
Risk Owner	The individual who is ultimately accountable for ensuring the risk is managed appropriately.
Risk Manager	Risk Manger is the one to communicate risk policies and processes for an organization.

References

- ❑ Zentyal Documentation: <https://doc.zentyal.org/en/>
- ❑ Microsoft Server 2008 Documentation: [https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-essentials-sbs/cc816569\(v=ws.10\)?redirectedfrom=MSDN](https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-essentials-sbs/cc816569(v=ws.10)?redirectedfrom=MSDN)
- ❑ Google Workspace Documentation <https://support.google.com/a/users/?hl=en#topic=11499463>
- ❑ ForensIT User Profile Wizard Documentation: <https://www.forensit.com/Downloads/User%20Profile%20Wizard%20Professional%20User%20Guide.pdf>
- ❑ Australian Government: - Business Risk Management
- ❑ Harvard Business: - Managing Risk
- ❑ Project Charter
- ❑ Project Management Plan
- ❑ Risk Register
- ❑ Risk Matrix
- ❑ Business Case

Risk Management Procedures

Risk management procedures in project management involve identifying, assessing, prioritizing, and mitigating potential risks that could impact the successful completion of a project. By proactively addressing risks, project managers can minimize their negative effects and improve the overall project outcome. Here are some key steps and procedures taken to manage the possible risk in this project.

1. **Risk Identification:** The first step is to identify potential risks. This involves analysing the project scope, objectives, and stakeholders' requirements to identify any events or conditions that could have adverse effects on the project. Risks can be internal (within this project), such as code issues, and data security, or external (outside the project's control), like vendor issues.
2. **Risk Assessment:** Once risks are identified, they need to be assessed to determine their potential impact and probability of occurrence. This assessment helps prioritize risks and allocate appropriate resources for mitigation. Risks are typically assessed based on their severity, likelihood, and detectability.
3. **Risk Prioritization:** After assessing risks, project managers prioritize them based on their potential impact on the project's objectives. High-priority risks are those with significant consequences and a higher probability of occurrence. Prioritization helps in allocating resources and attention to the most critical risks.
4. **Risk Mitigation Planning:** Once risks are prioritized, mitigation strategies are developed to reduce their impact or likelihood. These strategies can include contingency plans, risk avoidance, risk transfer, risk sharing, or risk acceptance. Mitigation plans outline specific actions, responsibilities, timelines, and required resources for each risk.
5. **Risk Monitoring and Control:** Throughout the project lifecycle, risks need to be continuously monitored to ensure the effectiveness of mitigation measures and identify any new risks that may arise. Regular project status updates, meetings, and risk reviews help in monitoring risks. If a risk occurs or its severity changes, appropriate adjustments to the mitigation plan are made.
6. **Risk Response Execution:** When a risk materializes, the planned mitigation strategies are put into action. This may involve activating contingency plans, involving subject matter experts, or implementing alternate approaches. Effective communication and collaboration among project stakeholders are crucial during this phase.
7. **Lessons Learned:** At the end of the project, a comprehensive review is conducted to capture lessons learned from the risk management process. This includes evaluating the effectiveness of the risk mitigation strategies, identifying areas for improvement, and documenting best practices for future projects.

In addition to these steps, it is important to establish a risk management framework that defines roles, responsibilities, and communication channels for effective risk management.

Risk Management Deliverables

- ❑ Backup Test Report: Documenting the results of the trial backup and restoration.
- ❑ Trial Google Workspace Account: Documented setup and user information for the trial account.
- ❑ Risk Register: Maintained document listing identified risks, their likelihood, impact, and mitigation strategies.
- ❑ Risk Assessment Reports: Periodic reports detailing the status of identified risks and mitigation efforts.

Risk Register									
Project Name: Captivate Connect Date: 10/10/23 Last updated: 11/10/23 Version: 1.1									
ID No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Status
R1	1	Extreme	Server hardware failure during migration	Technical	Ageing hardware components and potential wear and tear leading to hardware failure.	Unusual noises from server hardware, sudden performance degradation, error messages indicating hardware issues.	Replace or upgrade ageing hardware components before migration.	IT-Team	Ongoing Monitoring Continuous monitoring of server hardware condition and performance.
R2	2	Medium	Data loss during server migration	Technical	Inadequate backup procedures or failure in data transfer protocols during migration.	Failed data transfer attempts, error messages during migration process, inaccessible data after migration.	Conduct thorough data backups before initiating the migration process.	IT-Team	Mitigation in Progress Active measures in place to mitigate the risk of data loss during migration.
R3	3	Medium	Compatibility issues with SBS 2008 data	Technical	Incompatibility between SBS 2008 data structure and the target platform, potentially due to outdated software.	Error messages during data transfer or migration process, data corruption or loss after migration.	Conduct thorough compatibility testing in a controlled test environment.	IT-Team	Mitigation in Progress Compatibility testing and mitigation efforts underway.
R4	4	Low	Limited familiarity with Zentyal platform	Personnel	Lack of prior experience and training with Zentyal, leading to potential delays or errors during implementation.	Difficulty in performing specific tasks on the Zentyal platform, prolonged setup or configuration times.	Provide comprehensive training sessions for IT team members on the Zentyal platform.	IT-Team	Mitigation in Progress Training sessions and mentorship programs initiated to address familiarity with Zentyal.
R5	5	Medium	Inadequate testing of Google Workspace	Technical	Insufficient testing environment or inadequate test cases for Google Workspace integration.	Critical errors or malfunctions during Google Workspace integration testing, unexpected behaviour in test environments.	Establish a dedicated testing environment for Google Workspace integration.	IT-Team	Testing Phase Google Workspace testing environment established, and testing procedures underway.
R6	6	Medium	Google Workspace authentication issues	Technical	Potential conflicts or mis-configurations in the authentication process with Google Credential Provider.	Users unable to log in with Google credentials, error messages during authentication attempts.	Conduct extensive testing of Google Credential Provider to ensure seamless authentication.	IT-Team	Testing Phase Google Workspace authentication processes are being thoroughly tested.
R7	7	Medium	Data loss during Google Workspace migration	Security	Inadequate backup procedures or potential data integrity issues during the migration process.	Inaccessible data after migration, error messages indicating data integrity issues.	Implement data validation checks post-migration to ensure data integrity.	IT-Team	Mitigation in Progress Measures in place to mitigate the risk of data loss.
R9	9	High	Data security breach during migration	Security	Insufficient security measures, potential vulnerabilities, or unauthorised access points in the migration process.	Unusual network activity, unauthorised access attempts, unexpected system logs indicating potential security breaches.	Monitor network traffic for suspicious activity and implement intrusion detection systems.	IT-Team	Mitigation in Progress Security measures and controls are being actively implemented.

FIGURE 4

Risk Probability, Impact, Criteria

Risk probability, impact, and criteria have been essential components to assess the risk of this project. They help to evaluate and prioritize risks based on their likelihood of occurrence and potential consequences. The factors are:

1. **Risk Probability:** Risk probability is the likelihood or chance that a specific risk event will occur during the project. It is assessed on a scale, such as low, medium, or high. The assessment is based on data, expert analysis, and statistical analysis.

For example, if the project involves integrating asset management software with existing databases, the probability of encountering integration issues may be considered high due to the lack of available resources.

2. **Risk Impact:** Risk impact refers to the consequences or effects the risk has on the project. The impact can vary in terms of scope, magnitude, duration, and severity. Assessing risk impact helps in understanding the potential damage or disruption a risk event can cause to project objectives, such as schedule, cost, quality, or stakeholder satisfaction.

For instance, if there is a change in the scope, the impact could be significant in terms of delayed deliverables, reduced productivity, cost overrun, and increased workload on team members.

3. **Risk Criteria:** Risk criteria are the predefined guidelines or thresholds used to categorize risks based on their probability and impact levels. These criteria serve as a reference for ranking risks and determining appropriate response strategies. Risk criteria can be customized based on the project's objectives, industry standards, organizational policies, and stakeholder expectations.

For example, the project has established risk criteria where risks with a high probability and high impact are classified as "critical," requiring immediate attention and aggressive mitigation efforts. Risks with low probability and low impact may be categorized as "negligible" and may not require immediate action.

Probability and impact categories are used to further classify risks based on their likelihood of occurrence and the potential consequences they may have on a project. These categories help in prioritizing risks and determining appropriate responses.

Probability Categories:

1. **Low Probability:** Risks in this category have a relatively low likelihood of occurring. They may be considered rare events or have a very low chance of happening during the project.
2. **Medium Probability:** Risks in this category have a moderate likelihood of occurring. They are neither highly probable nor highly improbable and may occur occasionally during the project.
3. **High Probability:** Risks in this category have a high likelihood of occurring. They are likely to happen at some point during the project, and their occurrence is expected.

Impact Categories:

1. Low Impact: Risks in this category have minimal or negligible consequences on project objectives. They may cause minor disruptions or have a limited impact on schedule, cost, quality, or stakeholder satisfaction.
2. Medium Impact: Risks in this category have a moderate impact on project objectives. They can cause noticeable disruptions, delays, and additional costs, or affect the quality of deliverables to a certain extent.
3. High Impact: Risks in this category have significant consequences on project objectives. They can result in major disruptions, substantial delays, substantial cost overruns, severe quality issues, or significant dissatisfaction among stakeholders.

By combining the probability and impact categories, risks are classified into different levels of priority or severity, which helps project managers determine the appropriate course of action.

The Risk of this project has been divided by considering their probabilities and their impacts in a table also known as the Risk Matrix. The main purpose of this matrix is to clearly display risk into different categories.

Revision procedures

The Risk Management Plan will be reviewed and updated:

- ☐ After significant project milestones.
- ☐ In response to emerging risks or major changes in project scope.
- ☐ When new information becomes available that impacts risk assessments.

PROPOSED SOLUTION ARCHITECTURE

Proposed System

Zentyal

Google Workspace

Linode

Implementation Report

PROPOSED SYSTEM

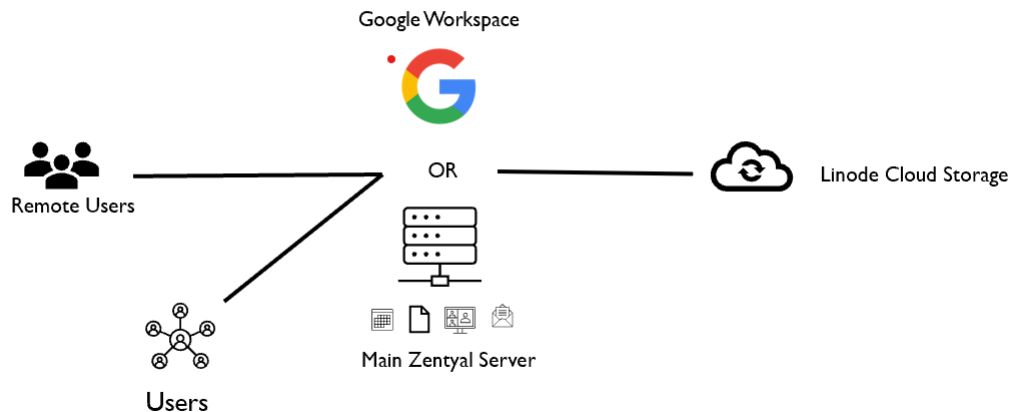


FIGURE 5

For Captivate Connect, an organisation specialised in providing advanced telephone on hold recordings and telephone systems for SMEs and government offices, the proposed system is a comprehensive upgrade and modernization initiative. The aim is to address essential challenges that exist within the organization's existing infrastructure, improve data management, and keep high quality audio services available.

Current Challenges and Opportunities:

Captivate Connect currently faces several challenges:

- 1. Aging Infrastructure:** There is an aging of the IT infrastructure, including servers, which does not meet the organisation's operational needs. You have a Debian 12 Server and Samba installation that's no longer supported, leading to compatibility problems.
- 2. Data Integrity and Backup:** A robust backup service is missing from the current configuration, which could lead to data loss or corruption.
- 3. Cloud Integration:** The Organization recognises the need for data to be migrated to a new and safer cloud-based system in order to achieve greater flexibility and accessibility.
- 4. Demo Server Improvement:** The Captivate Connect team would like to improve the way clients are sharing audio demonstrations so that they can be more effective and interactive.
- 5. Cost- Effective Solutions:** The organisation has a firm commitment towards the implementation of cost-efficient solutions with lasting value, as they seek to improve.

Proposed System Objectives:

The proposed system seeks to achieve the following objectives:

- 1. Infrastructure Modernization:** To increase the processing power, capacity to store and data redundancy, you are acquiring and deploying new server infrastructure.
- 2. Data Management and Backup:** In order to safeguard data and ensure their availability, implement a comprehensive backup solution.
- 3. Cloud Integration:** Evaluate cloud storage solutions like Google Workspace for data storage, addressing synchronization challenges.

- 4. Demo Server Enhancement:** Develop a user portal where customers will be able to get and give feedback on Audio Demos as well as the tracking and Analytics features.
- 5. Security and Compliance:** Strengthen data security and encryption, as well as compliance with industry and regulatory standards.
- 6. Skills Development:** Provide instructions for team members to enhance their knowledge of Linux systems and command line operations.
- 7. Collaborative Approach:** Ensure that the approach to solving problems is one of collaboration and innovation.
- 8. Cost Effective Solution:** To review technology options to make sure that they are in line with an organisation's budget.

The suggested solution is a big step in the direction of realising Captivate Connects goal of giving callers on wait entertaining audio. It is made to be more affordable while improving data accessibility, infrastructure dependability, and administration. By taking advantage of the available possibilities and challenges, the organisation hopes to position itself for long-term growth and success in the audio services and telecommunications sectors by offering a more effective, dependable, and secure audio service.

IMPLEMENTATION REPORT

Project Overview

This project is being built for Captivate Connect®, a company based in Perth, Western Australia which specializes in delivering broadcast audio, music, and data content.

The aim of this project is an upgrade and migration from an outdated 2008 Windows Small Business Server into a much more efficient and business friendly server system and integrate Google Workspace for Active Directory and user management and implement Linode as a data backup solution.

The purpose of this upgrade/ migration is to enhance the system performance, reliability, security, and scalability and as well as optimize business operations for Captivate Connect®.

Main Objectives:

1. Research about potential replacement server options
2. Finalize a new server option.
3. Setup server for testing
4. Integrate Google Workspace as an Active Directory
5. Implement Linode as a backup solution.
6. Connect all components together.
7. Testing and Monitoring
8. Ensure seamless communication and collaboration between components and organization.

1. Determining the Major Components of the Solution

1.1. Server Option

After careful and extensive research and approval from the Client Representative Mr. Lawrence Bayly of Captivate Connect®, the team finalized Zentyal to replace and takeover the current Windows SBS 2008 server.



FIGURE 6

Zentyal Server is an open-source Linux Distribution designed to act as an all-in-one server solution. It provides a comprehensive suite of services, including file sharing, mail server, network management and security features.

Zentyal carries a critical role in this implementation and serves as the core infrastructure for hosting services and applications.

Key Features:

- Directory Services (LDAP)
- E-mail and Collaboration
- Network Services
- Security
- Integration Capabilities

1.2. Google Workspace and Active Directory

Google Workspace is a cloud-based productivity suite that offers a range of applications for communication, collaboration, and document management. It includes Google Calendar, Gmail, Google Drive and much more.



FIGURE 7

Google Workspace once integrated into the business system will serve as the primary solution for email, document storage, user management/ authentication and collaboration.

Key Features:

- E-mail and Calendar
- Document Management and Storage
- Video Conferencing
- User Management and Authentication
- Security and Compliance

1.3. Linode as a Backup

Linode is a cloud infrastructure provider offering cloud computing and storage solutions. It provides scalable virtual machines and storage options.



FIGURE 8

Linode is implemented as a backup solution to ensure data redundancy and disaster recovery capabilities. It acts as an off-site storage location for critical data.

Key Features:

- Scalable Storage
- Automated Backup and Recovery
- Data Integrity and Reliability
- Cost-Efficient Disaster Recovery

Business Value and Benefits

1.4. Challenges being faced by the company.

- The current infrastructure being used is outdated.
- Microsoft had officially ended support for Windows SBS 2008 which means, no more security updates, patches or bug fixes which potentially leaves the system vulnerable to security threats.
- Integration with recent and updated technology can be complex and sometimes could be impossible.
- Hardware maintenance and finding replacement parts can be expensive and difficult.

1.5. Benefits of switching to proposed solution

- Zentyal and Google Workspace are based off modern technology and offer new technology and services to boost productivity and security.
- Both systems are actively maintained, and support and updates are regularly provided reducing the risk of security breaches.
- Flexible payment plans boosting cost efficiency.
- Active users' community to receive support and discuss issues.

- Allows cloud integration to reduce on premise hardware which in turn reduces maintenance costs.

1.6. Things to consider post migration.

- The full migration can be time consuming and complex.
- Guaranteed backups need to be set up to ensure data recovery if anything fails during migration.
- Ensure the new setup integrates with existing system and applications.
- Existing employees will have to be trained and familiarized with new system to ensure productivity and return on Investment.
- The proposed system must comply with industry-specific laws and company regulations.

The suggested solution has the potential to make the business more effective and help everyone work better. It can also save money if it's used the right way. However, to make sure everything goes smoothly, the team and the company need to do thorough testing before making any big changes.

This testing phase is important. It helps detect any problems before they become big issues. This careful testing will make sure that the new solution fits seamlessly into the company's day-to-day work. This way, the chances of a successful transition are much higher.

2. Implementing Zentyal

- Zentyal holds a significant importance as a multifunctional server solution. Its centralized management capabilities streamline the administration of critical services like LDAP, DNS and DHCP, simplifying the configuration and maintenance of network resources. Zentyal is considered to be a foundation that holds the other components in the solution in place.
- **Setup and Configuration**
 - Zentyal can be downloaded as an ISO from the official website and can be installed based on functionality requirements.
 - More details on installation and configuration can be found in Product Documentation or Official Zentyal Documentation.
- **Testing and Validation**

- Compatibility Testing was performed to initiate a connection between the existing system SBS 2008 to confirm that a migration is possible without major issues.

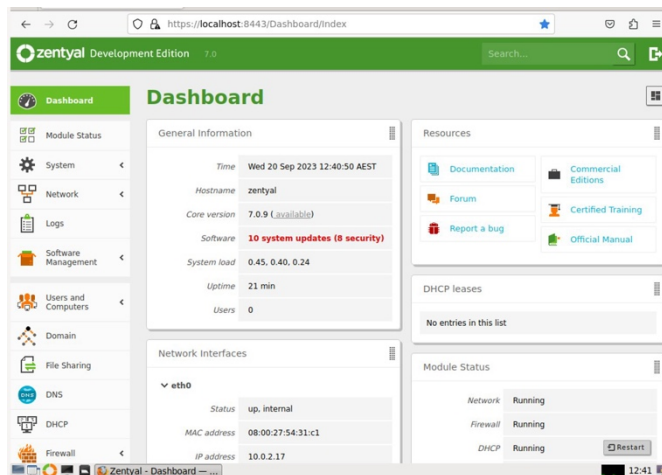


FIGURE 9

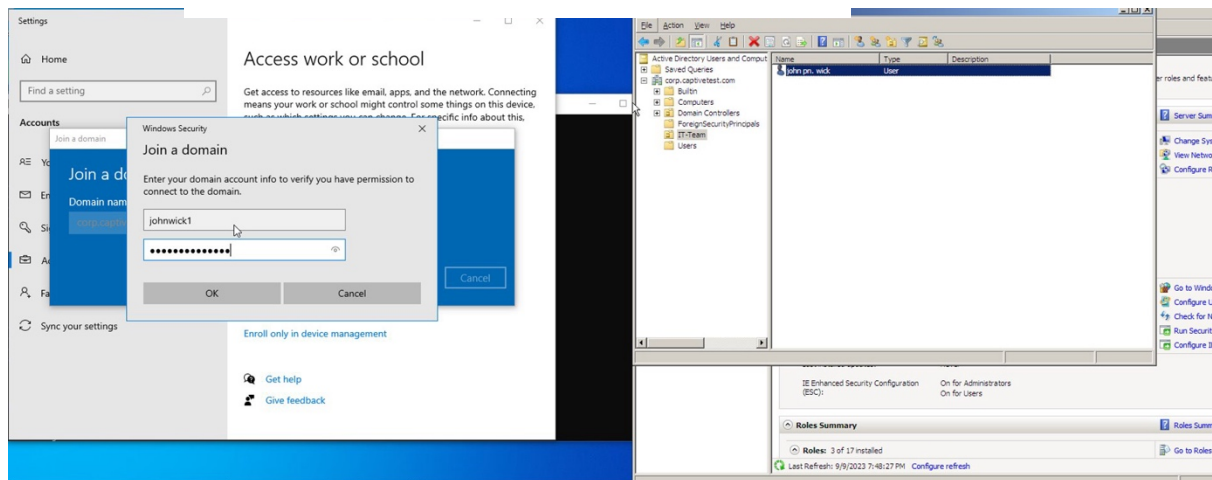


FIGURE 10

3. Implementing Google Workspace

Google workspace plays a crucial role in the system by serving as the primary cloud-based productivity platform for the organization.

Google workspace provides features such as Email/Calendar, Document management and collaboration, Meetings, and conferences and many more features to improve and boost productivity within the business.

□ Setup and configuration

- The team created and configured a domain to represent the business for practice purposes and used this domain as a network to create users and perform testing to evaluate the performance and check compatibility.
- Further details on set up and configuration can be found in the product documentation.

□ Testing and Validation

- We tested the Google Workspace configuration by creating a user account within the Google powered workspace and we logged into the account using the credentials provided by the Google Credential Provider App successfully through a Windows 10 machine connected to the server.

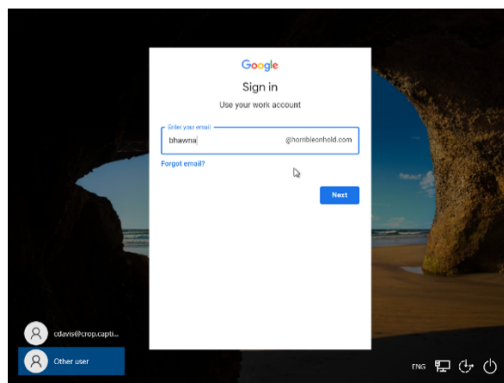


FIGURE 11

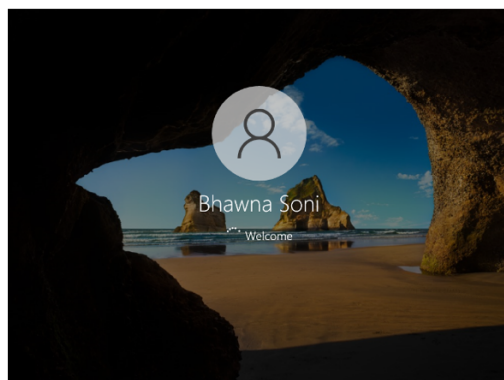


FIGURE 12

4. Implementing Linode as a backup strategy

Linode carries a critical role in this project as the selected backup solution. Its primary function is to provide a secure and reliable off-site storage location for crucial data, ensuring data redundancy and facilitating disaster recovery.

□ Setup and configuration

- Create a Linode account by signing up.
- Navigate to Linode Manager
- Create a Compute Instance
- Connect to Instance
- Configure settings and Firewall.

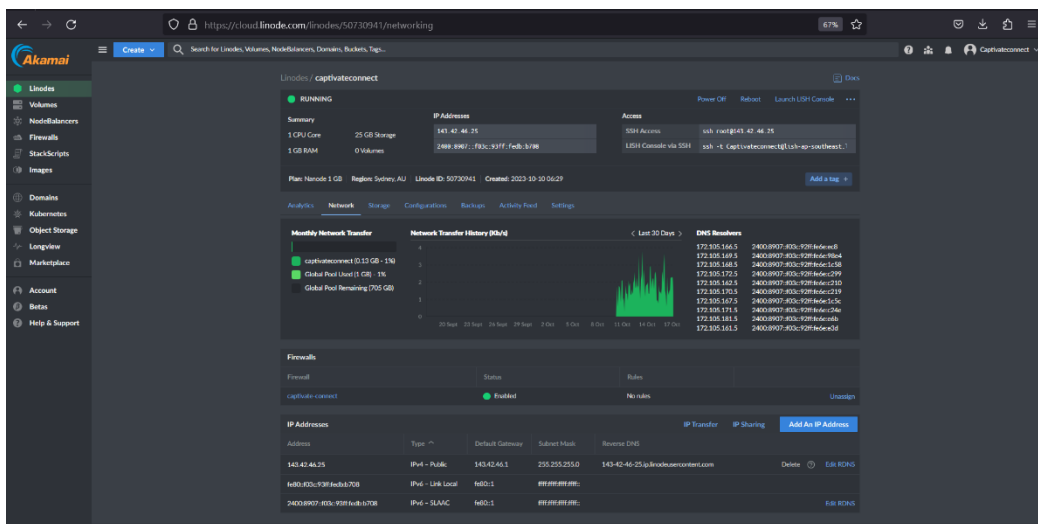


FIGURE 13

- More information and detailed steps can be found on the official product documentation made by our team and as well as the Linode Company provided documentation.

PRODUCT DOCUMENTATION

ZENTYAL



FIGURE 14

1. Product Overview

1.1 Introduction

Zentyal is a Linux based open-sourced network management platform that provides a range of services to efficiently manage a business network and its users. Zentyal is a much easier and cost-efficient alternative to Windows Server.

1.2 Use Cases and target audience.

Small and Medium level business owners can use Zentyal to set-up their business network and navigate through their business requirements and needs with ease. Zentyal provides a GUI Dashboard as a control panel to set up and manage the server.

1.3 Key features

- Unified GUI Interface
- Networking services
- Firewall Management
- Mail server services
- File Sharing and storage
- Active Directory and User/Group Management
- Active community and available support

2. Features in Detail

List and describe the main features of Zentyal 7.0. Organize this section in a bullet point format for easy readability. Include both basic and advanced functionalities. For example:

- **Unified Graphical Interface:** Visually appealing dashboard that allows the administrator to control all functions and services within one screen. The Dashboard will allow activating and deactivating features such as anti-virus checker, DHCP and many more.

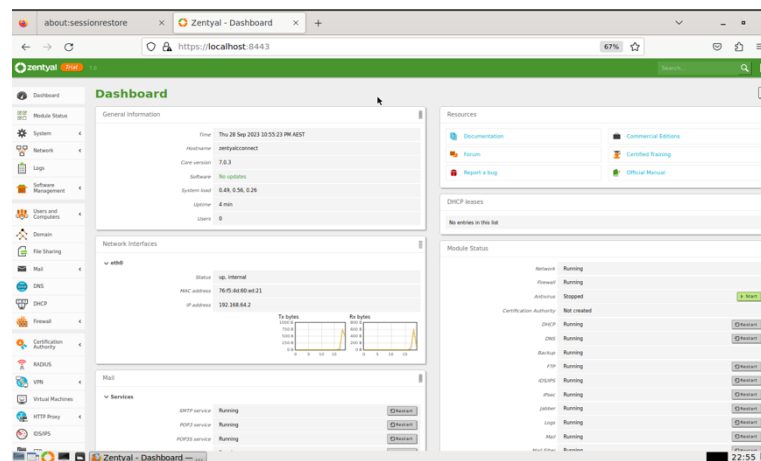


FIGURE 15

- **Network Management:** Zentyal allows you to manually configure network settings, including DHCP, DNS, and firewall from the dashboard which helps the network administrator look after the server and the system with ease.
- **User and Group Management:** Create, modify, and delete user accounts and groups within the server and manage accounts from the click of a mouse.

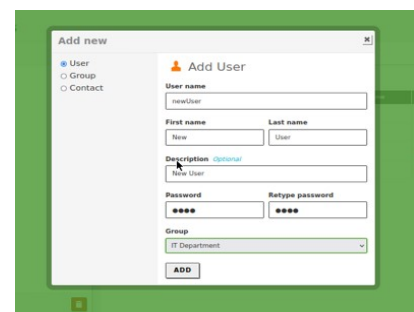
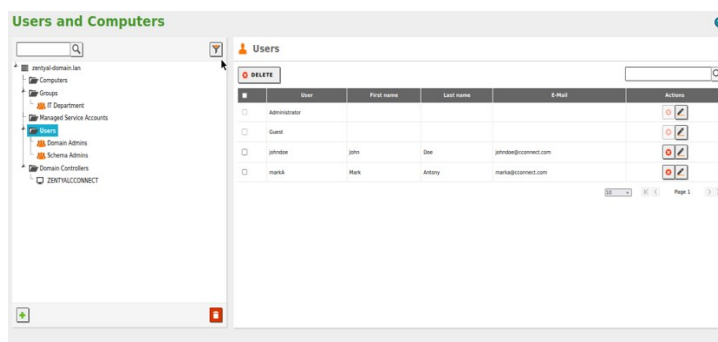


FIGURE 16

- Mail Server: Set up and manage email services with features like POP3, IMAP, and SMTP.

FIGURE 17

- **2.1 File Sharing**
- File Sharing: Provide shared folders with access control and permissions.

FIGURE 18

- **Configure File Sharing:**

- In the Zentyal web interface, navigate to "Shared Folders" and create a new folder.
 - Set the folder permissions, access control, and user/group permissions as needed.
 - Configure share settings like share name and description. ○ **Access the File Share from Windows 10:**
 - On your Windows 10 machine, open File Explorer. ○ in the address bar, enter \\<Zentyal_Server_Name>\<Share_Name>, replacing <Zentyal_Server_Name> with the Zentyal server's hostname or IP address and <Share_Name> with the name of the shared folder.
 - You'll be prompted to enter domain credentials; use your Windows Server 2008 domain credentials.
- VPN and Remote Access: Allow secure connections to the network from external locations.

3.Installation Guide

- **Visit the Zentyal website (<https://www.zentyal.com/>) and download the latest version of Zentyal.**
- **Boot your server from the Zentyal installation media and follow the prompts on the screen to choose which type of installation you require. Zentyal provides 3 types of installations:**
 1. Install Zentyal 7.1-development (delete all disk)
 2. Install Zentyal 7.1-development (expert mode)
 3. Install Zentyal 7.1-development with GUI (expert mode)

At the initial installation stage, we will be able to select main language and register a username for the server administrator and a password as well which can be changed later with permissions.

Zentyal will also require configuring the language, keyboard layout and time zone access during the installation process. We will be able to select disk space/ partitioning and memory usage depending on the type of installation we execute.



FIGURE 19

The first option (delete all disk) will only need specification for keyboard layout/ language and system user and the rest such as disk space and memory usage will be configured automatically.

The Second option (expert mode) will allow you to specify keyboard layout/ language and as well as the disk and memory configurations.

The final option (expert mode with GUI) is very similar to the second option, but this version of installation comes with a Graphical User Interface as well. Further details and support can be found at the official Zentyal website and documentation.

<https://doc.zentyal.org/en/installation.html#installation-guide>

Configure Network settings.

- Assign a static IP address to your Zentyal server, ensuring it is on the same network as your client machines.

Complete Installation

- After the installation, remove the installation media and reboot the server.

Using Zentyal as a Primary Domain Controller

- Before making any major changes ensure there is a complete backup of all the critical data and configurations on both the windows Server and Zentyal.
- Transfer FSMO Roles:
 - On the Existing server, transfer the FSMO (Flexible Single Master Operations) roles to Zentyal. These roles include PDC emulator, RID Master, Infrastructure Master, Schema Master and Domain Naming Master.
 - Use the following steps:
 - Open PowerShell or Command Prompt with administrative privileges.
 - Use the *ntdsutil* tool to transfer each role.

```
ntdsutil roles  
connections
```

```
connect to server <Zentyal_Server_Name> q
```

```
transfer <Role_Name>
```

```
q
```

- **Verify FSMO Role Transfer:**
 - Confirm that Zentyal now holds all the FSMO roles using the following PowerShell command:
 - **graphql**
 - **netdom query fsmo**

- **Demote Windows Server** o Once all FSMO roles are successfully transferred to Zentyal, demote the Windows Server using the following steps:
 - Open the Server Manager
 - Click on the flag icon at the top right corner and select “Promote this server to a domain controller”
 - Select “Remove Active Directory Domain Service”.
- **Adjust DNS and DHCP settings (Optional)**
 - o If the windows server was also acting as DNS and/or DHCP server, ensure that these services are properly configured on Zentyal.
- **Update DNS Settings on Clients:**
 - o Make sure that client machines, including the Windows 10 machine, are using Zentyal as their DNS server.
- **Verify Domain Functionality:** o Test domain logins, access to shared resources, and any other critical services.
- **Monitor and Troubleshoot:**
 - o Monitor the domain for any issues and address them promptly.
- Please note that this process involves significant changes to your network configuration, so it's crucial to plan and test thoroughly before making any changes in a production environment. Additionally, consider consulting with an IT professional experienced with Windows Server and Zentyal if you're unsure about any steps.
- **“<Role_Name>” refers to the specific FSMO (Flexible Single Master Operation) role that you want to transfer. Here are the role names and their descriptions:**
- **PDC Emulator:**
 - o The PDC Emulator role handles password changes and account lockouts in a Windows domain. It also serves as the authoritative time source for the domain.
- **RID Master:**
 - o The RID Master role is responsible for allocating unique Relative Identifier (RID) numbers to each domain controller in a domain.
- **Infrastructure Master:**
 - o The Infrastructure Master role ensures that cross-domain group memberships are correctly handled.
- **Schema Master:**
 - o The Schema Master role manages updates and modifications to the Active Directory schema.
- **Domain Naming Master:**

- The Domain Naming Master role manages the addition or removal of domains in the forest.
- When transferring a role, you would replace “<Role_Name>” with the name of the specific role you're transferring. For example, if you want to transfer the PDC Emulator role, you will use: ***transfer pdc***
- **Remove Windows Server 2008 from the Domain**
 - **Demote Windows Server 2008:**
 - On your Windows Server 2008, use Server Manager to demote the server from its domain controller role.
 - Follow the prompts to complete the demotion.
 - **Shutdown Windows Server 2008:**
 - After demotion, shut down the Windows Server 2008 machine.
- **Test User Logins**
 - **Test User Login:**
 - § On your Windows 10 machine, log in using the user credentials created in Active Directory on Zentyal.
 - **Verify Access:**
 - Ensure that you can access files on the Zentyal file share using domain credentials.
 - Also, verify that the Zentyal server is correctly handling domain authentication.
- These steps should help you transition from Windows Server 2008 to a Zentyal based domain controller. Please note that the actual configurations and settings may vary depending on your specific requirements and network environment.

4. Pricing & Support

Zentyal provides Yearly Subscription or Perpetual License packages, and the prices differ based on the required usage and features.

The number of users and Support tickets are the factors that differ the most in between the packages. The pricing scheme from the official website provides more detail about costs and available features.

PRICES

Yearly subscription

Your server always up to date for a yearly payment

[See the Perpetual License prices](#)

	Micro 195 € per server	Small 495 € per server	Medium 995 € per server	Enterprise 2995 € per server
Users	up to 25	up to 75	up to 300	Unlimited
Features	All	All	All	All
Software & security updates	✓	✓	✓	✓
Upgrades to new versions	✓	✓	✓	✓
Technical support	1 ticket	3 tickets	5 tickets	Unlimited
Duration	1 year	1 year	1 year	1 year
	BUY	BUY	BUY	BUY

FIGURE 20

What is the support response time?

The maximum support response time depends on the type of subscription and the severity level of your incident established by the Support Team.

	MICRO, SMALL Y MEDIUM	ENTERPRISE
S1	1 business day	4 business hours
S2	1 business day	4 business hours
S3	2 business days	1 business day
S4	2 business days	2 business days

Maximum response times on business days.

Severity levels: S1 = Critical, S2 = High, S3 = Normal, S4 = Low.

FIGURE 21

The one-time payment license however does not provide any technical support or usable tickets with the license, but these support features can be purchased at any time during the server’s lifetime.

Automatic upgrades to new versions are also disabled in the Perpetual License package.

PRICES

Perpetual licenses

Buy once, use it forever

[See the Yearly Subscription prices](#)

	Micro	Small	Medium	Enterprise
	195 € per server	495 € per server	995 € per server	2995 € per server
Users	up to 25	up to 75	up to 300	Unlimited
Features	All	All	All	All
Software & security updates Until the end of life of the purchased version	✓	✓	✓	✓
Upgrades to new versions	✗	✗	✗	✗
Technical support	✗	✗	✗	✗
Duration	Perpetual	Perpetual	Perpetual	Perpetual
	BUY	BUY	BUY	BUY

FIGURE 22



Business Ticket

MAXIMUM RESPONSE TIME
2 business days

DURATION
1 year

245 €
per ticket

[BUY](#)



Express Ticket

MAXIMUM RESPONSE TIME
4 business hours

DURATION
1 year

495 €
per ticket

[BUY](#)

FIGURE 23

GOOGLE WORKSPACE



FIGURE 24

INTRODUCTION:

Google has created a whole set of cloud-based productivity and collaboration tools called Google Workspace. The way people, companies, and organisations collaborate in the digital era has been significantly changed because of it. Here is a quick overview of Google Workspace that includes its background and significance in the field of cloud-based productivity tools:

History:

Google originally released Google Workspace, formerly known as G Suite, in 2006. It is signalling Google's entry into the market for commercial and enterprise applications. Google Workspace has developed and broadened its offers over time, including additional functions, programmes, and security upgrades. Google's dedication to enhancing workplace efficiency, communication, and collaboration has been the driving force behind this progress.

Importance in the World of Cloud-Based Productivity Tools:

Google Workspace is becoming an important player in the world of cloud productivity tools for several reasons:

1. **Cloud-Centric Approach:** Google Workspace's based on a Cloud centric Model, which means that applications and data will be stored across the Internet. This approach will allow users to access their work wherever they are, on any device with an Internet connection. It will reduce the demand for on-premises infrastructure, which makes it more accessible and easier to use.
2. **Collaboration and Real-Time Editing:** Google Workspace's applications are renowned for their real time collaboration features, such as Google Docs, Sheets and Slides. Multiple users can work at the same time and contribute to efficient teamwork and reduce friction caused by traditional email collaboration, enabling a concurrent editing of documents, spreadsheets, or presentations.
3. **Integrated Communication:** Google Workspace effortlessly combines video conferencing (Google Meet), chat (Google Chat), and email (Gmail). Whether teams

are based locally or remotely, our unified communication suite makes it simpler for them to connect and work in real-time.

4. **Reliability and Security:** Google Workspace offers the highest levels of protection, such as spam filters, data encryption and two factor authentication. It is ensured that the Google Global Infrastructure provides high availability and reliability.
5. **Scalability:** Google Workspace is a cloud service designed for organizations of all sizes, from start-ups to the largest enterprises. Once your organization has grown or evolved, you can easily add or remove users.
6. **Cross-Platform Compatibility:** Google Workspace is compatible with a variety of operating systems and devices, including Windows, Mac, Android, and iOS. That flexibility is designed to provide consistency in the experience on different platforms.
7. **Customization and Integration:** The Google Workspace has a variety of customization options and integration with third party applications and services. This will enable organisations to configure the pack according to their needs and processes.
8. **Cost-Efficiency:** The Google Workspace is available with competitive pricing options, making it more cost effective for businesses and organisations. It'll cut the cost of expensive on-premises software and infrastructure.

USE CASES AND TARGET AUDIENCE:

Google Workspace aims to appeal to a wide range of users, such as SMEs, big businesses, higher education institutions, NGOs, and individuals. It is a useful instrument for many types of organizations and individuals, thanks to its flexibility and adaptability. In Google Workspace, there are some of the main target audiences and common use cases:

1.Small Businesses:

Use Case: To set up a professional web presence with specific domain email addresses, such as www.YouYourBusiness.com, smaller businesses frequently use Google Workspace to do so. To be able to have easy access to important data and documents, they take advantage of the use of email communication, document collaboration as well as cloud storage.

Additional Use Cases: Sharing and collaborating on marketing materials, managing appointments with Google Calendar, and conducting video meetings with Google Meet.

2. Large Enterprises:

Use Case: With its scalability and strong set of tools, Google Workspace is a powerful tool for larger organizations. They'll be able to set up more user accounts, implement the level of security and compliance features for their enterprises, as well as simplify communication between them both internally and externally.

Additional Use Cases: Centralized document management, enterprise email with enhanced security, and custom app development using Google Workspace APIs.

3. Educational Institution:

Use Case: Google Workspace is used by education organizations to improve remote and in person learning, e.g., schools and universities. It ensures the use of collaborative tools, a secure communications platform and easy access to educational resources for students, teachers, and staff.

Additional Use Case: Management of the classroom, collaborative research projects, and the creation of a digital ecosystem for eLearning.

4. Non-Profit Organization:

Use Case: Non-profit organizations can avail themselves of Google Workspace's discounts for non-profits. It shall be used for the management of operations, as well as to facilitate cooperation between volunteers and staff with a view to promoting their missions.

Additional Use Cases: Fund raising and donor management, sharing of resources and information with interested parties, and coordination of events.

5. Individuals and Freelancers:

Use Case: To create custom email addresses as well as access tools such as Google Drive for the storage and collaboration of files, people or businesses who want to make a professional presence on the Internet can take advantage of Google Workspace.

Additional Use Case: Resume building and job applications, personal project management, and using Google Sites for personal websites or portfolios.

6. Remote and Hybrid Workforces:

Use Case: Google Workspace is an excellent solution for teams who need to work from a variety of locations due to the growing adoption of Remote and Hybrid Work Arrangements. It offers support for remote work by means of communication and collaboration tools.

Additional Use Case: Virtual team meetings, recording of collaboration from different times zones and secure access to company resources anywhere in the world.

7. Creative and Design Team:

Use Case: Google Workspace enables creative and design teams to work together on projects in the field of graphic design, advertising campaigns or multimedia production. Working together on visual and media content in Google Workspace is easy.

Additional Use Case: In Google Drive, you can create and share design prototypes, real time design review sessions, and manage creative assets.

KEY FEATURES:

The Google Workspace offers a complete suite of cloud-based productivity and collaboration tools, each with its own set of features. A breakdown of the key features of Google Workspace is given below:

1. Email (Gmail):

- ☐ Professional mail services with a custom domain email address.
- ☐ A great amount of storage space for your email.
- ☐ To protect against threats, spam filtering and robust security measures.

2. Documentation Collaboration (Google Docs):

- ☐ Realtime collaboration in editing and commenting on documents.
- ☐ You can choose from a wide variety of formatting and style options.
- ☐ Version history and file recovery for tracking changes.

3. Cloud Storage (Google Drive):

- ☐ Secure cloud storage of files, documents, photographs, and videos.
- ☐ File sharing and collaboration features with customizable access permissions.
- ☐ To make it easy to access, you can integrate with other Google Workspace apps.

4. Spreadsheet (Google Sheets):

- ☐ An online spreadsheet software to analyse and visualize data.
- ☐ Realtime collaboration of several users.
- ☐ Advanced formula and charting capabilities.

5. Presentation (Google Slides):

- ☐ Presentation software to create slideshows and graphical content.
- ☐ Collaboration on the creation and editing of presentations in real time.
- ☐ Built-in templates and themes for professional designs.

6. Online Forms (Google Forms):

- ☐ Tool for the creation of survey, questionnaire, and data collection forms.
- ☐ For analysis, responses will be automatically collected and stored in Google Sheets.
- ☐ Customizable forms based on the different types of questions.

7. Calendar (Google Calendar):

- ☐ Videoconferencing and a platform for meetings on the internet.
- ☐ High-quality video and audio, suitable for small to large meetings.
- ☐ To schedule easily, you can integrate with Google Calendar.

8. Video Conferencing (Google Meet):

- ☐ A platform for videoconferencing and online meetings.
- ☐ The quality of the video and sound is excellent for small to large meetings.
- ☐ To schedule easily, you can integrate with Google Calendar.

9. Real-time Chat (Google Chat):

- ☐ Realtime chat and text application to communicate with the team.
- ☐ Allows direct mail, group conversation and thread conversations to be used.
- ☐ A seamless video conferencing integration with Google Meet.

10. Cloud-Based Website Building (Google Sites):

- ☐ Web building and publishing tool that enables you to create your own or public websites.
- ☐ The interface for designing Web pages is user friendly, drag and drop.
- ☐ Integrates to the rest of Google Workspace applications and services.

11. Note-Taking (Google Keep):

- ☐ A tool for taking notes and organising reminders, to-do lists, and schedules.
- ☐ Integration for simple access with other Google Workspace apps.

12. Admin Console:

- ☐ Centralized management and control of Google Workspace.
- ☐ Management of users and equipment, security settings, and reporting.
- ☐ Access to the billing and account information.

SETTING UP GOOGLE WORKSPACE

Google's workspace service can help you with a lot of things, but one of the most notable features is that it hosts your email for you and lets you use the Gmail interface with your own domain name (e.g., you@yourdomain.com), thus giving you a professional email address.

That makes it a great option as a email hosting solution for your WordPress site, though this step-by-step guide is by no means limited to just WordPress users.

Every step that you need to take to get google workspace working with the website and running with the professional email address.

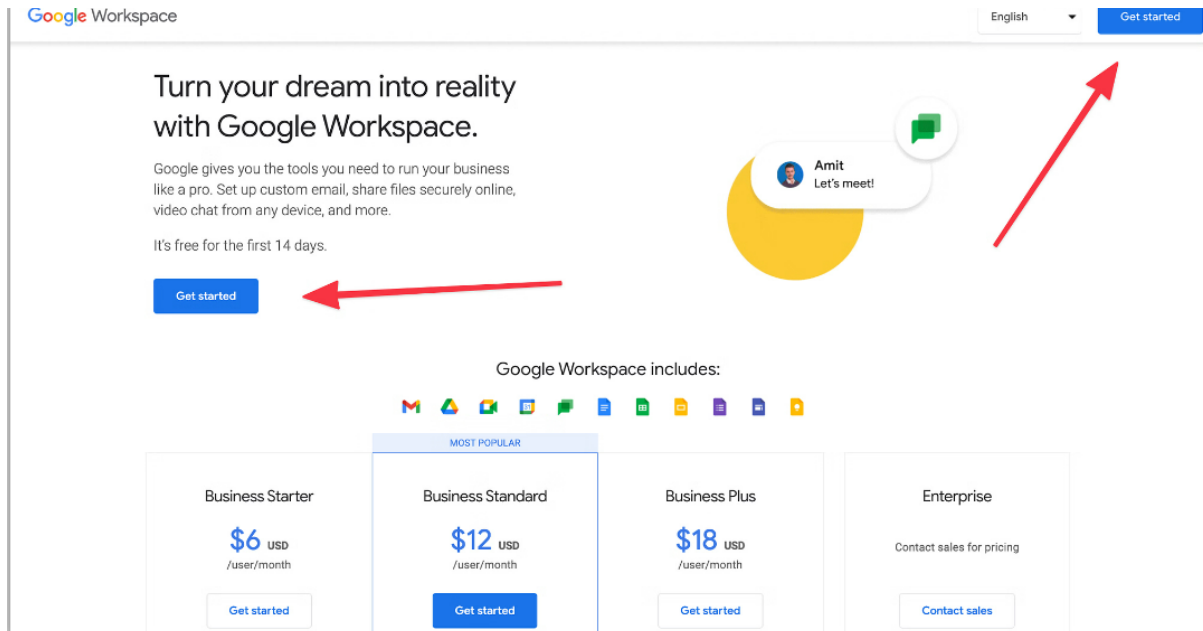
By the end, you'll have a working email account hosted through Google Workspace, as well as access to all the other Google workspace functionalities (Google apps, like Google Hangouts, separate Google drive storage for Docs, Sheets, etc.).

A nicely set up G Suite/ Google Workspace is equally useful for small businesses with only a small number of employees – even non-profits.

HOW TO SET UP GOOGLE WORKSPACE (G SUITE) in 4steps (with screenshots)

Here's a quick overview of the steps that you'll need to take to configure Google workspace with your website and professional email address:

FIGURE 25



1. Step 1: Register an account and go through the Google workspace setup wizard.
2. Step 2: Add other users to your account
3. Step 3: Verify domain ownership by adding a TXT record to your DNS records.
4. Step 4: Set up email by adding MX records.

STEP 1: COMPLETE THE GOOGLE WORKSPACE (G Suite) ACCOUNT SETUP WIZARD

To get started, you'll need to create your own Google Workspace account.

To do that, head here and click the big Get started button:

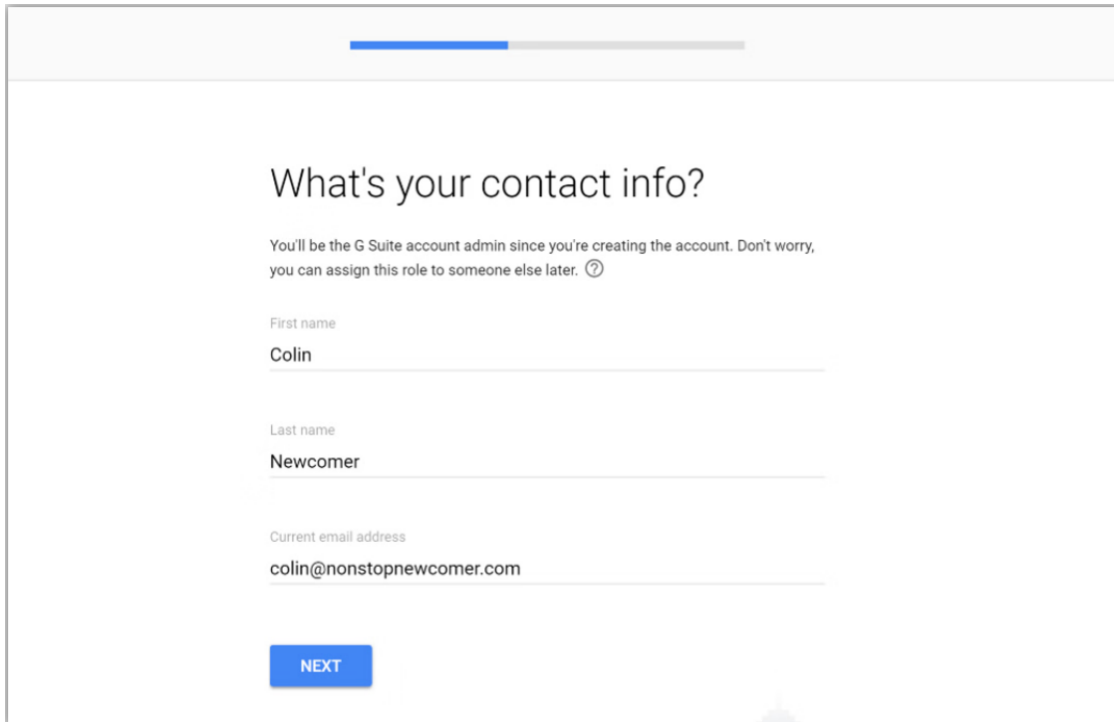
This will launch the account setup wizard, which will prompt you to provide Google with some information.

First, enter some basic information about your account.

Then, click **Next**:

On the next screen, you'll need to enter the information for your Google Workspace admin account.

If you've already set up email through your domain host, you use that email in the current email address box. Otherwise, you can enter a personal email address.

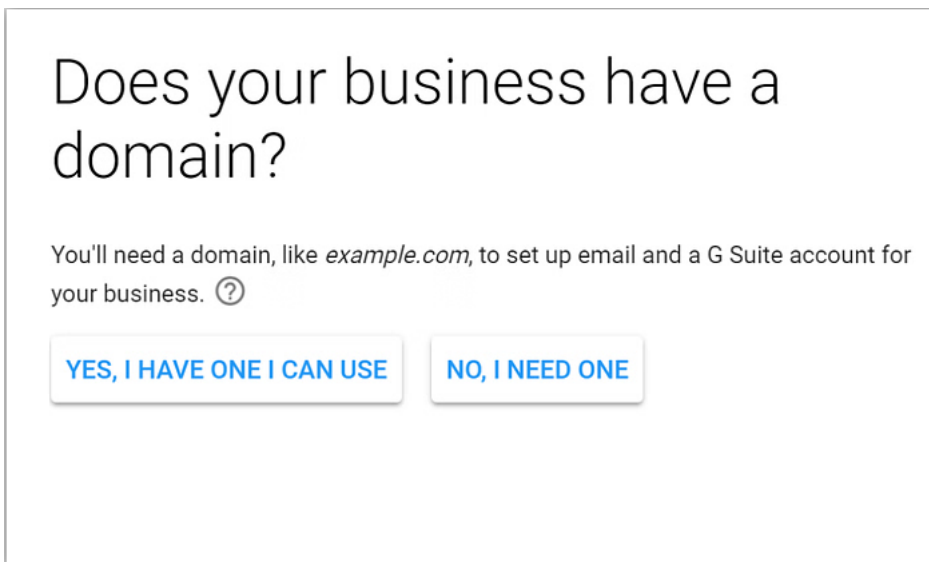


The screenshot shows a web form titled "What's your contact info?". Below the title is a sub-header: "You'll be the G Suite account admin since you're creating the account. Don't worry, you can assign this role to someone else later. ?". The form contains three input fields: "First name" with the value "Colin", "Last name" with the value "Newcomer", and "Current email address" with the value "colin@nonstopnewcomer.com". At the bottom of the form is a blue button labeled "NEXT".

FIGURE 26

Next, indicate whether your business already has a domain name. If you already have a website that you want to use with Google Workspace, you should choose

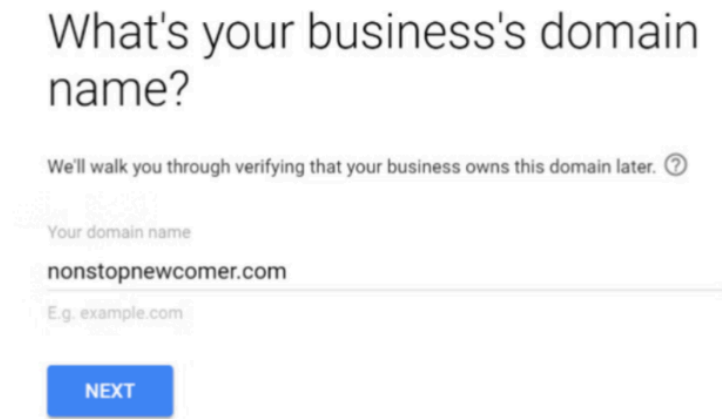
Yes, I Have One I Can Use:



The screenshot shows a web form titled "Does your business have a domain?". Below the title is a sub-header: "You'll need a domain, like *example.com*, to set up email and a G Suite account for your business. ?". At the bottom of the form are two buttons: "YES, I HAVE ONE I CAN USE" and "NO, I NEED ONE".

FIGURE 27

Then, enter your existing domain name in the box and click **Next**:



What's your business's domain name?

We'll walk you through verifying that your business owns this domain later. ?

Your domain name

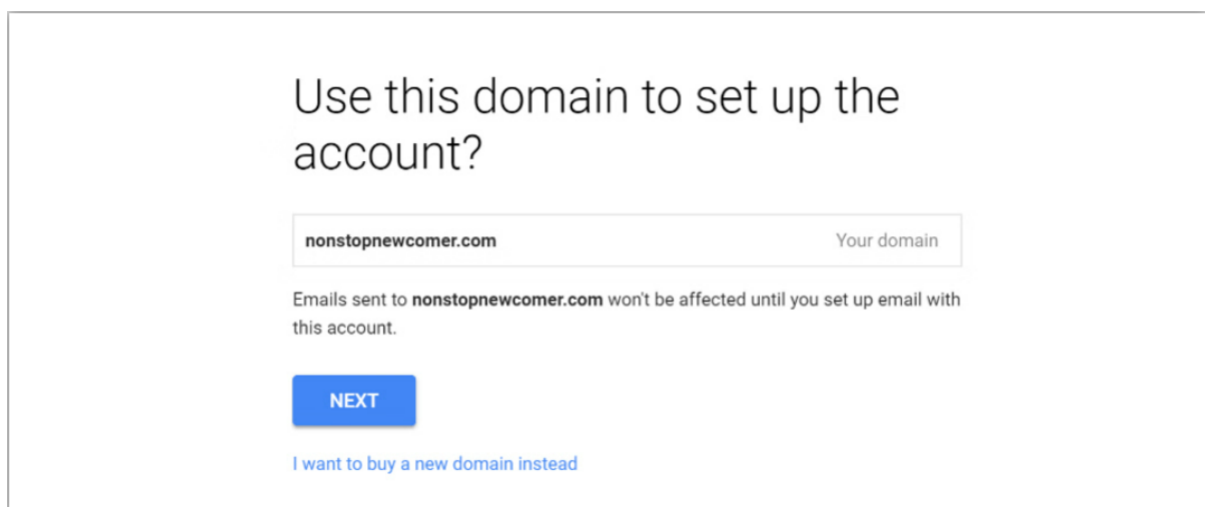
nonstopnewcomer.com

E.g. example.com

NEXT

FIGURE 28

Then, click Next again to confirm that you want to use that domain name:



Use this domain to set up the account?

nonstopnewcomer.com Your domain

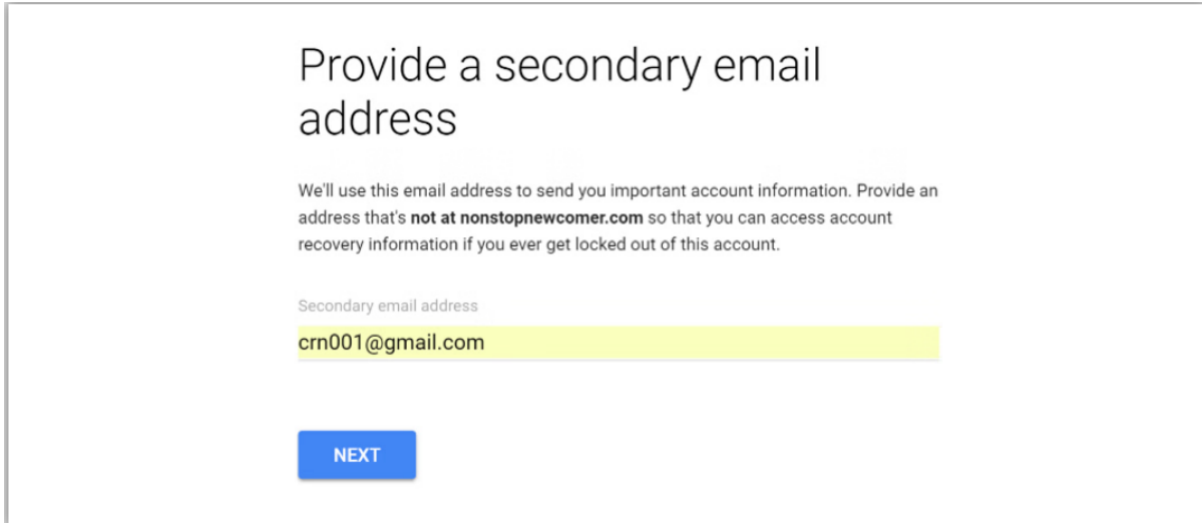
Emails sent to nonstopnewcomer.com won't be affected until you set up email with this account.

NEXT

[I want to buy a new domain instead](#)

FIGURE 29

After that, you'll need to enter a secondary recovery email address, which you'll use in case you can't access your primary email address. For example, you could enter your personal Gmail account here, if you have one:

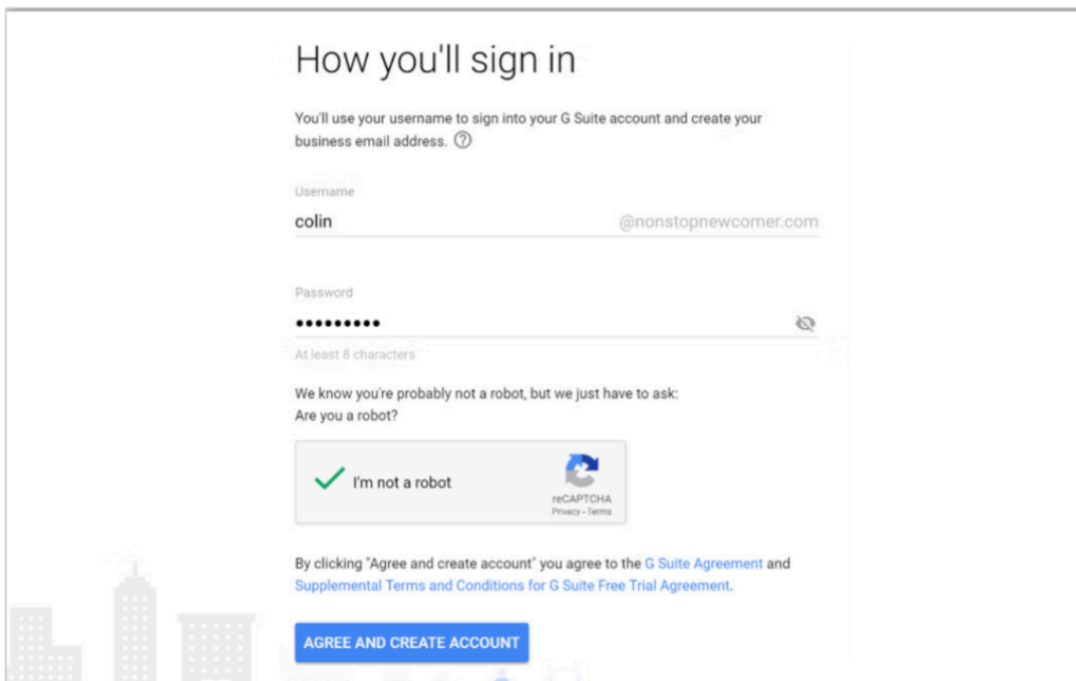


The screenshot shows a web form titled "Provide a secondary email address". Below the title, a message states: "We'll use this email address to send you important account information. Provide an address that's **not at nonstopnewcomer.com** so that you can access account recovery information if you ever get locked out of this account." Below this message is a label "Secondary email address" and a text input field containing "crn001@gmail.com". At the bottom of the form is a blue button labeled "NEXT".

FIGURE 30

Then, you'll need to enter a username and password that you'll use to sign into Google Workspace. Your username will be your business email address by default (i.e., username@yourdomain.com), so you should keep that in mind when choosing which username to use.

Then, click **Agree and Create Account** to finish the process:

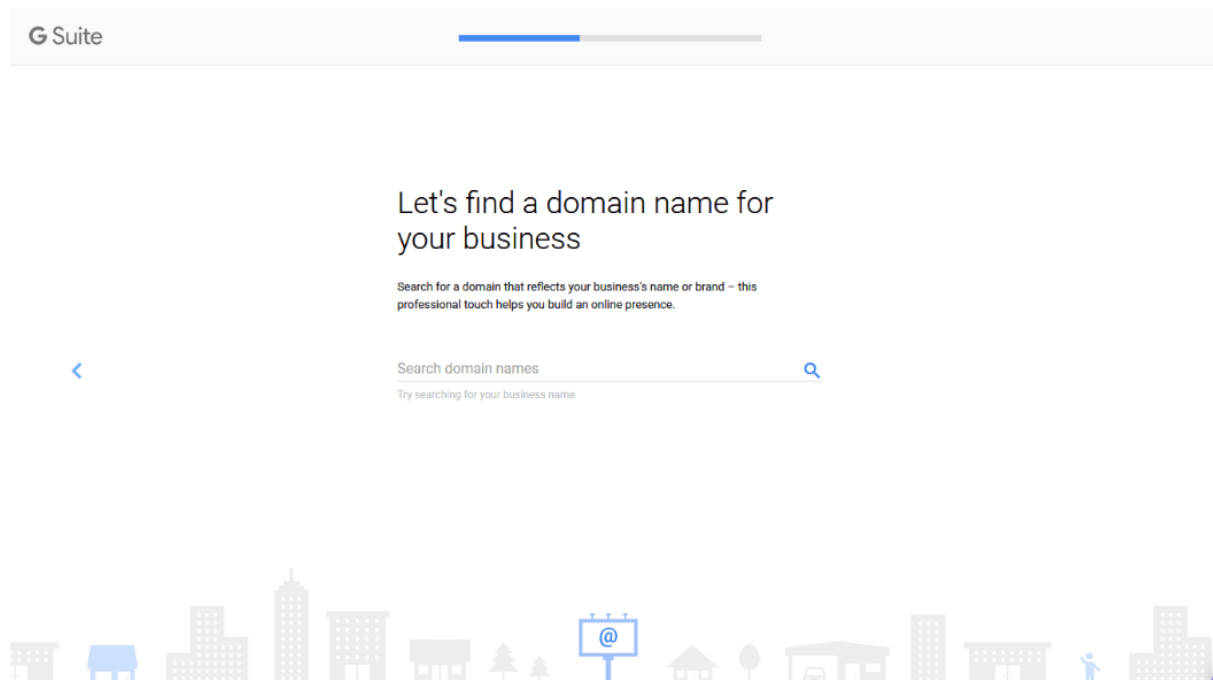


The screenshot shows a web form titled "How you'll sign in". Below the title, a message states: "You'll use your username to sign into your G Suite account and create your business email address. ?". Below this message are two input fields: "Username" with the value "colin" and "@nonstopnewcomer.com", and "Password" with a masked value "••••••••". Below the password field is a note "At least 8 characters". Below this is a CAPTCHA section with the text "We know you're probably not a robot, but we just have to ask: Are you a robot?" and a checkbox labeled "I'm not a robot" which is checked. To the right of the checkbox is a reCAPTCHA logo and links for "Privacy" and "Terms". Below the CAPTCHA section is a message: "By clicking 'Agree and create account' you agree to the [G Suite Agreement](#) and [Supplemental Terms and Conditions for G Suite Free Trial Agreement](#)." At the bottom of the form is a blue button labeled "AGREE AND CREATE ACCOUNT".

FIGURE 31

OR IF YOU DON'T HAVE DOMAIN THEN YOU HAVE ONE, CLICK ON I DON'T HAVE ONE.

Let find a domain name for your business



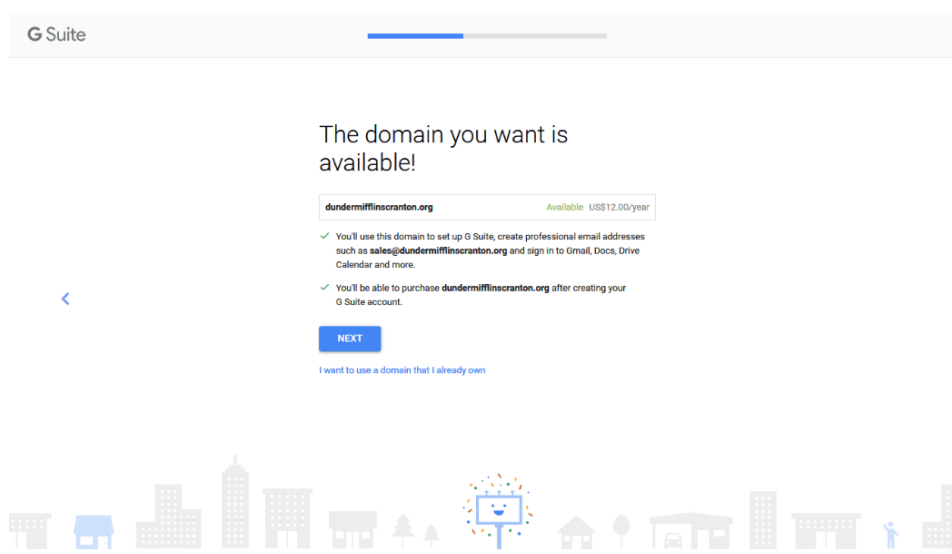
The screenshot shows the G Suite domain search interface. At the top, the 'G Suite' logo is on the left, and a blue progress bar is in the center. Below this, the heading 'Let's find a domain name for your business' is centered. Underneath the heading is a subtext: 'Search for a domain that reflects your business's name or brand – this professional touch helps you build an online presence.' A search bar with the placeholder text 'Search domain names' and a magnifying glass icon is positioned below the subtext. A small link 'Try searching for your business name' is located directly under the search bar. At the bottom of the page, there is a decorative row of various building and house icons.

FIGURE 32

Here, enter a domain name you want to buy. Domains can cost as little as \$12 a year to as much as \$50 a year. That's because every domain has its own value, and therefore, no fixed price.

Also, note that not all domain names that you type in will be available. So, it might take a few tries until you find the perfect name.

Once you find it, click on it and you'll reach this page:

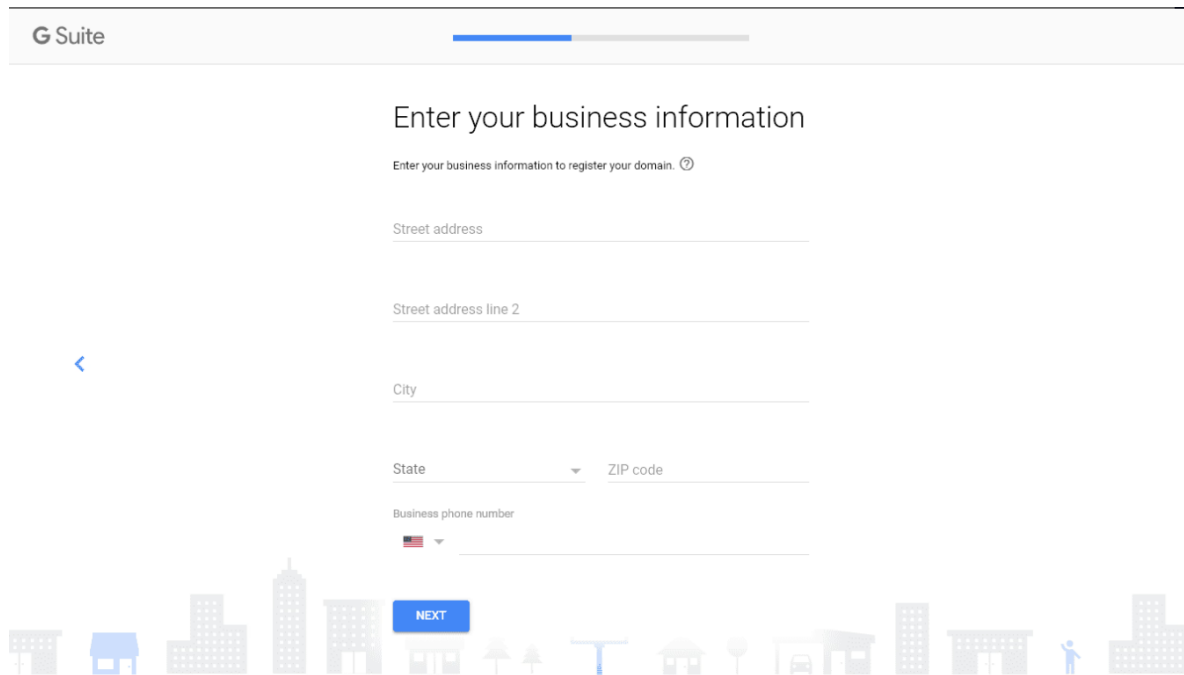


The screenshot shows the G Suite domain availability confirmation page. At the top, the 'G Suite' logo is on the left, and a blue progress bar is in the center. Below this, the heading 'The domain you want is available!' is centered. Underneath the heading, the domain 'dundermifflinscranton.org' is displayed in a box, with 'Available US\$12.00/year' to its right. Below the domain box, there are two checkmarks with text: '✓ You'll use this domain to set up G Suite, create professional email addresses such as sales@dundermifflinscranton.org and sign in to Gmail, Docs, Drive Calendar and more.' and '✓ You'll be able to purchase dundermifflinscranton.org after creating your G Suite account.' A blue 'NEXT' button is positioned below the checkmarks. A link 'I want to use a domain that I already own' is located at the bottom left of the main content area. At the bottom of the page, there is a decorative row of various building and house icons.

FIGURE 33

Click 'Next'. You'll land on page where you'll have to enter the exact address of your business. This is required to register your domain.

ENTER YOUR BUSINESS INFORMATION TO REGISTER YOUR DOMAIN

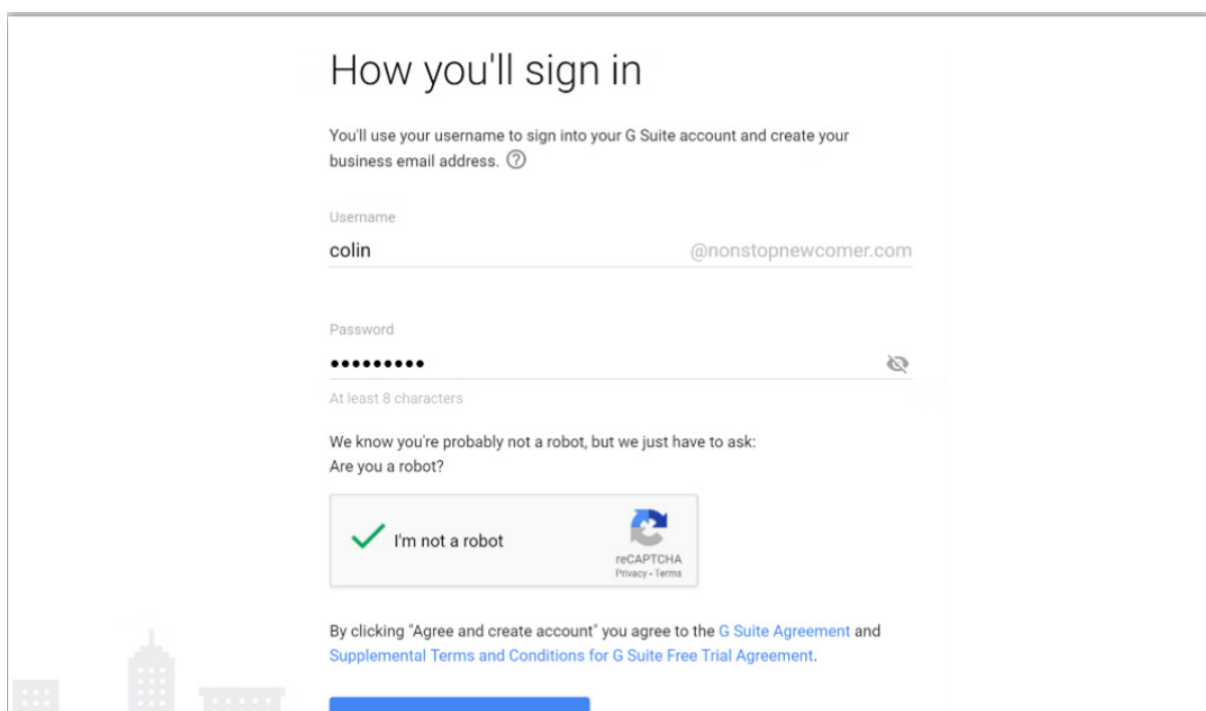


The screenshot shows the 'G Suite' header with a progress bar. The main heading is 'Enter your business information'. Below it, a subtext says 'Enter your business information to register your domain.' followed by a help icon. The form includes input fields for 'Street address', 'Street address line 2', 'City', 'State' (a dropdown menu), and 'ZIP code'. There is also a 'Business phone number' field with a country code dropdown (currently showing the US flag). A blue 'NEXT' button is positioned above a decorative city skyline illustration at the bottom of the form.

FIGURE 34

Fill the form and Click 'Next':

The next page will ask you to create your first G Suite username:



The screenshot shows the 'How you'll sign in' page. The heading is 'How you'll sign in'. Below it, a subtext says 'You'll use your username to sign into your G Suite account and create your business email address.' followed by a help icon. The form has two main sections: 'Username' and 'Password'. The 'Username' section shows 'colin' in the input field and '@nonstopnewcomer.com' as a suggestion. The 'Password' section shows a masked password field with a strength indicator 'At least 8 characters' and a toggle icon. Below the password field, there is a text prompt 'We know you're probably not a robot, but we just have to ask: Are you a robot?' followed by a reCAPTCHA widget showing a green checkmark and the text 'I'm not a robot'. At the bottom, there is a link to the 'G Suite Agreement' and 'Supplemental Terms and Conditions for G Suite Free Trial Agreement', and a blue 'AGREE AND CREATE ACCOUNT' button.

FIGURE 35

Choose any username you like. It can be your name, the department in which you work or even something as simple as hello.

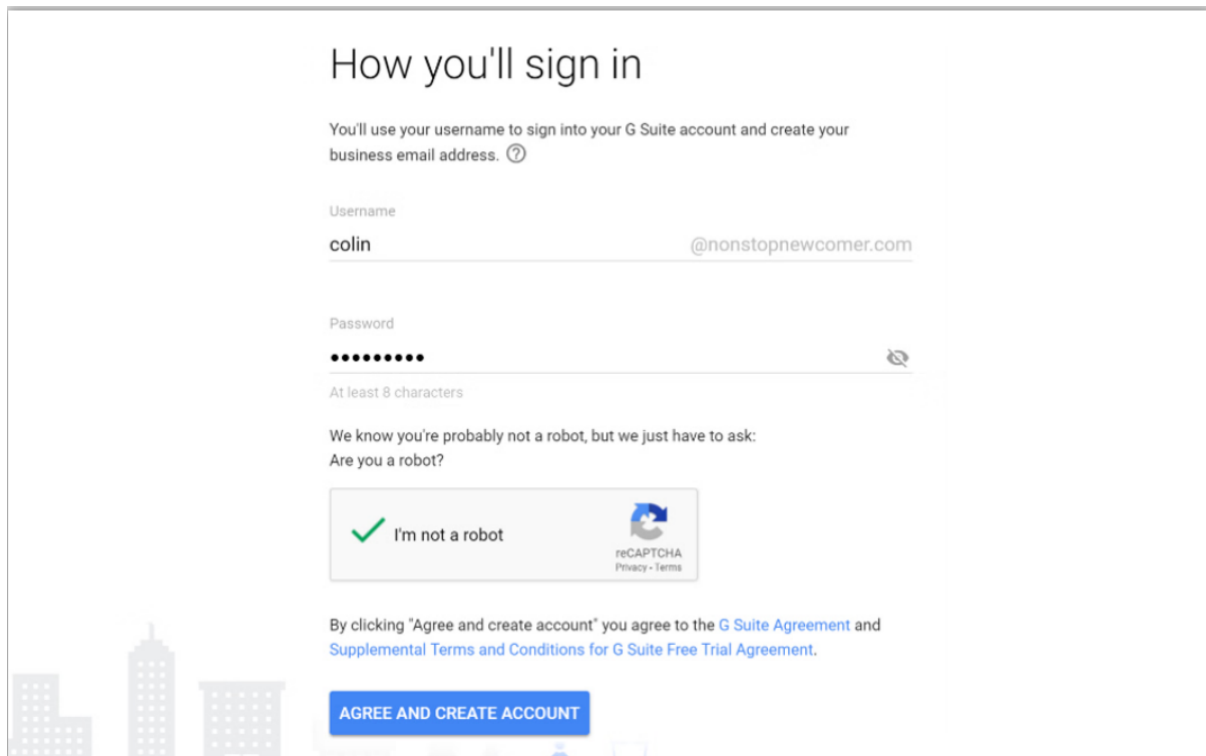
The screenshot shows the Google account creation interface. At the top, it says "How you'll sign in". Below this, a message states: "You'll use your username to sign into your G Suite account and create your business email address." followed by a help icon. The "Username" field contains "colin" and the domain "@nonstopnewcomer.com" is shown to the right. The "Password" field is filled with dots, with a "Show/Hide" icon to its right. Below the password field, it says "At least 8 characters". A message follows: "We know you're probably not a robot, but we just have to ask: Are you a robot?". Below this is a reCAPTCHA widget with a green checkmark and the text "I'm not a robot". At the bottom, a blue button reads "AGREE AND CREATE ACCOUNT". Above this button, a line of text says: "By clicking 'Agree and create account' you agree to the [G Suite Agreement](#) and [Supplemental Terms and Conditions for G Suite Free Trial Agreement](#)." The background features a faint city skyline illustration.

FIGURE 36

and a number in your password to make it as safe as possible.

Click on 'I'm not a 'robot' – and click on 'Agree and Continue'.

That's it!

Sign into your account.

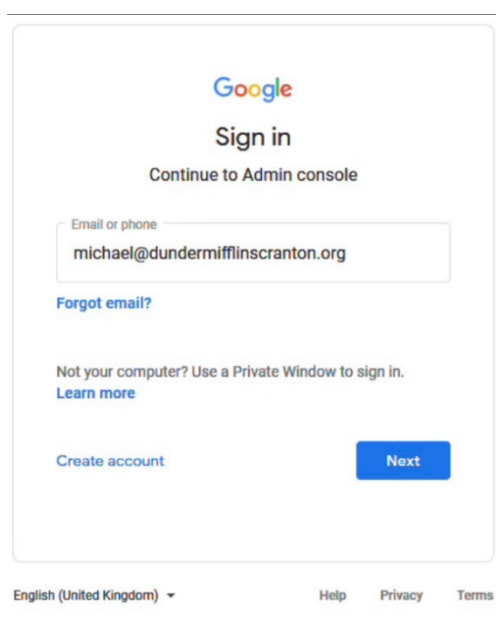
The screenshot shows the Google "Sign in" page. At the top is the Google logo, followed by "Sign in" and "Continue to Admin console". Below this is a text input field labeled "Email or phone" containing the email "michael@dundermifflinscranton.org". A link "Forgot email?" is positioned below the input field. Further down, a message says: "Not your computer? Use a Private Window to sign in." with a "Learn more" link. At the bottom left is a "Create account" link, and at the bottom right is a blue "Next" button. The footer contains "English (United Kingdom)" with a dropdown arrow, and links for "Help", "Privacy", and "Terms".

FIGURE 37

You'll land on the familiar sign-in screen. Sign into your account. Once you sign in, you'll be asked to review your G Suite plan, Confirm, and click next.

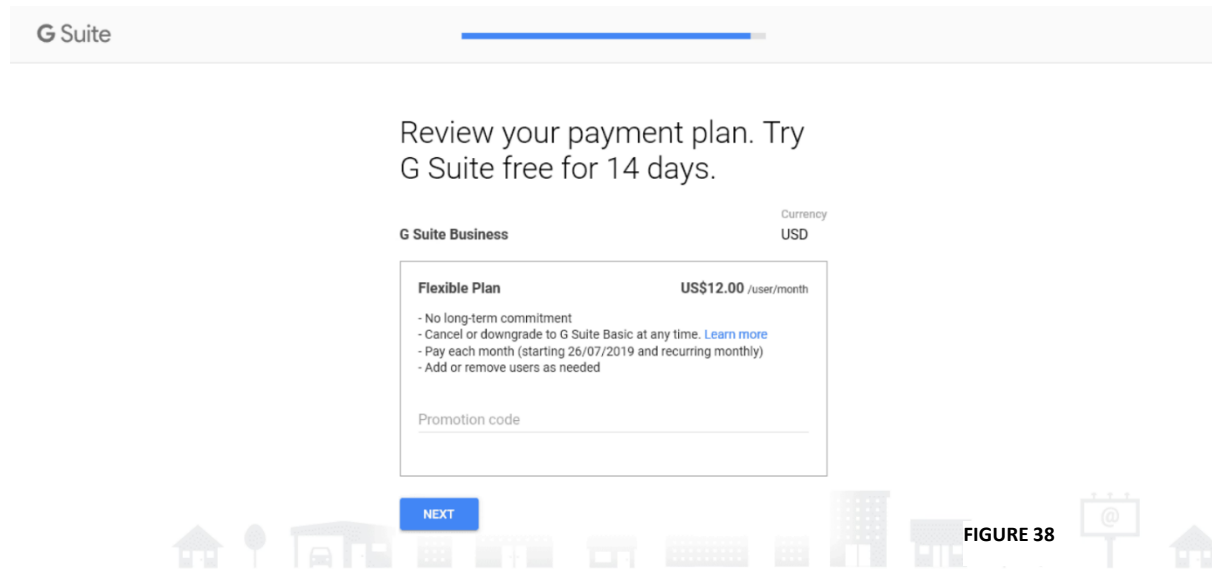


FIGURE 38

You'll see the checkout page. Here, enter your credit card details, review your information, and click next. You'll see your account has been created and can go to the setup screen.

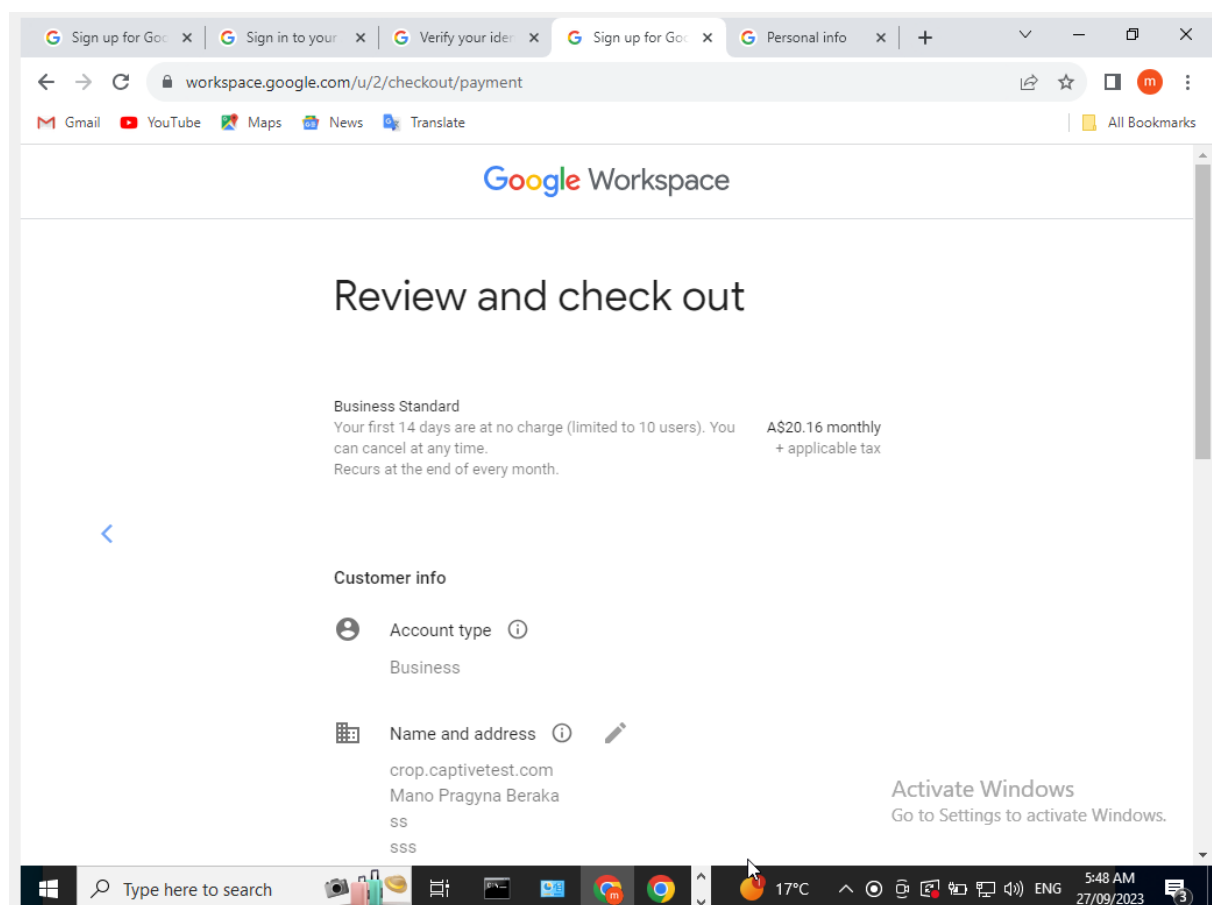
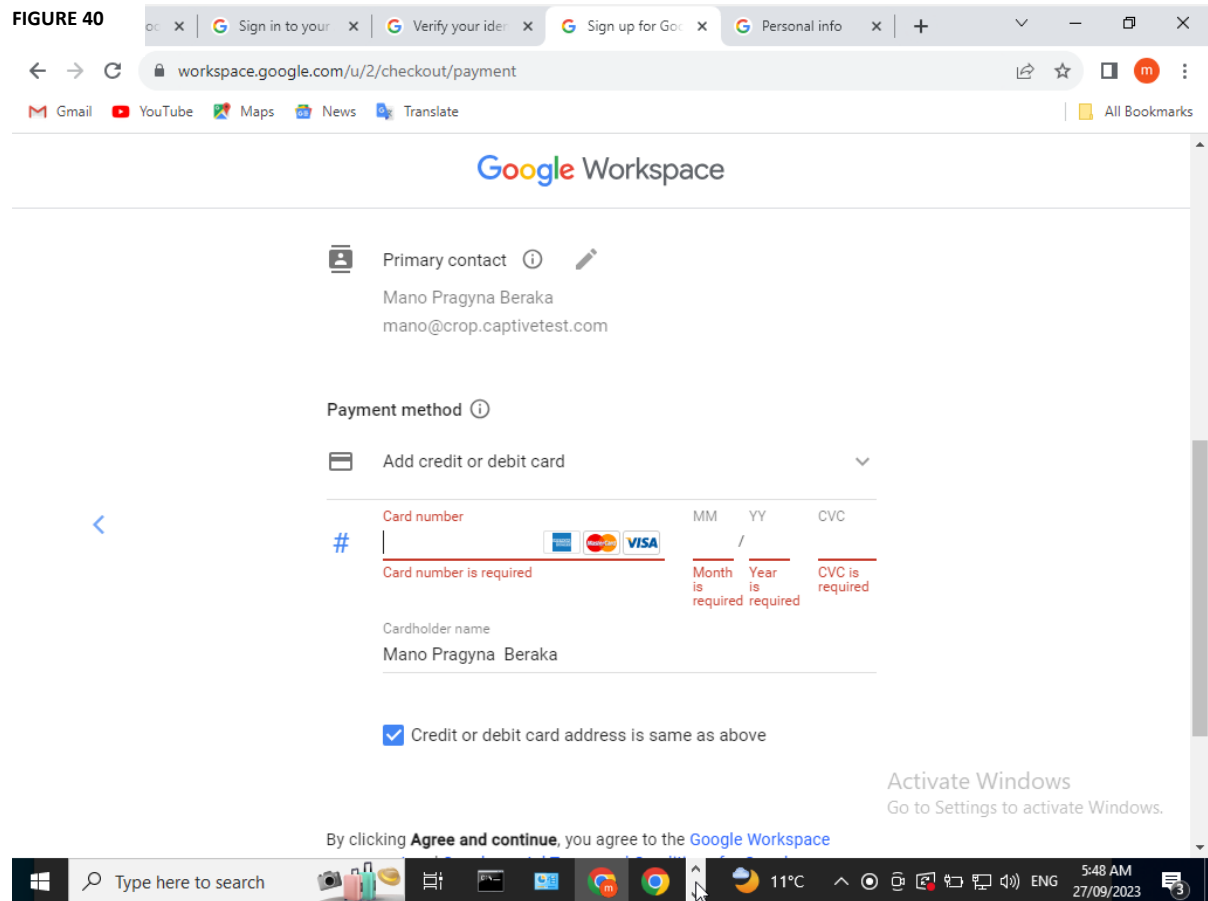


FIGURE 39



Once you've signed up for a free trail, you'll land on the setup page which will look something like this.

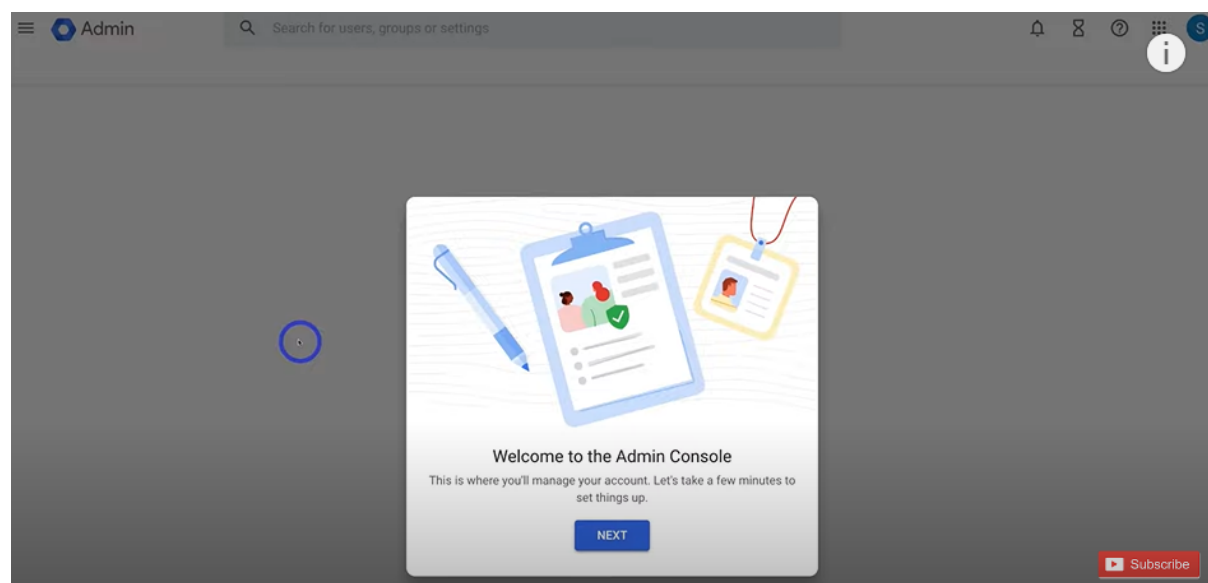


FIGURE 41

STEP 2: ADD OTHER PEOPLE TO GOOGLE WORKSPACE (OPTIONAL)

Once you finish the wizard above, you'll see a conformation that your account was created, as well as a button to Go to Setup. Go ahead and click that button:

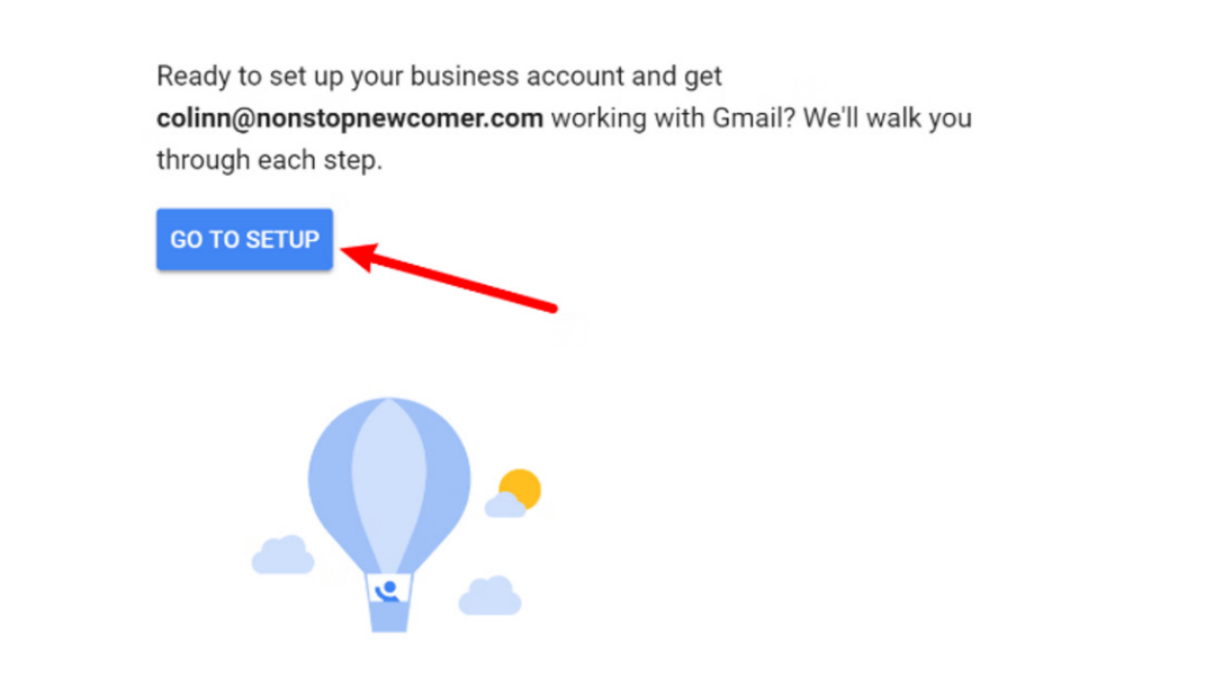


FIGURE 42

If you want to give other people access to your shared Google Workspace account, click Start next to Add people to your Google Workspace account. Then, you'll be able to add additional users.

If you're the only person who will use this Google Workspace account, just check the box for I added all user email... and click Next:

FIGURE 43

Add your users to give them access to G Suite. [Learn more](#)

- Add up to 10 users below. If you want to add more users, use [Manual Setup](#).
- Groups like [info@nonstopnewcomer.com](#) or [sales@nonstopnewcomer.com](#) can be added later at no additional cost.

Already receiving email at [nonstopnewcomer.com](#)? Be sure to add everyone who has an existing email address with [@nonstopnewcomer.com](#). This ensures that they continue receiving email when you switch to G Suite. [Learn more](#)

First name Last name

username @nonstopnewcomer.com

Colin Newcomer (You)
colinn@nonstopnewcomer.com

☒ I added all user email addresses currently using @nonstopnewcomer.com.

Adding users

User List - Admin Console

admin.google.com/ac/users?action_id=ADD_USER&journey=1

× Add new user HELP

User Information

First name * Surname *

Primary email * @horribleonhold.com
This will be the email that the user signs in with

Secondary email Phone number
An email (like a personal email) where you can send the user initial sign-in instructions

[Manage user's password, organisational unit and profile photo](#)

This user will be assigned a Google Workspace Business Standard licence (\$20.36 AUD per user/month)

Type here to search

2:50 PM 12/10/2023

FIGURE 44

Adding the user details

User List - Admin Console

admin.google.com/ac/users?action_id=ADD_USER&journey=1

Add new user [HELP](#)

User Information

First name *
Bhawna

Surname *
Soni

Primary email *
bhawna@horribleonhold.com

This will be the email that the user signs in with

Secondary email
manopragynaberakaau@gmail.com

An email (like a personal email) where you can send the user initial sign-in instructions

Organisational unit *
Captivate Connect

[UPLOAD PROFILE PHOTO](#)

Password

☒ Automatically generate a strong password with 16 characters
You'll be able to view and copy the password in the next step

☐ Create password

FIGURE 45

New user added.

User List - Admin Console

admin.google.com/ac/users?journey=1

New user added [HELP](#)

Your new user can start using Google Workspace within 24 hours. In most cases, it should just take a few minutes.

Bhawna Soni

Username: bhawna@horribleonhold.com

Password: NQ>43b24L8A76aV%

[COPY PASSWORD](#) [PRINT](#)

Send sign-in instructions

The email will provide a link to set the password and sign in to Google Workspace

[PREVIEW AND SEND](#)

The user has been assigned Google Workspace Business Standard licence (\$20.16 AUD per user/month). The current month's payment will be prorated. [View billing](#)

[ADD ANOTHER USER](#) [DONE](#)

FIGURE 46

Users, Username and Password.

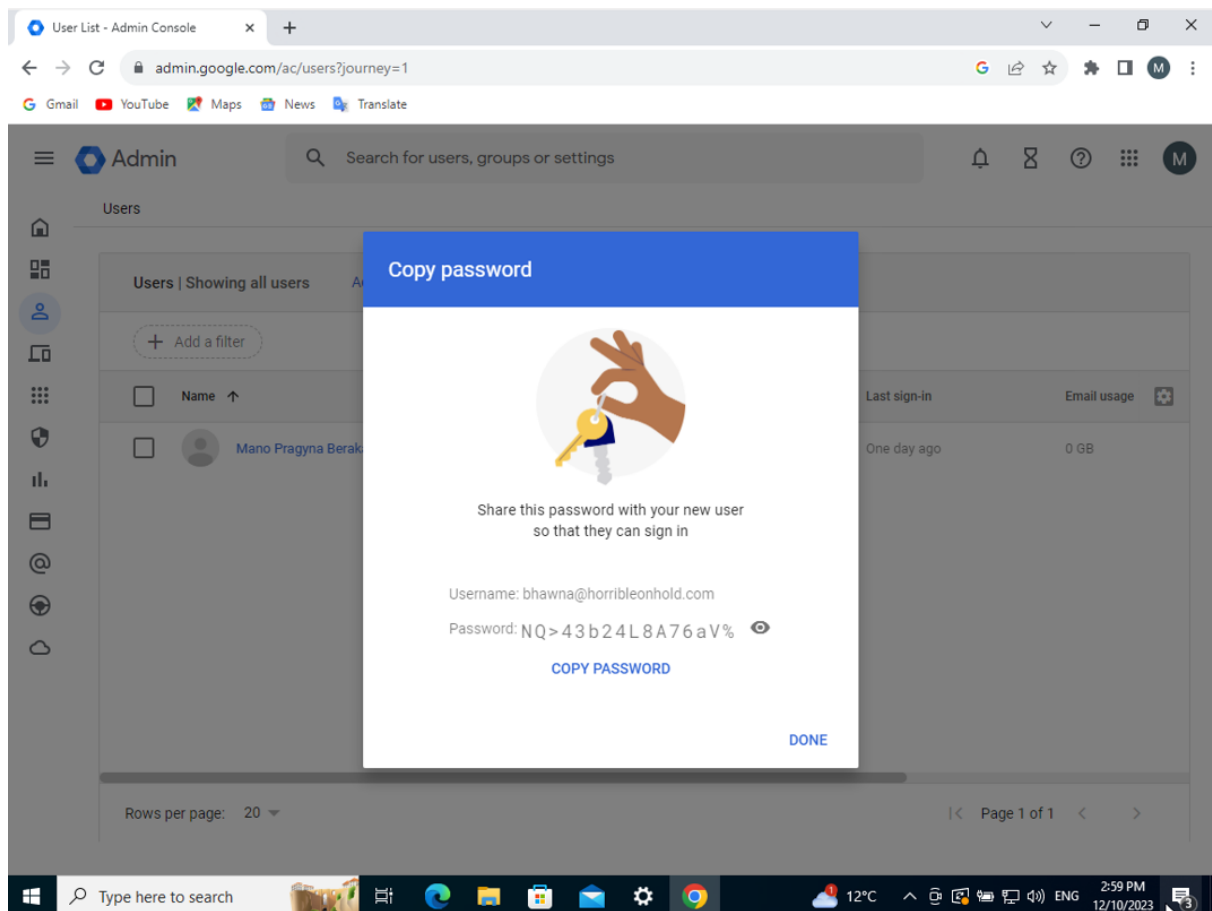


FIGURE 47

STEP 3: VERIFY YOUR DOMAIN NAME WITH GOOGLE

Now is when you start getting into the more technical aspects of how to set up Google Workspace (G Suite).

First, you'll need to verify your domain ownership with Google by adding something called a TXT record.

To help you do this, Google will try to detect where your domain is hosted and provide instructions. For example, Google detected that my example site is hosted at Site Ground.

These instructions are helpful, so they may be all you need.

But to give you a little extra help, I'll show you how to do things using cPanel, which is the hosting dashboard (usually part of your hosting account) that most web hosts use (especially budget web hosts). If your host doesn't use cPanel, you might need to consult your host's support if Google's instructions aren't enough by themselves.

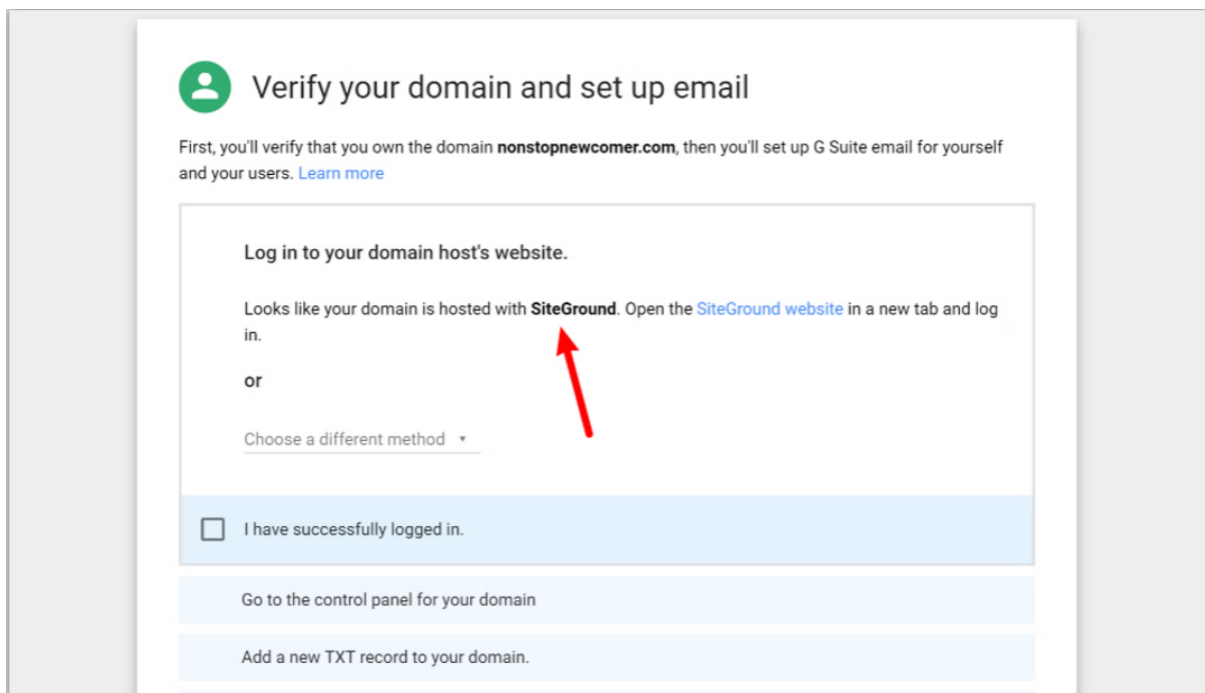


FIGURE 48

Zone Editor Tool:

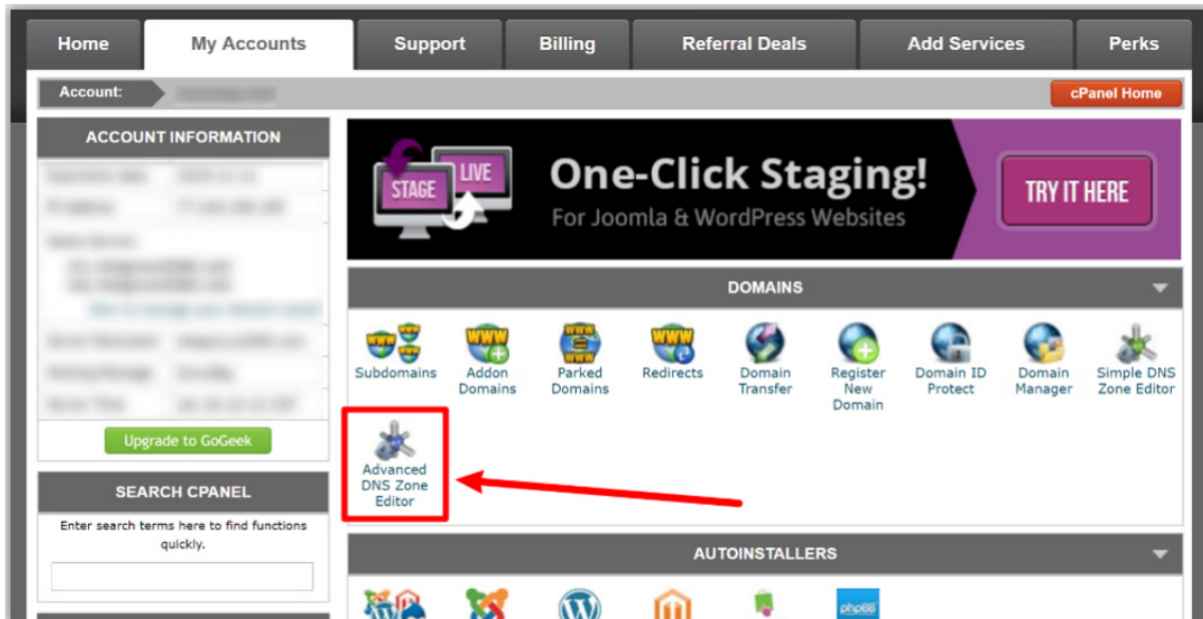


FIGURE 49

Next, select your domain name from the drop-down.

Then, you need to use the form to add TXT record that contains the information from the google Workspace (G suite) Website:

1. Name – your domain name
2. TTL – (The sort of expiration date that is put on a DNS record.) 86400.

TTL is measured in seconds. The typical default value is usually 12 hours (43200 seconds) or 24 hours (86400 seconds). This means that if, for example, a website moves to a new server or you add a new URL to your server, it takes 12 or 24 hours for the new DNS changes to take effect.

3. Type – TXT
4. TXT data – copy and paste from the Google workspace (G Suite) interface (click below if you're not sure where to find this)

STEP 4: ADD MX RECORDS FOR YOUR PROFESSIONAL EMAIL ADDRESS.

Next, you need to complete one more technical step and add something call MX Records. These are the things that allow Google workspace (G Suite) to handle email for your new domain name.

Again, I'll show you to do this using cPanel. But if your host doesn't use cPanel, you might need to reach out to your host's support staff.

To get started, go back to main cPanel dashboard and final the MX Entry tool:

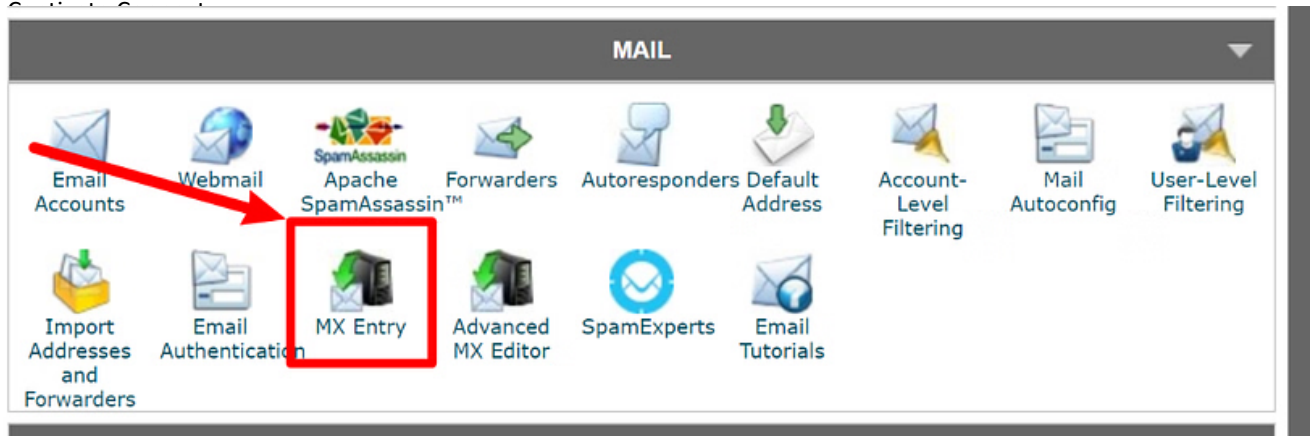


FIGURE 50

Then, select your domain name from the drop-down. After that, you should see a pre-made button for Set Google MX. That's all you need to click! No need to do things manually:

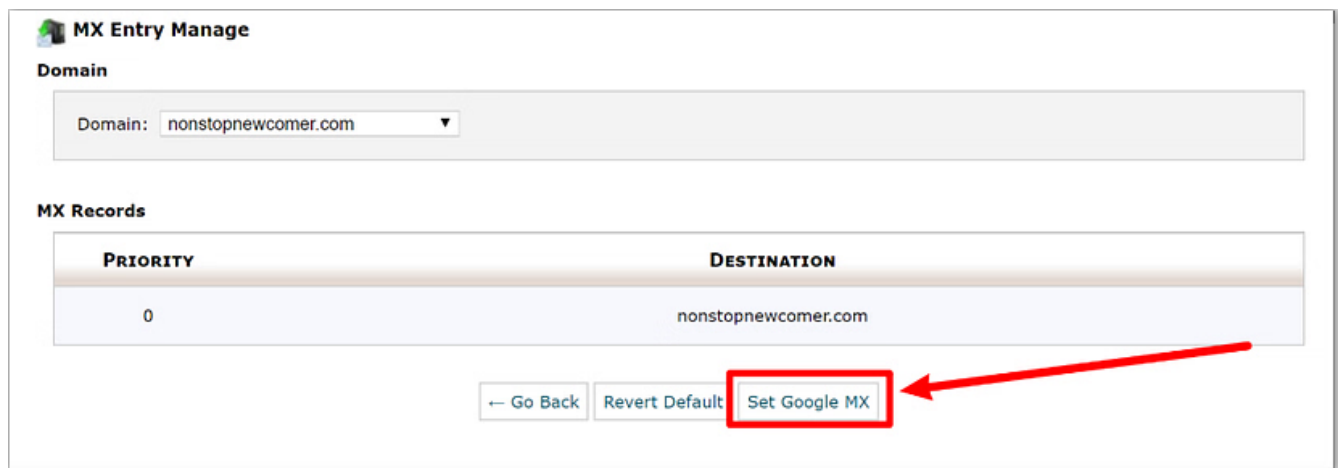


FIGURE 51

If you don't see that pre-made option for Google account, you can open the Advanced MX Editor from your main cPanel dashboard.

Then, you can manually delete the existing entries and use the form to add the following entries:

PRIORITY	DESTINATION
1	ASPMX.L.GOOGLE.COM
5	ALT1.ASPMX.L.GOOGLE.COM
5	ALT2.ASPMX.L.GOOGLE.COM
10	ALT3.ASPMX.L.GOOGLE.COM
10	ALT4.ASPMX.L.GOOGLE.COM

FIGURE 52

+ Click if you can't find the text for the TXT Data field

Select a domain.

Domain: nonstopnewcomer.com

Reset a Zone File

Reset the entries in your zone file to a default state. [more »](#)

Add a Record

1 Name: nonstopnewcomer.com. ✓

2 TTL: 86400 ✓

3 Type: TXT

4 TXT Data: google-site-verification=0dl ✓

Add Record

FIGURE 53

Add a New Record

Priority: 1 ✓

Destination: ASPMX.L.GOOGLE.COM ✓

Add New Record

MX Records

PRIORITY	DESTINATION	ACTIONS
5	ALT1.ASPMX.L.GOOGLE.COM	Edit Delete
5	ALT2.ASPMX.L.GOOGLE.COM	Edit Delete
10	ALT3.ASPMX.L.GOOGLE.COM	Edit Delete
10	ALT4.ASPMX.L.GOOGLE.COM	Edit Delete

FIGURE 54

Once you've finished adding the entries, go back to the Google Workspace (G Suite) interface and click the Verify Domain and Set Up Email Button:

✓ I have opened the control panel for my domain.

✓ I created the new MX records.

✓ I have deleted existing MX records.

✓ I saved the MX records.

VERIFY DOMAIN AND SET UP EMAIL

FIGURE 55

And if you did everything correctly, you should get a success message:

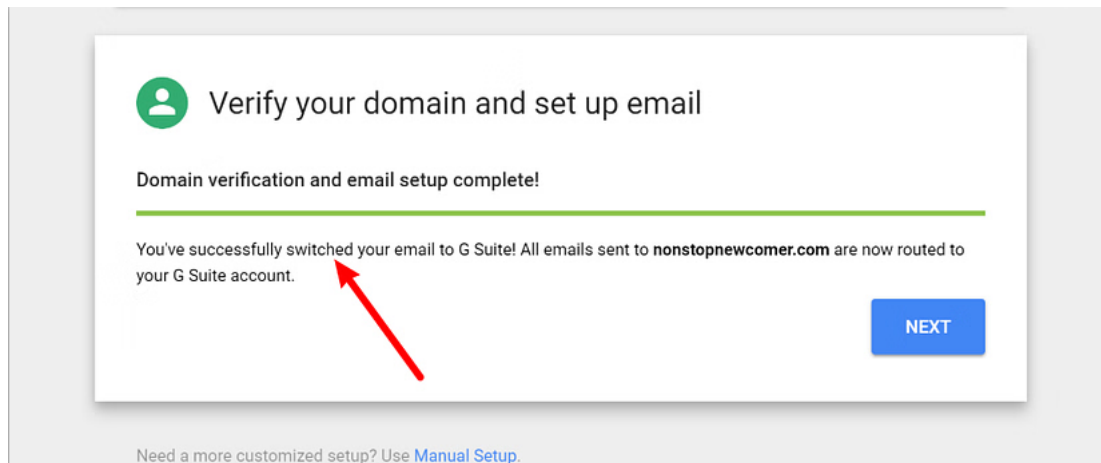


FIGURE 56

You just set up Google Workspace (G Suite) and your email should start working soon – enjoy! Note, it might take a few hours before your email starts working, so don't worry if you don't receive new emails right away.

WRAP UP + HOW TO MANAGE YOUR NEW GOOGLE WORKSPACE (G Suite) ACCOUNT

At this point, you've set up Google Workspace (G Suite) and your professional email address should be working. You can switch to your new Google Workspace (G Suite) account by sign into admin console.

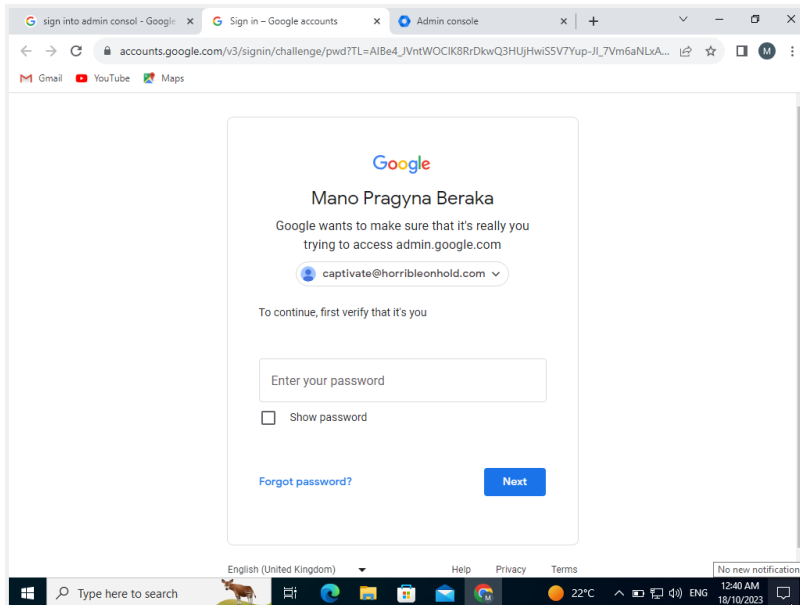


FIGURE 57

Admin Console:

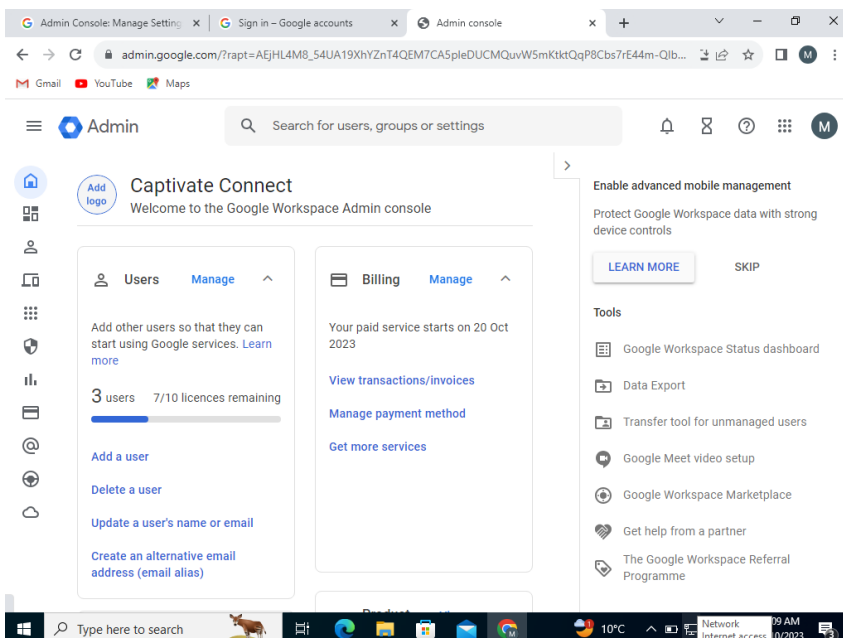


FIGURE 58

And if you to manage your Google Workspace (G Suite) – like adding new users or managing apps – you can do that from the Google Workspace (G Suite)

LINODE

What is Linode?

A trusted and well-known cloud hosting company, Linode focuses on hosting virtual private servers (VPS). Linode has been the go-to option for developers, companies, and people looking for dependable and scalable cloud infrastructure solutions ever since it launched in 2003. It has grown in popularity over time thanks to its simplicity, transparency, and developer-friendly approach. At its core, Linode enables users to create, customize, and manage virtual servers hosted in data centers around the world. These virtual servers, or Linode, can run a wide range of applications, including web servers, databases, content management systems, and more. Linode simplifies cloud infrastructure, making it accessible to both newcomers and experienced professionals in the field. With more than 800,000 clients from 196 countries and data centers in 11 different locations, it had developed over the years into an international firm that was widely known among developers, hobbyists, and small businesses alike. Before being acquired, Linode was known as one of the biggest independent cloud infrastructure companies. Akamai Technologies paid \$900 million for Linode's acquisition in February 2022. On February 14, 2023, Akamai stopped marketing the Linode name after introducing its "Akamai Connected Cloud" product line, which was built using the Linode cloud computing technology and Akamai's current content delivery network operations.



FIGURE 59

Key Features

Linode offers a wide range of services and features to meet the diverse needs of its user base:

1. **Virtual Private Servers (VPS):** At the core of Linode's offerings are virtual private servers. These VPS instances are virtualized cloud servers that provide users with dedicated computing resources, including CPU, RAM, and storage. Linode users can choose from various server plans based on their performance and capacity requirements.
2. **Data Centers:** Linode has a global presence with data centers strategically located in multiple cities worldwide. This geographic diversity allows users to deploy their servers in proximity to their target audience, minimizing latency and enhancing performance. As of my last knowledge update in September 2021, Linode had data centers in regions including North America, Europe, and Asia.
3. **Block Storage:** Linode provides scalable block storage options that can be easily attached to your VPS. This feature enables users to expand their storage capacity without the need to upgrade their entire server.
4. **Managed Services:** Linode offers managed services to assist users with server maintenance and security. Managed services include automated server backups, node balancers (a load balancing solution), and Longview (a server performance monitoring tool).
5. **Kubernetes Hosting:** Recognizing the growing popularity of containerization and microservices, Linode offers managed Kubernetes services. This simplifies the deployment and management of containerized applications, allowing users to leverage the power of Kubernetes without the complexity.
6. **Object Storage:** Linode's Object Storage is designed for efficiently storing and serving large volumes of unstructured data, making it suitable for various use cases, including content distribution, backups, and archival storage.
7. **Developer-Friendly:** Linode is known for its user-friendly control panel and management interface. It's designed with developers in mind, and its straightforward approach makes it accessible to both experienced and novice users. The Linode Manager, as it's called, provides easy access to all essential management and configuration options.

8. **Community and Support:** Linode has cultivated an active and supportive user community. The Linode Community Forum serves as a valuable resource where users can seek advice, share knowledge, and troubleshoot issues. Additionally, Linode offers responsive customer support to assist users with technical problems and inquiries.

Get Started

Platform - Getting Started on the Linode Platform

1. Create Linode Account

- ☐ To use Linode services, you must first create a Linode account.
- ☐ Go to login.linode.com/signup in your web browser to sign up.
- ☐ Enter a unique username, a secure password, and your email address. You can also register by using an existing Google or GitHub account.
- ☐ Soon after, a confirmation email is sent to your email address. To verify your email address, click the link in that email.
- ☐ In the Security section of the signup process, input your phone number. Linode uses phone numbers as a measure to verify identity and prevent fraud. Once a phone number is provided, a verification code is sent to that number as an SMS text message. Enter this code into the signup form.
- ☐ Within the Billing section, select payment method and enter the billing address details. Linode accepts credit/debit cards, Google Pay, and PayPal.
- ☐ Once the account has been created, consider enabling 2FA to prevent unauthorized access.

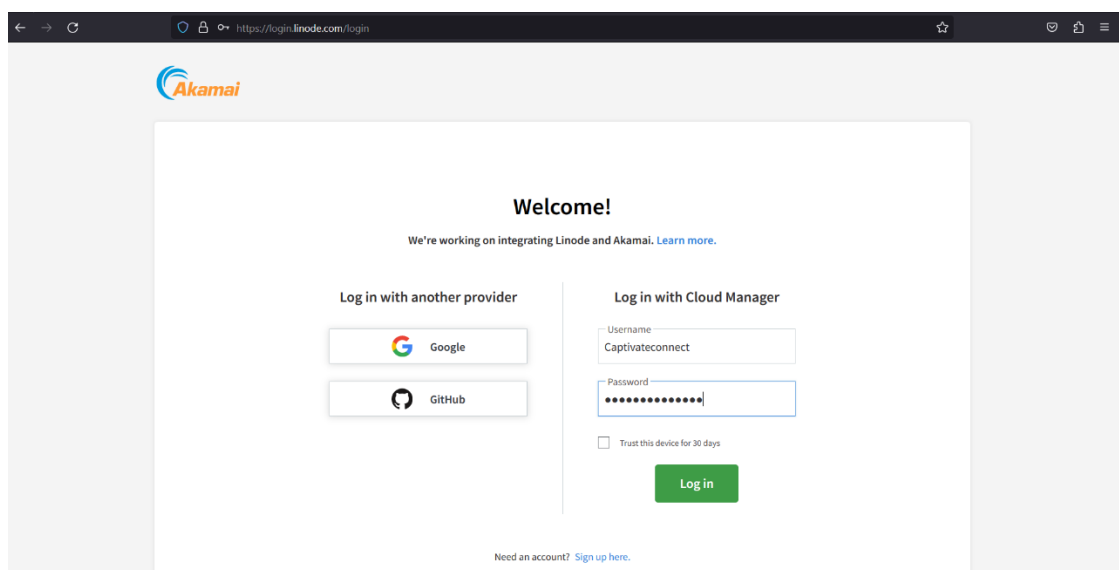


FIGURE 60

Login Page

2. Navigate to the Linode Manager

Log in to the Linode Manager using your account credentials. The Linode Manager is the control panel for managing your Linode and other services.

- ❑ **Linode:** View a list of your Compute Instances. Clicking on one takes you to its details page, where you can power it on or off, reboot it, resize it, access the console, change its configuration, attach Block Storage volumes, and lots more.
- ❑ **Account:** View your account, billing information, payment methods, and previous invoices. You can also add additional users to your account.
- ❑ **Help & Support:** Search through our guides, find answers on the Community Site, or open a ticket to reach our 24/7 award-winning Support Team.

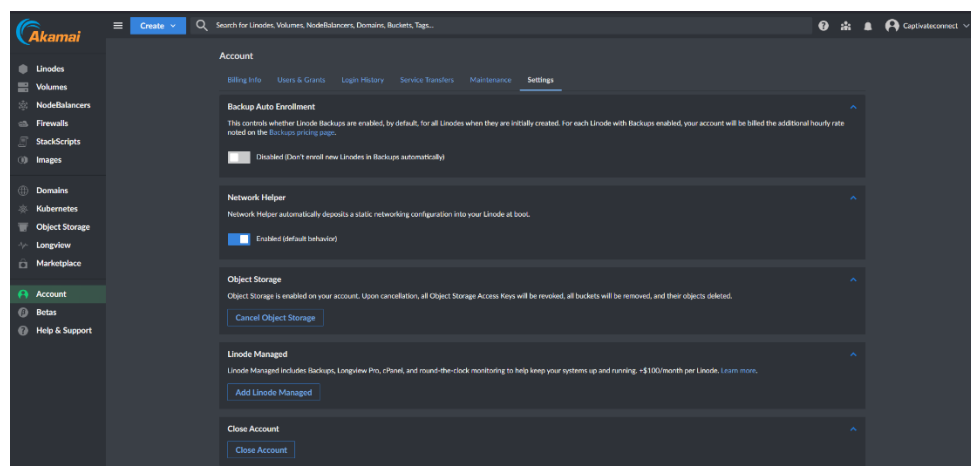


FIGURE 61

Linode Account Manager

3. Create a Compute Instance

- ❑ Log in to the Cloud Manager, click the Create dropdown menu on the top bar, and select Linode.
- ❑ Select other requested options like Distribution, Region, Linode Plan, Linode Label, Root password, SSH keys, VLAN and Add-ons.
- ❑ After that, click on Create Linode.
- ❑ Once Linode is created, it will look something like this.

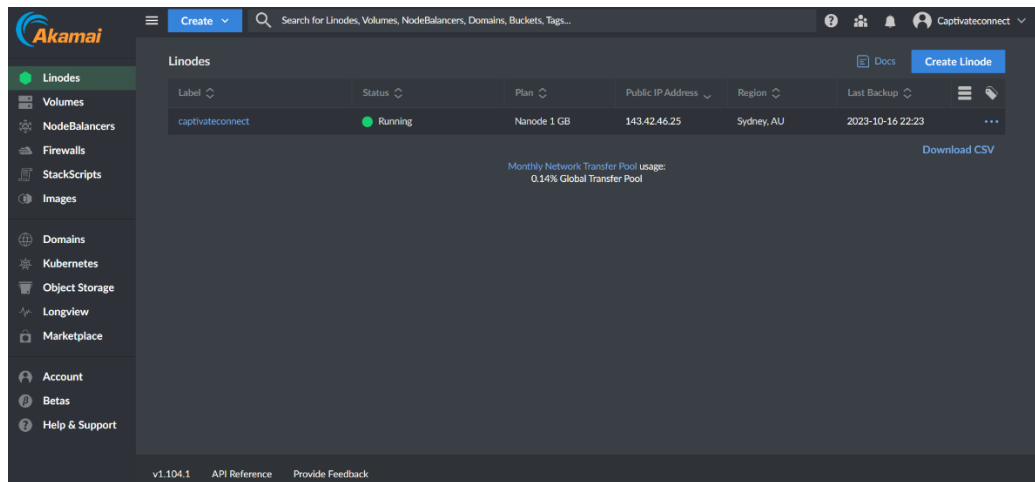


FIGURE 62

4. Getting Started After Deployment

There are a few initial configuration steps we should carry out on the new Linux system after we have successfully built a Compute Instance. This involves updating your system, defining a custom hostname, adding a restricted user, hardening SSH to prevent unauthorized access, and configuring a firewall. These steps ensure that our instance is up-to-date, secure, and ready to use.

- In the cloud manager, you can see your instance. Here it opens the instance's details page, where we can find and view critical information about our instance.

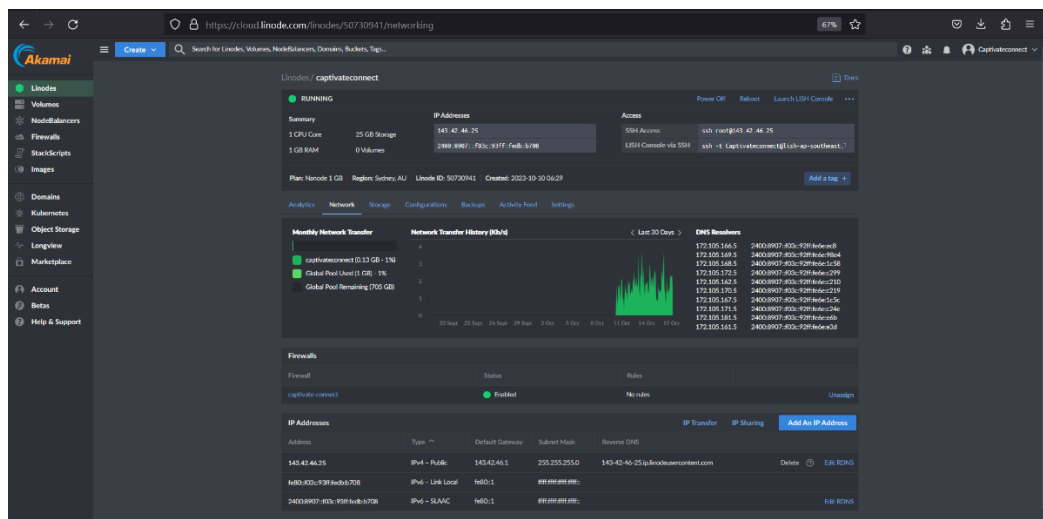


FIGURE 63

- **Connect to Instance**

The SSH (Secure Shell) protocol is often used to connect to your instance, however you can use the Lish Console to bypass SSH and connect directly to your instance.

The Lish Console can be accessed by a web browser (Weblish) or command line SSH.

Here we have used SSH.

- First we copied the command from SSH Access field under Access section on the compute instance's detail page and paste it into our windows cmd with the ip address of our instance i.e. captivate connect.

```
ssh root@192.0.2.17
```

- Lish (via SSH): Copy the command from the LISH Console via SSH field under the Access section on the Compute Instance's detail page and paste it into your local computer's terminal. The command should look similar to the one below, only with your username, data center, and Compute Instance label.

```
ssh -t user@lish-newark.linode.com example-instance
```

- Perform System Updates

```
apt update && apt upgrade
```

- Set the Timezone

```
dpkg-reconfigure tzdata
```

- Check the time

After setting up the time zone, you can check it by following command.

```
root@localhost:~# date
```

```
Thu Feb 16 12:17:52 EST 2018
```

- Configure the custom hostname

A hostname is used to identify your Compute Instance using an easy-to-remember name. It can be descriptive and structured or a generic word or phrase.

This hostname can be used as part of a FQDN (fully qualified domain name) for the system (ex: web-01-prod.example.com).

```
hostnamectl set-hostname example-hostname
```

- Update Your System's host File

The hosts file creates static associations between IP addresses and hostnames or domains which the system prioritizes before DNS for name resolution.

1. Open the hosts file in a text editor

nano /etc/hosts

2. Add a line for your Compute Instance's public IP address. You can associate this address with your instance's Fully Qualified Domain Name (FQDN) if you have one, and with the local hostname you set in the steps above. In the example below, 203.0.113.10 is the public IP address, example-hostname is the local hostname, and example-hostname.example.com is the FQDN.

File: /etc/hosts

127.0.0.1 localhost.localdomain localhost

203.0.113.10 example-hostname.example.com example-hostname

3. Add a line for your Compute Instance's IPv6 address. Applications requiring IPv6 will not work without this entry:

File: /etc/hosts

127.0.0.1 localhost.localdomain localhost

203.0.113.10 example-hostname.example.com example-hostname

2600:3c01::a123:b456:c789:d012 example-hostname.example.com example-hostname

☐ Add Limited User Account

We have accessed our compute instance as the root user, which has unlimited privileges and can execute any command—even one that could accidentally disrupt your server. Administrative tasks will be done using `sudo` to temporarily elevate your limited user's privileges so you can administer your server.

1. Create the user, replacing `example_user` with your desired username. You'll then be asked to assign the user a password:

adduser example_user

2. Add the user to the `sudo` group so you'll have administrative privileges:

adduser example_user sudo

☐ Log in as the New User

1. After creating limited user, disconnect from compute instance

exit

2. Log back in as your new user. Replace example_user with your username, and the example IP address with your instance's IP address:

ssh example_user@192.0.2.17

5. Harden SSH Access

By default, password authentication is used to connect to your Compute Instance via SSH. A cryptographic key-pair is more secure because a private key takes the place of a password, which is generally much more difficult to decrypt by brute-force.

- Create & Upload SSH Key

To protect your user account with public key authentication, you first need to create an SSH key pair and upload the public key to your server.

- 1 We located our existing SSH key. If you have an existing SSH key, find the public key on your local machine. SSH keys are typically stored in a hidden .ssh directory within the user's home directory.

Linux: **/home/username/.ssh/**

macOS: **/Users/username/.ssh/**

Windows: **C:\Users\Username\.ssh**

```
root@Captivate:~# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa): jack
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in jack
Your public key has been saved in jack.pub
The key fingerprint is:
SHA256:g3Fv4AmDyR4/Gpx9bYozeoBvC1/Jlv/c9EvY0J9iJvY root@Captivate
The key's randomart image is:
+---[RSA 3072]-----+
|
| . o
| = + o
| o = B =
| . = S =
| . + * =
| ....X . o .
| oo+.+ .oo=* .
| .+o .oo=E+* .
+---[SHA256]-----+
root@Captivate:~#
```

FIGURE 64

- 2 Upload the public key to your Compute Instance. Replace example_user with the name of the user you plan to administer the server as and 192.0.2.17 with your instance's IP address.
On your Compute Instance (while signed in as your limited user):

sudo nano /etc/ssh/sshd_config

From your local computer:

scp C:\Users\MyUserName\.ssh/id_rsa.pub example_user@192.0.2.17:~/.ssh/authorized_keys

- 3 Finally, you'll want to set permissions for the public key directory and the key file itself. On your Compute Instance, run the following command:
sudo chmod -R 700 ~/.ssh && chmod 600 ~/.ssh/authorized_keys
- 4 Now exit and log back in to Compute Instance

□ SSH Daemon Options

Lastly, edit the SSH configuration file to disallow root login and disable password authentication over SSH.

- I. Open the SSH configuration file on your Compute Instance using a Linux text editor, such as nano or vim:
 - a. **sudo nano /etc/ssh/sshd_config**
- II. Disallow root logins over SSH. This requires all SSH connections be by non-root users. Once a limited user account is connected, administrative privileges are accessible either by using sudo or changing to a root shell using su –
 - a. **# Authentication:**
 - b. **...**
 - c. **PermitRootLogin no**
- III. Disable SSH password authentication. This requires all users connecting via SSH to use key authentication. Depending on the Linux distribution, the line PasswordAuthentication may need to be added, or uncommented by removing the leading #.
 - a. **# Change to no to disable tunnelled clear text passwords**
 - b. **PasswordAuthentication no**
- IV. Listen on only one internet protocol. The SSH daemon listens for incoming connections over both IPv4 and IPv6 by default. Unless you need to SSH into your Compute Instance using both protocols, disable whichever you do not need. This does not disable the protocol system-wide, it is only for the SSH daemon. Depending on the Linux distribution, the line AddressFamily may need to be added, or uncommented by removing the leading #
 1. Use the option:
 2. AddressFamily inet to listen only on IPv4.
 3. AddressFamily inet6 to listen only on IPv6.
 - a. **# Port 22**
 - b. **AddressFamily inet**

V. Restart the SSH service to load the new configuration.

6. Configure a Firewall

Using a firewall to block unwanted inbound traffic to your Compute Instance provides a highly effective security layer. By being very specific about the traffic you allow in, you can prevent intrusions and network mapping. A best practice is to allow only the traffic you need, and deny everything else. See our documentation on some of the most common firewall applications:

- `nftables` or its predecessor, `iptables`, is the controller for `netfilter`, the Linux kernel's packet filtering framework. One of these utilities is included in most Linux distributions by default.
- `firewalld` is a firewall management tool that serves as a frontend to `nftables` or `iptables`. It is preinstalled on the RHEL family of distributions (and others), including CentOS, AlmaLinux, Rocky Linux, Fedora, and openSUSE Leap.
- `UFW` is another firewall management tool that operates as a frontend to `nftables` or `iptables`. It is used by default on Ubuntu and is also available on other Debian-based distributions.

Introduction to Restic

Restic is an open-source backup and restore software designed for secure and efficient handling of backup data. It's written in Go and is known for its simplicity and robustness. Restic provides features like data deduplication, strong encryption, and support for multiple storage backends, making it a versatile and reliable choice for data backups.

Understanding Linode Object Storage:

Linode Object Storage is a scalable and cost-effective cloud storage service provided by Linode. It's based on the S3-compatible API, making it compatible with various S3 client applications. Object storage is designed for storing and managing vast amounts of unstructured data, such as images, videos, backups, and archives. Unlike traditional file systems, object storage does not organize data into a hierarchy of folders but instead uses a flat structure with objects that are uniquely identified by their keys.

Advantages of Object Storage

1. **Scalability:** Object storage is highly scalable, allowing organizations to easily expand their storage capacity as data requirements grow. It can handle vast amounts of unstructured data, making it ideal for big data and multimedia applications.
2. **Durability and Redundancy:** Object storage systems are designed to be highly durable. Data is typically replicated across multiple servers or even data centers, ensuring that it remains available even in the case of hardware failures.
3. **Cost-Effective:** Object storage often follows a pay-as-you-go pricing model, making it cost-effective for businesses. You only pay for the storage you use, without the need for significant upfront investments.
4. **Data Accessibility:** Objects are typically stored with associated metadata, making it easy to retrieve and search for specific data. This is especially useful for managing and accessing large datasets.
5. **Data Security:** Object storage systems typically offer robust security features, including access controls, encryption, and authentication mechanisms to protect data from unauthorized access.
6. **Geographical Distribution:** Many object storage systems support geographic distribution, allowing data to be stored in multiple locations. This improves data availability and reduces latency for users in different regions.
7. **Versatility:** Object storage is versatile and can be used for a wide range of applications, including data backup, content distribution, archival storage, and more.
8. **Simple API:** Object storage systems often provide simple and standardized APIs for developers to interact with the storage, making it easy to integrate with various applications and services.

Setting up Linode Object Storage:

- ☐ To set up Linode Object Storage, follow these steps:

1. Log In to Linode: Sign in to your Linode account.
2. Create an Object Storage Service: In the Linode Cloud Manager, navigate to the "Object Storage" section and create a new Object Storage service. You'll need to choose a region and create your Linode Object Storage access key and secret key.
3. Configure the S3-Compatible Endpoint: Make note of the S3-compatible endpoint provided by Linode for your object storage. This is the URL where your data will be stored, and you'll need it to configure Restic.

❑ **Creating an Object Storage Bucket:**

Once your Linode Object Storage is set up, you can create a bucket to organize your data. Buckets are like containers for your objects. To create a bucket:

1. Log in to your Linode Object Storage account.
2. Navigate to the "Buckets" section and click on "Create Bucket."
3. Provide a unique name for your bucket, select the region where it will be hosted, and set any access controls if needed.
4. After creating the bucket, you can start uploading objects (files) to it using the S3-compatible API or client tools like Restic.

Install Restic

1. Download the latest version of Restic from the Github Releases page

wget

https://github.com/restic/restic/releases/download/v0.15.2/restic_0.15.2_linux_amd64.bz2

2. Extract the downloaded file:

bzip2 -d restic_0.15.2_linux_amd64.bz2

3. Move the extracted file to your system's \$PATH and make it executable for all users:

```
sudo mv restic_0.15.2_linux_amd64 /usr/local/bin/restic  
sudo chmod ugo+x /usr/local/bin/restic
```

4. You can now run Restic using the command restic:

```
restic version
```

5. You should see a similar output:

```
restic 0.15.2 compiled with go1.20.3 on linux/amd64
```

Create the Restic Repository

1. Configure Restic to use your Object Storage access key pair and to use the bucket you created in the Before You Begin section of this guide. Replace your-key, your-secret, and us-east-1.linodeobjects.com/your-bucket-name with your own values.

```
AWS_ACCESS_KEY_ID=your-key AWS_SECRET_ACCESS_KEY=your-secret  
restic -r s3:us-east1.linodeobjects.com/your-bucket-name init
```

2. Following the prompt, set a password to encrypt your repository's data. Enter your desired password twice, and you will see a similar output confirming that your repository has been created:

```
enter password for new repository:
```

```
enter password again:
```

```
created restic repository c3ffbd1ea6 at s3:us-east-1.linodeobjects.com/restic-  
backups-example
```

(Please note that knowledge of your password is required to access the repository. Losing your password means that your data is irrecoverably lost.)

Store the access key and secret

Your access key, secret key, and password are required every time Restic communicates with your repository. To make it easier to work with your repository, create a shell script containing your credentials.

1. To keep your credentials secure, using a text editor, create the example script in the root user's home directory, and run all your Restic scripts as the root user. The example uses the Nano text editor.

```
sudo nano /root/restic_params
```

Copy and paste the example file's content and replace your-key, and your-secret with your own Object Storage account's access key credentials.

```
export AWS_ACCESS_KEY_ID=your-key
```

```
export AWS_SECRET_ACCESS_KEY=your-secret
```

2. Create a password file to hold your Restic password:

```
sudo nano /root/restic_pw
```

Enter your Restic password and save the file.

```
YourPasswordGoesHere
```

Backup all Files

This section's example script instructs Restic to backup all files on the filesystem, starting from the root filesystem. If you have mounted block storage devices, these will also be backed up.

1. Create a file in your /usr/local/bin directory:

```
sudo nano /usr/local/bin/backup_files
```

2. Copy the following contents into the file:

```
#!/bin/bash
```

```
PATH="/usr/local/bin:$PATH"
```

```
source /root/restic_params
```

```
restic -r s3:us-east-1.linodeobjects.com/your-bucket-name -p /root/restic_pw -q  
backup --exclude={"/dev,/media,/proc,/run,/sys,/tmp,/var/tmp"} /
```

3. Make the script executable:

```
sudo chmod u+x /usr/local/bin/backup_files
```

4. Run your first backup using the script you created:

```
sudo backup_files
```

5. Executing the script creates a snapshot in your Restic repository. Use Restic's snapshot command to view it:

```
sudo /bin/bash -c "source /root/restic_params; restic -r s3:us-east-  
1.linodeobjects.com/your-bucket-name -p /root/restic_pw snapshots"
```

Set the Automated Backups

Linux has several ways of running a job on a defined schedule. This section outlines several common methods that you can use to configure the backup script to run periodically. Read through the methods and select one that suits your needs.

1. Cron

System Cron jobs exist as entries in the `/etc/crontab` file. Open your systems crontab file for editing with the following command:

```
sudo crontab -e
```

Add a line pointing to your backup script. This example runs the backup at 12am every day. See the [Schedule tasks with Cron](#) article for additional scheduling options.

```
0 0 * * * /usr/local/bin/backup_files > /tmp/backup-log.txt 2>&1
```

2. Systemd

Systemd can run commands (known as units) on a periodic basis using timers. You can use systemd commands to monitor when the timers and commands last ran, and the output from running the commands.

To schedule a command, you need two configuration files: the service file which includes the commands to run and a timer file which defines when to run the service.

Create the service configuration file and copy and paste the contents of the example:

```
sudo nano /etc/systemd/system/backup-files.service
```

```
[Unit]
```

```
Description=Backup local filesystems
```

```
[Service]
```

```
ExecStart=/usr/local/bin/backup_files
```

```
Environment=USER=root HOME=/root
```

Create the timer configuration file and copy and paste the contents of the example. The `OnCalendar` line instructs Systemd when to execute the service file's commands. In the example, the service file's commands are run at 12am every day.

```
sudo nano /etc/systemd/system/backup-files.timer
```

```
[Unit]
```

```
Description=Backup local filesystems
```

```
[Timer]
```

```
OnCalendar=*-*-* 00:00:00
```

```
[Install]
```

```
WantedBy=timers.target
```

When you are satisfied with your timer's configurations, enable the timer:

```
sudo systemctl enable --now backup-files.timer
```

You can monitor all your system's timers with the following command:

```
sudo systemctl list-timers
```

Finishing Up

Log into your Linode Cloud Manager account and view the Object Storage bucket you created to store your Restic backups. You should see a set of files like the ones displayed in the screenshot below. These files collectively make up the Restic repository; you will not see your individual files.

To explore the backups and files held within the Restic repository, you must issue the `restic` command from the machine that you installed Restic on when following the steps in the Before You Begin section.

Authentication and Security Considerations

☐ Authentication

1. **Linode Object Storage Access Key:** You'll need an access key to connect to your Linode Object Storage. Ensure this key is kept secure and not shared unnecessarily.
2. **Restic Password:** Restic requires a password for encrypting your backups. Keep this password secure and do not store it in plaintext on your Linode.

☐ Security

1. **Encryption:** Restic encrypts your backups by default, but you can add an additional layer of security by using server-side encryption on your Linode Object Storage.
2. **Firewalls:** Restrict access to your Linode and Object Storage to only trusted IPs.
3. **Regularly Review Access:** Regularly review who has access to your backups and Object Storage. Remove access for users who no longer require it.

Best Practices for Recovery

Backups are only as good as your ability to restore them. Here are some best practices for a smooth recovery:

1. **Test Your Backups:** Regularly perform test restores to ensure that your backups are working correctly.
2. **Documentation:** Keep detailed documentation of your backup and restore procedures. This documentation should be easily accessible to your team.
3. **Backup Verification:** Use `restic check` to verify the integrity of your repository.
4. **Incident Response Plan:** Create a plan for restoring data in case of an incident. Ensure that relevant personnel are aware of the plan and know how to execute it.

Retention Policy:

A clear retention policy is essential for managing your backups effectively. The specifics of your retention policy should be determined by your organization's needs, compliance

requirements, and the resources available for backup storage. Here's a guideline for defining a retention policy:

1. Determine Backup Retention Periods:
 - ☐ Daily Backups: Decide how long you want to keep daily backups. Typically, daily backups are retained for a limited period, such as 7-14 days, to provide quick access to recent data.
 - ☐ Weekly Backups: Weekly backups can be retained for a longer duration, such as 4-8 weeks, to cover the past month or two.
 - ☐ Monthly Backups: These are usually retained for several months to a year or more, depending on your specific needs. They serve as historical snapshots
2. Decide the Number of Versions:
 - ☐ Full Backups: The number of full backups you retain depends on how often you perform full backups. If you do a full backup weekly, you may retain the last 4-8 full backups, which provides coverage for the past month or two.
 - ☐ Incremental or Differential Backups: Incremental or differential backups are generally retained for a shorter period, typically in alignment with your daily or weekly backup retention. You may keep the last 7-14 daily incremental backups, for instance.
 - ☐ Archival Backups: For critical data that needs to be retained for an extended period, consider creating an archival policy. These backups can be stored separately and retained for several years or even indefinitely.
3. Compliance & Legal Requirements
If your organization operates within specific industries or geographic regions subject to data retention regulations, ensure that your retention policy aligns with these requirements. Consult with legal or compliance experts to stay in compliance.
4. Business Continuity
Consider your business continuity needs. Ensure that you retain backups for a period that allows you to recover from various types of data loss incidents, such as hardware failures, data corruption, or security breaches.
5. Testing & Documentation
Regularly test your retention policy to ensure that you can restore data as expected. Document the policy and make it easily accessible to relevant team members.

Backup Strategies & Best Practices:

1. Incremental Backups

Definition: Incremental backups only store changes made since the last backup, making them efficient in terms of storage space.

Advantages: Less storage space required, faster backup times.

Disadvantages: Longer recovery times, as you need to restore multiple backup sets.

2. Full Backups

Definition: Full backups copy the entire dataset every time they run.

Advantages: Faster recovery, as you only need one backup to restore from.

Disadvantages: Higher storage and bandwidth requirements.

3. Differential Backups

Definition: Differential backups store all changes made since the last full backup.

Advantages: Faster recovery compared to incremental backups, as you only need the last full and the last differential backup.

Disadvantages: More storage space and bandwidth required than incremental backups.

Advance Restic Features:

Restic offers various advanced features to enhance your backup process:

- ☐ Pruning: Use restic prune to remove old snapshots according to your defined retention policy. This helps in maintaining a manageable repository size.
- ☐ Tagging: You can tag your snapshots with useful metadata, making it easier to manage and locate specific backups.
- ☐ Cache: Implement a cache to improve restore performance and reduce the load on your Linode during restores.
- ☐ Mounting: Restic allows you to mount a snapshot as a filesystem, which is useful for restoring individual files.

Linode Support & Resources

Linode provides extensive support and resources to assist with your backup needs:

- ☐ Linode Object Storage Documentation: Linode offers comprehensive documentation on how to set up and use Object Storage.
- ☐ Linode Community: The Linode Community forum is a valuable resource for troubleshooting and sharing best practices.
- ☐ Linode Support: For technical issues and support, contact Linode's customer support.

USABILITY TESTING

Objectives

- ☐ To determine problems with the design and infrastructure.
- ☐ To determine if the solution meets the requirements.
- ☐ Identify and error in the proposed solution.

Participants

We conducted open day testing for 2 days on an online platform called Gather Town, where the participants were students from all the different projects. Some of the IT faculty of Latrobe also participated.



FIGURE 65



FIGURE 66

Metrics

The metrics is analyzing the feedback from google forms, based on the features we can compare between one of the advised solutions.

Methods

Participants will be required to ready a case study and participate in a survey conducted in google forms. The evaluation is based on the analysis the drawbacks of clients' current system infrastructure and advise a new and better alternative.

CASE STUDY

Current System:

Hardware/Software:

- Small Business Server 2008 (SBS 08),
- RAID 5 storage
- Nextcloud for file storage,
- Backblaze for remote access.

Proposed System

Hardware/Software:

- Zentyal server
- Google Workspace for collaboration
- Linode for robust backup

Main Considerations

Change	Benefit	Drawback
User interface Change	New interfaces are designed for better efficiency and ease of use	Users will need training
File Storage transition from onsite to cloud based	Ensures accessibility and data integrity	Requires an internet connection definitely
Remote access replaced with google workspace	Makes accessibility easier	The transition would take time to adapt to the business
New Ecosystem google workspace	Improves productivity	Employees will take time to adapt and be productive
Support and Maintenance	Regular updates and support always available	-
Cloud based solution	No requirement for hardware	-

SUMMARY

Do you have any experience working with Servers?

11 responses

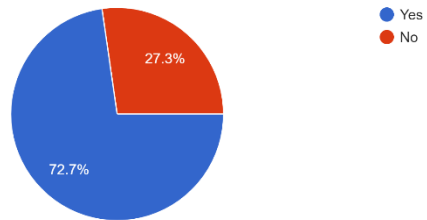


FIGURE 67

Do you know anything about the SBS 2008 Server ?

11 responses

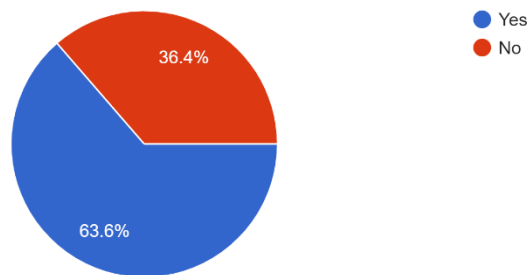


FIGURE 68

According to you, what challenges could the outdated server impose ?

11 responses

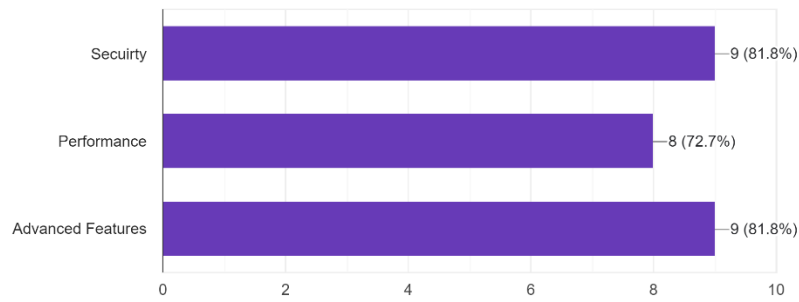


FIGURE 69

Which backup option is more liable ?

11 responses

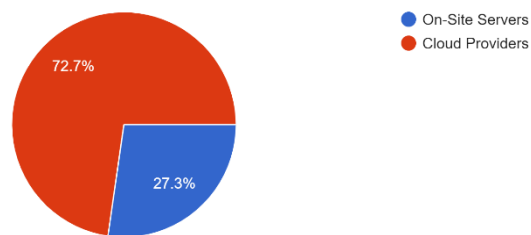


FIGURE 70

Which type of backup is advisable?

11 responses

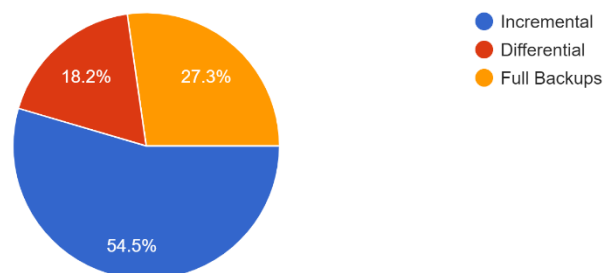


FIGURE 71

Have you tried working with open-sources Linux based Server ?

11 responses

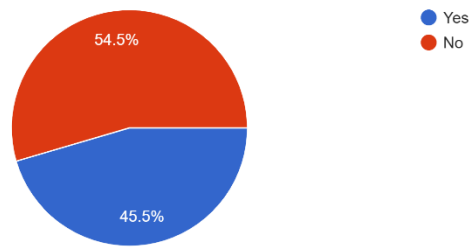


FIGURE 72

Would you like to move to a Linux-based Server and Active-Directory System?

11 responses

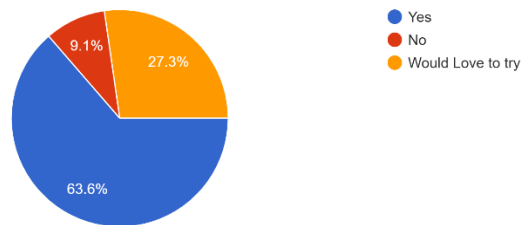


FIGURE 73

GLOSSARY TERMS

Terms	Description
Abstract	A Concise summary of the main points or key findings of a document, paper, or study, designed to provide a quick overview.
Introduction	The opening section of a document or research paper that provides an overview of the content, context, and purpose of the work.
Purpose Statement	This term is used to describe the specific goal, aim or purpose of the document or research study, The objective statement outlines what the author or researcher intends to achieve or address through their work.
Audience Description	Is a term used to describe the characteristics of a group of people who are the intended recipients of a message or communication.
System Overview	A system overview is typically used to provide stakeholders with a basic understanding of the system and how it works.
Applicability Statement	It is a document that describes which security controls from a given framework an organization has chosen to implement.
Document Usage Description	It is typically used to identify the audience for the document, the tasks that the document is intended to support, and the context in which the document is typically used.
Project Charter	It is a formal document that defines the scope, goals, objectives, and constraints of a project.
Business Case	A business case is a comprehensive document that outlines the rational ad justification for initiating a project or making a specific business decision.
Project Management Plan	A project management plan is a formal, comprehensive document outlines how a specific project will be executed, monitored, and controlled.
Work Breakdown Structure	A work breakdown structure is hierarchical and organization breakdown of a project into smaller, more manageable components or tasks.
Activity List	An activity list is a comprehensive and detailed enumeration of all the individual tasks, actions, or activities that need to be completed within a project.
Gantt Chart	A Gantt chart is a visual project management tool that display tasks or activities in timeline.
Risk Management Plan	A risk management plan is a structured document that outlines the approach and strategies for identifying, assessing, mitigation, and monitoring risks within a project or organization.
Zentyal	Zentyal is an open-source server platform designed for small and medium – sized businesses.
Server Distribution	A server distribution is a specialized operating system designed for running server applications and services. Zentyal is one such distribution that focuses on providing network services.
Network Services	Network services are software applications that provide various functionalities on a network, such as file sharing, user authentication, email, firewall, and more. Zentyal is known for offering a variety of network services in a n integrated package.

SMB (Small and Medium-Sized Business)	SMB refers to small and medium-sized businesses. Zentyal is tailored to meet the IT infrastructure needs of SMBs.
Integrated Package	Zentyal offers an integrated package of network services, meaning it provides multiple services in a single software platform, simplifying management and administration for businesses.
File Sharing	File sharing is network service that allows users to share files and documents across a network, facilitating collaboration within an organization.
User Authentication	User authentication is the process of verifying the identify of users to endure secure access to network resources. Zentyal provides authentication services to manage user access.
Email Services	Zentyal offers email services, which include email server functionality, allowing businesses to set up and manage their email communication.
Firewall	A firewall is security service that protects a network by controlling incoming and outgoing network traffic. Zentyal includes firewall capabilities to enhance network security.
Open Source	Zentyal is built on open-source software, meaning its source code is freely available and can be modified and redistributed by the community.
Linux Server	Zentyal is Linux – based server distribution, utilizing the Linux operating system as its foundation.
Google Workspace	A collection of cloud computing, productivity, and collaboration tools developed by google, including Gmail, google drive, google docs, google sheets, and more.
Domain Name	A human-readable web address used to identify a specific website or resource on the internet; like www.example.com
Domain Registrar	A company or organization that manages the registration of domain names on behalf of customers
TXT	A DNS resource record that stores text information associated with a domain.
DNS Records	Data associated with a domain name that specifies various setting, including the IP address, mail server and more.
MX Records (Mail Exchange Record)	A DNS resource that specifies the mail servers responsible for receiving email messages on behalf of a domain.
TTL (Time to Live)	A value in a TXT record that specifies how long the record should be cached by DNS resolvers, measured in seconds.
Admin Console	A web-based or software-based interface that allows administrators to configure and manage settings for a specific software, service, or system.
Linode	A cloud hosting provider that offers virtual private servers (VPS). Linode allows users to deploy and manage virtual server instances in data centres around the world.

Restic	An open-source, fast, secure, and efficient backup program. Restic is designed for data backup, restoration, and version control, offering encryption and deduplication features.
Linode Object Storage	Linode's cloud-based object storage service for storing and retrieving data. It provides scalable, durable, and cost-effective cloud storage solutions for businesses and individuals.
Repository	In the context of Restic, a repository is a storage location where backup data is stored. It can be a local directory, a network location, or cloud storage, such as Linode Object Storage.
Retention Policy	A defined set of rules that determine how long backups are kept and when they are deleted. The retention policy outlines the duration and number of versions of backups that are maintained.
Snapshot	A point-in-time copy of your data, created during the backup process. Snapshots capture the state of data at a specific moment, allowing you to restore files or systems to that particular point in time.
Cache	A local storage that stores frequently accessed data to improve performance. In the context of Restic, a cache is used to speed up restore operations by storing metadata and chunks of data that are frequently needed.
Tagging	Adding metadata to backups to make them easier to manage and search. Tags are labels or keywords associated with snapshots to categorize and identify specific backups for organization and retrieval.
Incident Response Plan	A documented procedure for responding to and recovering from data loss or other incidents. An incident response plan outlines the steps to be taken in the event of data breaches, data corruption, system failures, or other adverse events, with the goal of minimizing damage and downtime.

APPENDIX B: STATEMENT OF EFFORT



Department of Computer Science and Information Technology
CAPTIVATE CONECT: - SERVER ENHANCEMENT AND DATA MIGRATION

Section	Page	Members contributed
System Overview	9	Mano
User Story Dictionary	14	Loji
User stories Definition	15	Loji
Project Charter	18	Bhawna
Project management plan	22	Mano & Adib
Business Case	32	Mano
Work breakdown structure	34	Viraj
Activity list	35	Bhawna & Viraj
Gantt Chart	40	Loji & Madhura
Change Log	41	Bhawna
Risk Management plan	43	Adib
Proposed System	51	Mano
Implementation Report	53	Loji
Zentyal Configuration & Testing	N/A	Adib
Zentyal Documentation	61	Loji & Adib
Google Workspace Configuration and Documentation	71	Mano
Linode Configuration & Testing	N/A	Bhawna & Madhura
Linode Documentation	95	Madhura
Compiling And editing	N/A	Bhawna & Loji

Student Name	Student Id	Signature*	Date
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STATEMENT OF AUTHORSHIP



Department of Computer Science and Computer Engineering

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Please read the following instructions and associated University documents before completing and submitting this form with your item of assignment.

1. A Statement of Authorship must be completed for all items of assessment where the assessment is worth ten percent (10%) or more of the total for the unit.
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3. Statute 16 found at <http://www.latrobe.edu.au/legalservices/statutes.html>
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We each certify that our own contribution in the attached material is original work and that we have no knowledge of an act of plagiarism committed by any member of this group. We declare that no other person's work has been used without due acknowledgement. Except where we have clearly stated that we have used some of this material elsewhere, it has not been presented by us for examination in any other course or unit at this or any other institution.

We have read the Academic Misconduct Policy, Statute 16 and Regulation 16.2 relating to Student Discipline and Academic Misconduct, which are available on the University's web site and understand that we are bound by such Policy, Statute and Regulation and that we may be subject to student discipline processes in the event of an act of plagiarism by us.

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Family Name
(BLOCK LETTERS)

SONI

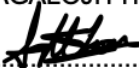
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Signature B.S. Date 29/10/2023

Family Name
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NAGARAJAH

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Signature  Date 29/10/2023

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HASHEMINEJADFARHANI

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Signature Adib H. Date 29/10/2023

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Unit Code: CSE3CAP Unit Name: CAPSTONE PROJECT

Lecturer/Tutor's Name: SCOTT MANN..... Assignment No. 3

Name of Document: STATEMENT OF AUTHORSHIP PART 1

For extra members please complete an additional form numbered 2.



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Signature

Date

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Signature..... Date

Unit Code: CSE3CAP Unit Name: CAPSTONE PROJECT

Lecturer/Tutor's Name: SCOTT MANN Assignment No. 3

Name of Document: STATEMENT OF AUTHORSHIP PART 2

1.2 Plagiarism

One form of academic cheating is plagiarism, the reproducing of someone else's words, ideas or findings and presenting them as one's own without proper acknowledgment.

There are many forms of plagiarism, including the following:

- (a) direct copying of sentences, paragraphs or other extracts from someone else's published work (including on the Internet and in software) without acknowledging the source;
- (b) paraphrasing someone else's words without acknowledging the source;
- (c) using facts and information derived from a source without acknowledging it;
- (d) using ideas directly derived from an identifiable author without acknowledging the source;
- (e) producing assignments which should be the student's own, independent work in collaboration with and/or using the work of other people (e.g. a student or tutor).

Assisting another person to plagiarise material may be punished as severely as is plagiarism itself. Assisting plagiarism may involve a student lending work (or by posting it on the Internet for sale) which is intended for submission for assessment, or which has already been submitted, so that it can be copied and handed in by another student as that student's own work.³

Students may find it helpful to discuss assignments with other students and their tutors. The University encourages students to communicate with one another in constructive ways about the learning process. Students may choose to assist each other, for example in discussing the approaches that might be taken to assignment topics, or helping with the availability of reading materials. They should, however, write their assignments independently, except when they are asked to work on a project as a member of a group which is to submit a joint report, as equal contributors. In this case, except where the final written work consists of sections for which particular individuals take sole responsibility, the group as a whole assumes responsibility for it and the proper acknowledgment of any use made of the words or ideas of people outside the group.

The increased vigilance of the University in regard to plagiarism is in response to the perceived growing incidence of plagiarism. The new recommended penalties reflect the value the University places on academic honesty and how seriously it regards plagiarism offences. The University will protect its reputation for academic integrity by ensuring that serious and serial offenders, who wish to obtain an unfair advantage through cheating, are given penalties proportionate to the offences committed.

A summary of the current legislation is attached in Appendix A. This legislation will be amended if the Working Party's recommendations are approved.

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